

CBWeb3 Platform — Scenario B User Manual

International Hub (FXAgreement + AMM + Bridge, with relay and circuit breaker)

2026-08-05

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About this document

This manual compiles the per-portal user manuals for **Scenario B — International Hub** into a single document, in the order used by the portal index: Bank, Governance, Treasury, Supervisor, NOC, and finally the cross-portal end-to-end walkthrough.

The chapters are the Markdown manuals kept in the repository under `docs/user-manuals/`. Each manual's own table of contents has been replaced by the single table of contents at the front of this document; the text is otherwise unchanged. Cross-references between manuals resolve to the corresponding chapter of this PDF.

Generated on 2026-08-05.

Implementation status — read this first

Every portal below is wired to a live backend through the API Gateway. The manuals only describe features that exist in the shipped product, and each manual states at the top exactly where its data comes from. A portal's screens are populated only when the backend services it calls are reachable; in an offline or demo environment with the backends down, screens show empty tables or zero counters rather than fabricated data.

Portal	Scenario A	Scenario B	Data source today
Bank	Live backend	Live backend	Real API (API Gateway)
Governance	Live backend (Dashboard + Registry exposed)	Live backend	Real API; Scenario A retains a development mock toggle
Treasury	Live backend (approvals exposed)	Live backend	Real API (API Gateway)
Supervisor	Live backend (connectivity-dependent)	Live backend (connectivity-dependent)	Real API; every screen calls live endpoints. See the note below for auth/data caveats
NOC	Live backend	Live backend	Real API — connects to the live NOC backend service

Important: The **Supervisor** portal's screens call real backend endpoints through the API Gateway — it is **not** a hardcoded mock-data or fake-login build, and its frontend holds no sample data. Two caveats remain, documented per screen in each manual:

- **Data freshness.** When the backend services are offline (as in most local/demo setups) screens show empty tables or zero counters, and in a development environ-

ment the endpoints may return demonstration data rather than a live production data set.

- **Authentication.** Scenario B authenticates against the platform's OIDC provider (Keycloak). In Scenario A, production Keycloak enforcement is not yet active, so the login is not a production security boundary.

Do not treat figures shown in an offline or development environment as authoritative for any real supervisory decision.

Conventions used in every manual

- **Route** — the in-app URL path (the part after the portal's base address).
- **Sidebar label** — the name as it appears in the portal's left navigation.
- **Status / state tables** — the exact status values you will see in the UI and what each one means.
- **Planned / not yet available** — a feature that exists in the design but is not exposed in the current build. These are called out explicitly and never documented as if they work.

Bank Portal — User Manual

(Scenario B)

Audience: Commercial bank operators participating in the International Hub. **Scenario:** Scenario B — International Hub (FXAgreement + AMM + Bridge). **Data source:** Live backend (real API Gateway) for all operational screens.

1. Overview

The Bank Portal is the day-to-day operational interface for a commercial bank operator in **Scenario B — International Hub**. Unlike Scenario A (which settles value bilaterally between two spokes using HTLC), Scenario B routes cross-border value through a shared **Hub** equipped with an **Automated Market Maker (AMM)** for FX. A commercial bank sends value through the **Bridge**, a single cross-currency operation over a liquidity pool that the Central Banks provision: the bank spends its own sovereign currency (tCeBM) and the beneficiary receives the pool counterpart currency on its spoke.

The end-to-end operational lifecycle is:

- 1. **Onboard** your institution with the Central Bank (one-time).
- 2. **Request tCeBM issuance** by submitting a fiat collateral proof (Issuance Requests page).
- 3. **Tokenise** the approved fiat reserve into tCeBM (Reserve Tokenisation page).
- 4. **Bridge** value cross-border by selecting a liquidity pool, obtaining a quote, and executing the swap (Bridge page).
- 5. **Redeem** tCeBM back to fiat when needed (Redeems page).

Actor	Portal	Role
Commercial Bank Operator	Bank Portal (this portal)	Manages tCeBM balances, issuance, tokenisation, redemption, and cross-currency bridge operations
Central Bank Governance Operator	Governance Portal	Approves onboarding KYC, issuance, tokenisation, and redeem requests; provisions and seeds liquidity pools

2. Access and Login

Route: /login

Open the Bank Portal URL provided by your administrator. You will land on the **Sign in to Bank Portal** screen.

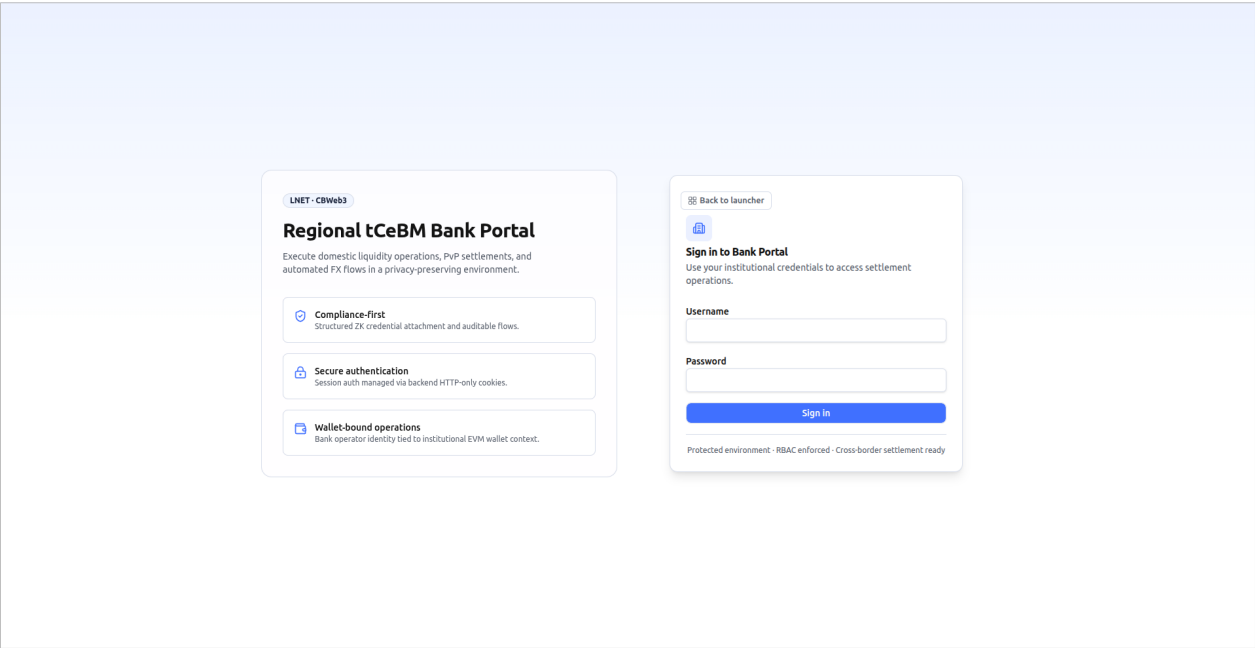


Figure 1: Login

Fill in both fields and click **Sign in**:

Field	Description
Username	The institutional service account identifier provided by the Central Bank (minimum 3 characters).
Password	The corresponding service account secret (minimum 6 characters).

Authentication is managed via backend HTTP-only session cookies. On success you are redirected to the Dashboard.

If sign-in fails immediately with correct credentials, confirm your institution’s onboarding status is **ACTIVE**. New institutions must complete onboarding before they can access operational screens.

3. Navigation

The left-hand sidebar lists all screens available in Scenario B:

Sidebar label	Route	Purpose
Dashboard	/	Real-time overview of balances, pool availability, and recent activity
Bridge	/bridge	Send a cross-currency payment over a Hub liquidity pool

Sidebar label	Route	Purpose
Bridge History	<code>/bridge/history</code>	Read-only history of your cross-currency bridge operations
Liquidity Pools	<code>/pools</code>	Read-only view of the pools your spoke can operate
Issuance Requests	<code>/deposits</code>	Request the Central Bank to issue tCeBM
Reserve Tokenisation	<code>/escrows</code>	Convert approved fiat reserve (fCeBM) into tCeBM
Redeems	<code>/redeems</code>	Convert tCeBM back to fiat
Onboarding	<code>/onboarding</code>	Institution registration wizard
Settings	<code>/settings</code>	Portal environment information

The **Compliance** page (`/compliance`) is accessible by direct URL but is not shown in the sidebar. It is described in section 4.10.

4. Screens

4.1 Dashboard

Route: `/` · Sidebar label: Dashboard

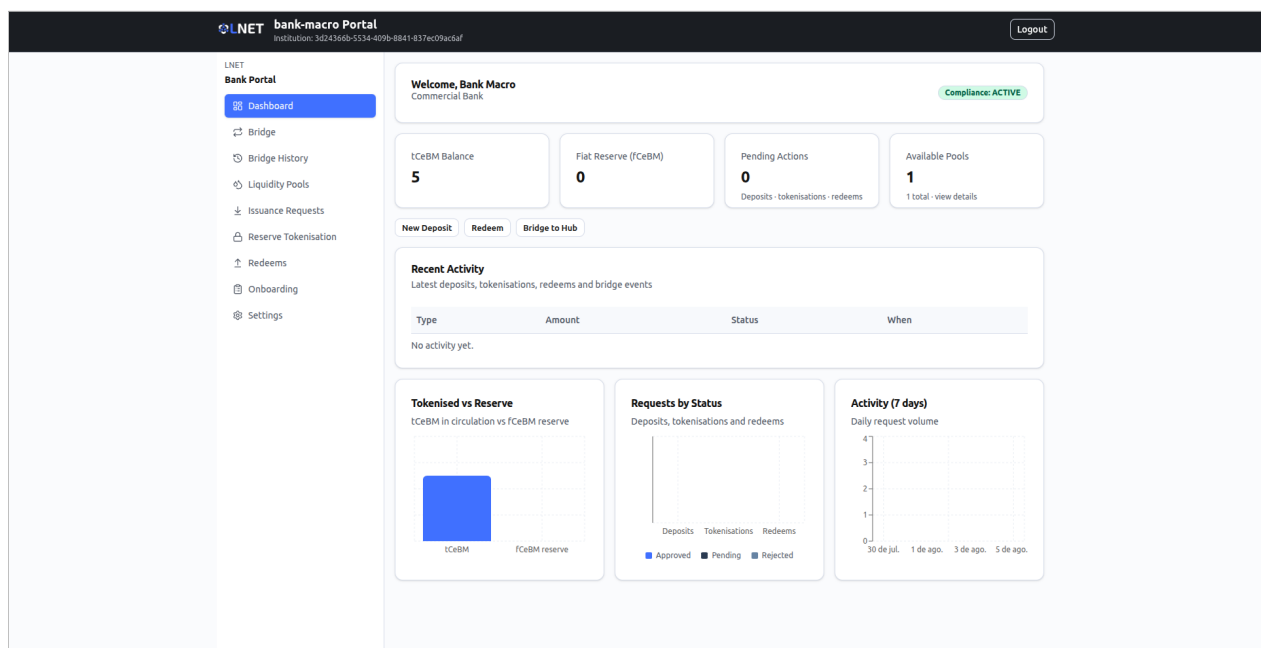


Figure 2: Dashboard

The Dashboard provides a real-time summary of your institution's position. It refreshes pool availability and bridge activity every 30 seconds automatically.

Identity / context card

Displays a welcome line with your institution name (or bank ID), your country, wallet address (truncated), roles, and a **Compliance** status badge derived from your onboarding state.

KPI row

Card	What it shows
tCeBM Balance	Your current on-chain tCeBM token balance.
Fiat Reserve (fCeBM)	Your current fCeBM (fiat-backed reserve token) balance.
Pending Actions	Combined count of deposits, tokenisations, and redeems in PENDING status.
Available Pools	Number of pools currently ACTIVE , with the total pool count. This card links to the Liquidity Pools page.

Quick actions

Shortcut buttons to: New Deposit, Redeem, and Bridge to Hub.

Recent Activity table

Chronological list (up to 8 records) of deposits, tokenisations, redeems, and bridge events, with type, amount, status badge, and timestamp.

Charts

Three bar charts summarising: **Tokenised vs Reserve** balance comparison, **Requests by Status** (Pending/Approved/Rejected) across deposits/tokenisations/redeems, and **Activity (7 days)** — daily request volume over the past 7 days.

4.2 Bridge (/bridge)

Route: /bridge · Sidebar label: Bridge

The screenshot displays the 'Bank Portal' interface for LNET. The sidebar on the left contains a 'Bank Portal' section with a 'Bridge' button highlighted. The main content area is divided into three steps:

- Step 1: Select Pool**: Includes a dropdown for 'Liquidity Pool' (BRL # ARS - W-BRL-W-ARS) and a 'Direction (fixed by sovereignty)' (BRL -> ARS). A green box indicates 'Pool W-BRL-W-ARS - ACTIVE' with reserves of 1005.025125 BRL / 995 ARS.
- Step 2: Get Quote**: Shows 'Amount Out (ARS)' as 160. A 'Get Quote' button is present.
- Quote Details**: Displays 'Effective Rate: 0.830825', 'Amount In: 192.579664 BRL', 'Amount Out: 160 ARS', and 'TTL: 15s'.
- Step 3: Execute Bridge**: Shows 'Max Amount In (BRL)' as 211.837631. A note indicates 'Suggested max from quote + slippage: 211.837631 BRL'.

Figure 3: Bridge

This is the primary screen for sending a **cross-currency payment** over a Hub liquidity pool. The bridge combines the bridge-in, swap, and bridge-out into one operation: your sovereign tCeBM

is spent on the source spoke, swapped at the Hub for the target currency, and delivered to the beneficiary's spoke.

Direction is **not** user-selectable. Your institution always spends its own sovereign currency (the source); the beneficiary receives the pool counterpart (the target). Only pools that include your sovereign currency are listed.

Step 1 — Select Pool

Field	Description
Liquidity Pool	Drop-down of ACTIVE pools that include your sovereign currency (for example BRL $\rightleftharpoons$ ARS).
Direction (fixed by sovereignty)	Read-only. Shows <source> $\rightarrow$ <target>; you spend your sovereign currency and the beneficiary receives the counterpart.

When a pool is selected, a status banner shows whether it is **ACTIVE** (with current reserves for each side) or another lifecycle state. The page polls pool status every 30 seconds.

Step 2 — Get Quote

Field	Description
Amount Out	The amount the beneficiary should receive, in the target currency (exact-output).

Click **Get Quote**. The Quote Details card appears showing:

Field	Description
Effective Rate	The implied exchange rate for this swap.
Amount In	How much of your source currency you will spend.
Amount Out	The target amount you entered.
TTL	Seconds remaining before the quote expires.

Quotes expire. If the TTL reaches zero before you execute, the Execute Bridge button becomes disabled — re-submit the Get Quote form to obtain a fresh quote. Quotes are also auto-refreshed in the background; when that happens a banner reads “*Quote was refreshed. Review and confirm again.*”

Step 3 — Execute Bridge

Once a valid quote is shown:

Field	Description
Max Amount In	Your slippage ceiling — the maximum source currency you will allow the swap to consume. Enter a value manually or click Use Suggested Max to fill in the quote amount plus slippage buffer.
Beneficiary Bank ID	The bank identifier of the recipient institution (for example <code>bank-macro</code>).

Click **Execute Bridge**. The button is disabled if the pool is not **ACTIVE**, if no quote is loaded, if the quote has expired, or if any required field is empty.

Bridge Progress

While the bridge is in flight, a four-step progress indicator is shown:

Step	Meaning
BRIDGE IN PROGRESS	Your tCeBM is being locked on your spoke.
SWAP IN PROGRESS	The AMM swap is executing on the Hub.
BRIDGE OUT PROGRESS	The swapped tokens are being delivered on the beneficiary's spoke.
COMPLETED	The bridge has settled end-to-end.

On completion, the **Bridge Completed** card shows the bridge transaction hash, amounts in and out, and a correlation ID. Click **New Bridge** to start another operation.

Bridge errors

If the operation fails, a **Bridge Failed** card shows a plain-language message and a **Try Again** button. Common causes:

Error	What it means	Action
Circuit breaker paused	The pool is paused by a Central Bank.	Wait for governance to resume, then retry.
Slippage limit exceeded	Price moved beyond your Max Amount In.	Re-quote and retry.
Bridge in failed	Funds could not be locked on the source network; the operation was rolled back and no funds were lost.	Retry.
Insufficient pool liquidity	Pool reserves too low for this amount.	Try a smaller amount or wait for the Central Bank to add liquidity.
Quote expired	Quote expired during submission; a new quote was auto-fetched.	Review the refreshed quote and confirm again.
Swap failed	The swap transaction failed on the Hub; no funds were moved.	Retry.
Daily transfer limit reached	Your Central Bank has set a volume cap for today.	Contact your Central Bank to adjust the limit.

4.3 Bridge History (/bridge/history)

Route: /bridge/history · **Sidebar label:** Bridge History

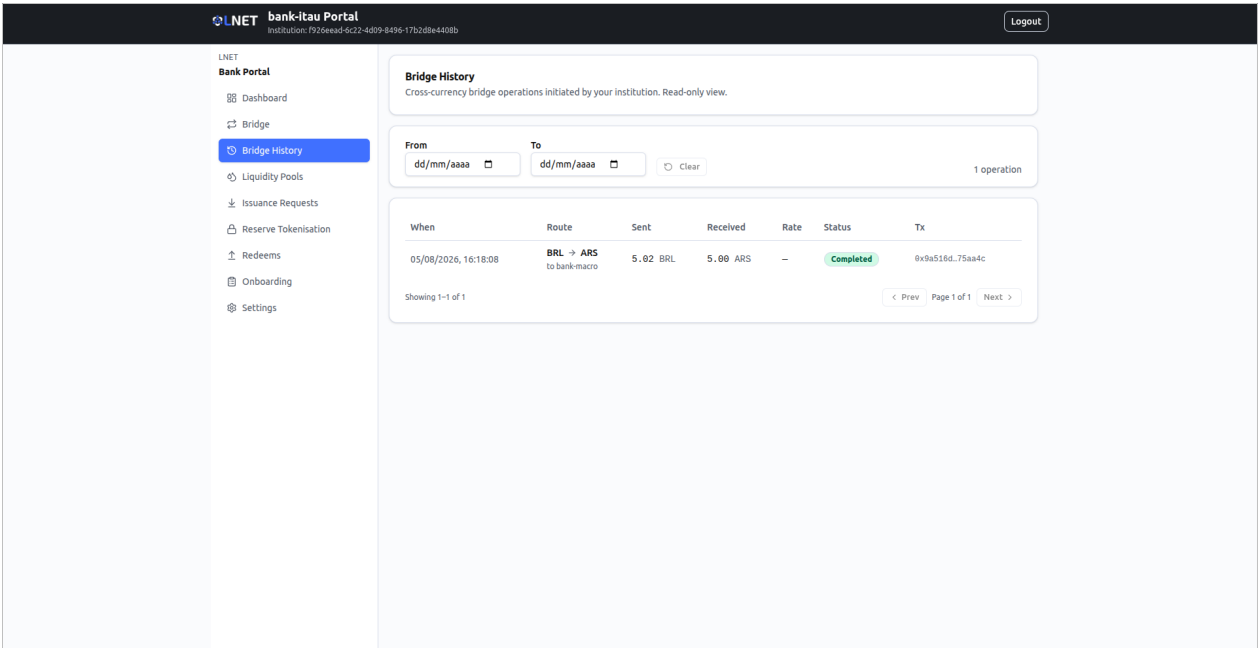


Figure 4: Bridge History

A **read-only** history of the cross-currency bridge operations initiated by your institution.

Filters

Control	Description
From / To	Date range. Changing either resets to the first page.
Clear	Resets both date filters.

A counter shows the total number of operations matching the current filter.

Operations table

Column	Description
When	Timestamp the operation was created.
Route	Source → target currency, with the beneficiary bank ID beneath it.
Sent	Amount spent, in the source currency.
Received	Amount delivered, in the target currency.
Rate	Effective exchange rate applied.
Status	Status badge (see below).
Tx	Bridge transaction hash (truncated).

Results are paginated (20 rows per page) with **Prev** / **Next** controls.

Status badges

Badge	Meaning
Completed	Operation settled end-to-end.

Badge	Meaning
Failed	Operation errored (hover the badge for the failure reason).
Quoting	A quote is being prepared.
Bridging in	Funds are being locked on the source spoke.
Swapping	The AMM swap is executing on the Hub.
Bridging out	Swapped tokens are being delivered on the beneficiary's spoke.

4.4 Liquidity Pools (/pools)

Route: /pools · Sidebar label: Liquidity Pools

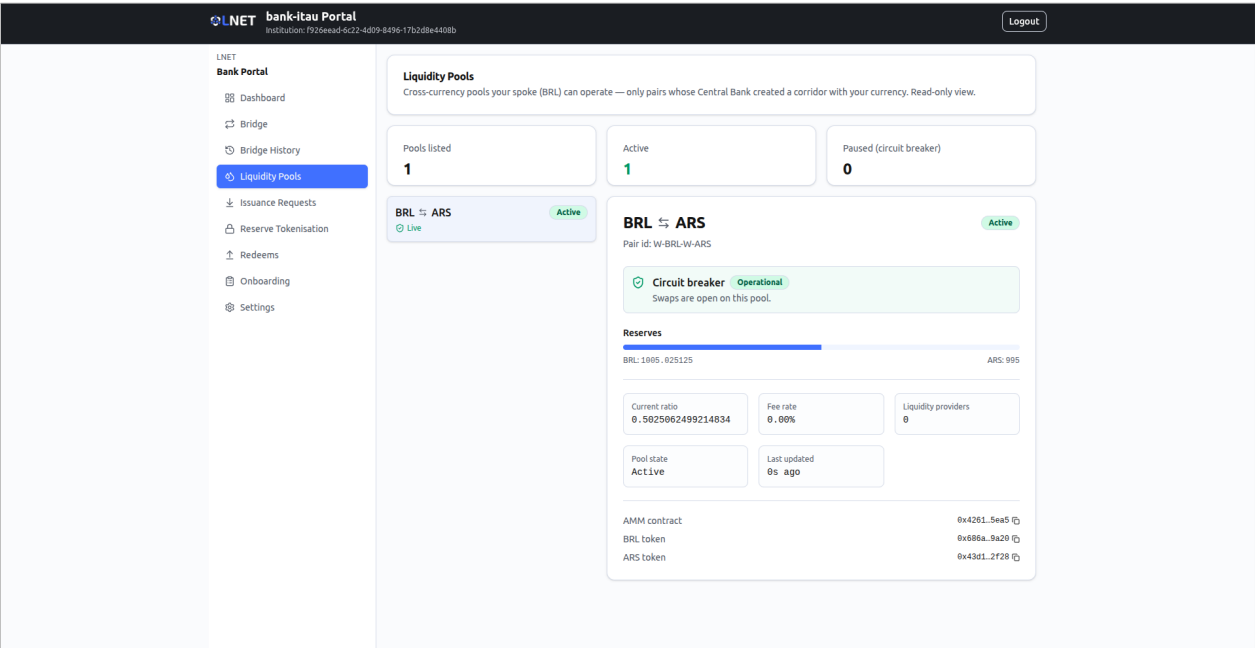


Figure 5: Liquidity Pools

A **read-only** view of the cross-currency pools your spoke can operate — that is, pools whose Central Bank created a corridor that includes your sovereign currency. The page polls live pool and circuit-breaker data every 15 seconds.

Summary cards

Card	What it shows
Pools listed	Number of pools your spoke can operate.
Active	Pools currently in ACTIVE status.
Paused (circuit breaker)	Pools whose circuit breaker is HALTED .

Pool list

A selectable list of pools, each showing the currency pair, a pool-status badge (**Active**, **Awaiting counterpart**, **Empty**, or the raw state), and a circuit-breaker indicator (**Live** / **Paused**).

Pool detail panel

Selecting a pool opens a detail panel:

- **Circuit breaker banner** — **Operational** (LIVE), **Paused** (HALTED, with reason and initiator where available), or **Resume pending** (RESUME_PENDING, awaiting the 2-of-N Central Bank quorum).
- **Reserves** — a balance bar and per-side reserve amounts, with an **Imbalanced** badge when the pool flags an imbalance.
- **Metrics** — Current ratio, Fee rate, Liquidity providers, Pool state, and Last updated.
- **On-chain references** — AMM contract address, both token addresses (click to copy), and the Proposer/Confirmer Central Bank identifiers where present.

4.5 Issuance Requests / Deposits (/deposits)

Route: /deposits · Sidebar label: Issuance Requests

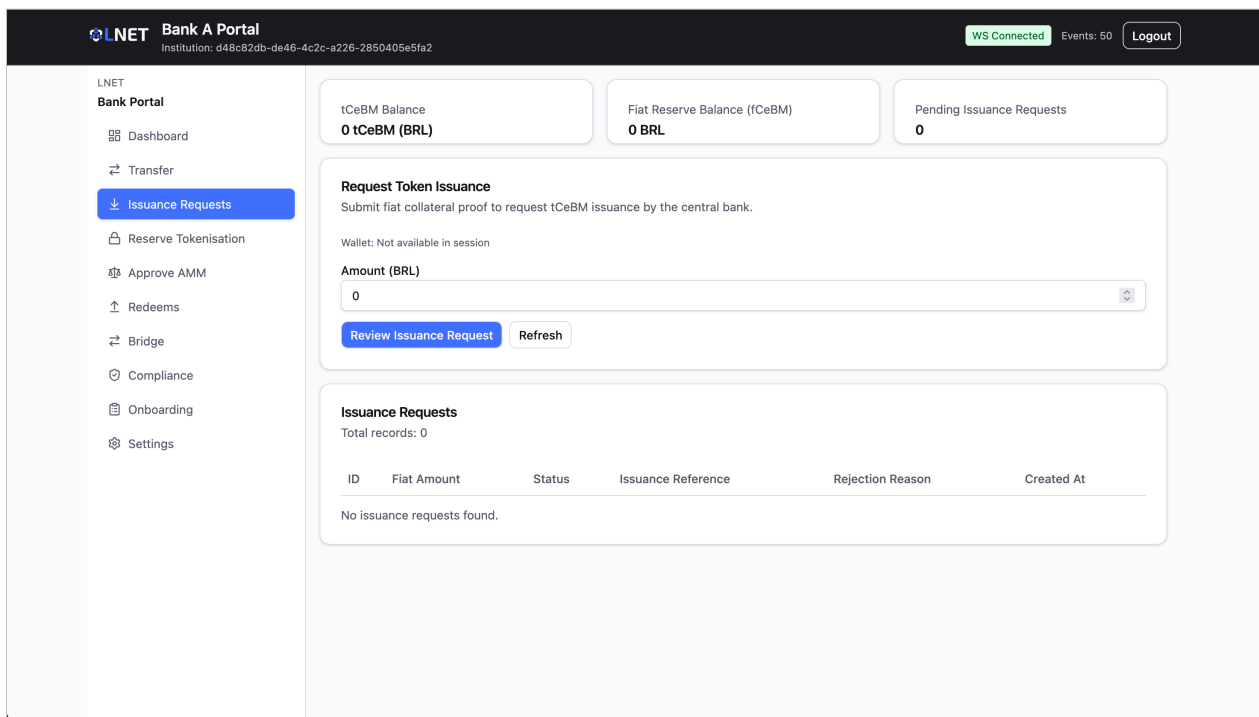


Figure 6: Deposits

Use this screen to request the Central Bank to **issue tCeBM** against your fiat collateral. The top of the screen shows your current tCeBM balance, fiat reserve (fCeBM) balance, and count of pending issuance requests.

Submitting a request

1. Enter the **Amount** in fiat currency units.
2. Click **Review Issuance Request**.
3. A confirmation card appears with the formatted amount. Click **Confirm Request** to submit, or **Cancel** to go back.
4. On success a toast notification shows the assigned request ID.

Issuance Requests table

Column	Description
ID	Unique request identifier.
Fiat Amount	The amount submitted.
Status	Current status badge (see Status Reference).
Issuance Reference	On-chain transaction hash once approved and minted (truncated).
Rejection Reason	Populated when the Central Bank rejects the request.
Created At	Submission timestamp.

tCeBM is minted only after the Central Bank approves the request from the Governance Portal. The page does not auto-refresh — click **Refresh** to update balances and statuses.

4.6 Reserve Tokenisation / Escrows (/escrows)

Route: /escrows · **Sidebar label:** Reserve Tokenisation

Figure 7: Escrows

After a deposit (issuance request) is approved, use this screen to **convert the approved fiat reserve (fCeBM) into tCeBM**. The top of the screen shows your fCeBM balance and count of pending tokenisation requests.

Submitting a tokenisation request

1. Enter or select the **Deposit ID** of an approved deposit (a drop-down datalist shows approved IDs with their amounts).
2. Enter the **Amount** to tokenise (in fiat currency units).
3. Click **Review Tokenisation Request**.
4. A confirmation card shows the fCeBM amount being converted and the equivalent tCeBM. Click **Confirm Request** to submit, or **Cancel** to go back.

Tokenisation Requests table

Column	Description
ID	Unique tokenisation request identifier.
Deposit ID	The parent deposit this request is linked to.
Amount	The fCeBM amount being converted.
Status	Current status badge.
Burn Tx (fCeBM)	Transaction hash of the fCeBM burn (truncated).
Mint Tx (tCeBM)	Transaction hash of the tCeBM mint (truncated).
Created At	Submission timestamp.

The Central Bank must approve the tokenisation request before the on-chain burn/mint executes. Click **Refresh** to update.

4.7 Redeems (/redeems)

Route: /redeems · Sidebar label: Redeems

Figure 8: Redeems

Use this screen to **convert tCeBM back to fiat**. The top of the screen shows your tCeBM balance and count of pending redeem requests.

Submitting a redeem request

1. Enter the **Amount** of tCeBM to redeem.
2. Click **Review Redeem Request**.
3. A confirmation card shows the tCeBM amount being submitted for fiat reserve release. Click **Confirm Request** to submit, or **Cancel** to go back.

- On success a toast notification shows the assigned redeem request ID.

Redeem Requests table

Column	Description
ID	Unique redeem request identifier.
Amount	The tCeBM amount submitted.
Status	Current status badge.
Redemption Tx	On-chain transaction hash of the redemption (truncated).
Rejection Reason	Populated when the Central Bank rejects the request.
Created At	Submission timestamp.

The Central Bank must approve the redeem from the Governance Portal before fiat is released. Click **Refresh** to update.

4.8 Onboarding (/onboarding)

Route: /onboarding · **Sidebar label:** Onboarding

LNEXT Bank A Portal
Institution: d48c82db-de46-4c2c-a226-2850405e5fa2

WS Connected Events: 50 Logout

LNEXT Bank Portal

- Dashboard
- Transfer
- Issuance Requests
- Reserve Tokenisation
- Approve AMM
- Redeems
- Bridge
- Compliance
- Onboarding**
- Settings

Commercial Bank Onboarding

1. Operator 2. Institution 3. KYC 4. Complete

Step 2: Institution Data
Submit your institution details to open the onboarding request.

Institution Name

Bank Code Country

Role Username

Email

Submit Onboarding Request

Figure 9: Onboarding

The Onboarding Wizard registers a new commercial bank with the Central Bank and provisions its on-chain wallet. This is a one-time process. Until onboarding completes and the institution status reaches **ACTIVE**, the operational screens are inaccessible.

For the full step-by-step onboarding procedure, see the [ONBOARDING-DOCS runbook](#).

Wizard stages

1. Operator → 2. Institution → 3. KYC → 4. Complete

Stage	Description
1. Operator Confirmation	Confirms the signed-in operator identity before proceeding. Click Start Onboarding .
2. Institution Data	Fill in Institution Name, Bank Code, Country, Role, Username, and Email. Click Submit Onboarding Request .
3. KYC Approval	Waiting state. The portal polls for Central Bank approval every 5 seconds. Displays Request ID, Wallet, and elapsed time. Click Refresh status to force an immediate check.
4. Complete	Finalization: PKI credential exchange and on-chain activation. Shows Wallet Address, Transaction Hash, and PKI login badge. Click Go to Dashboard .

4.9 Settings (/settings)

Route: /settings · **Sidebar label:** Settings

The Settings screen displays read-only environment information for the portal under an **Environment** card:

Item	Description
API mode	Indicates whether the frontend is running against mock services or the live backend.
Authentication	Backend-managed HTTP-only session cookies.
WebSocket	Relay event source used to drive UI flow.

There are no editable settings on this screen.

4.10 Compliance (/compliance) — not in sidebar

Route: /compliance · **Not shown in the sidebar** (accessible by direct URL)

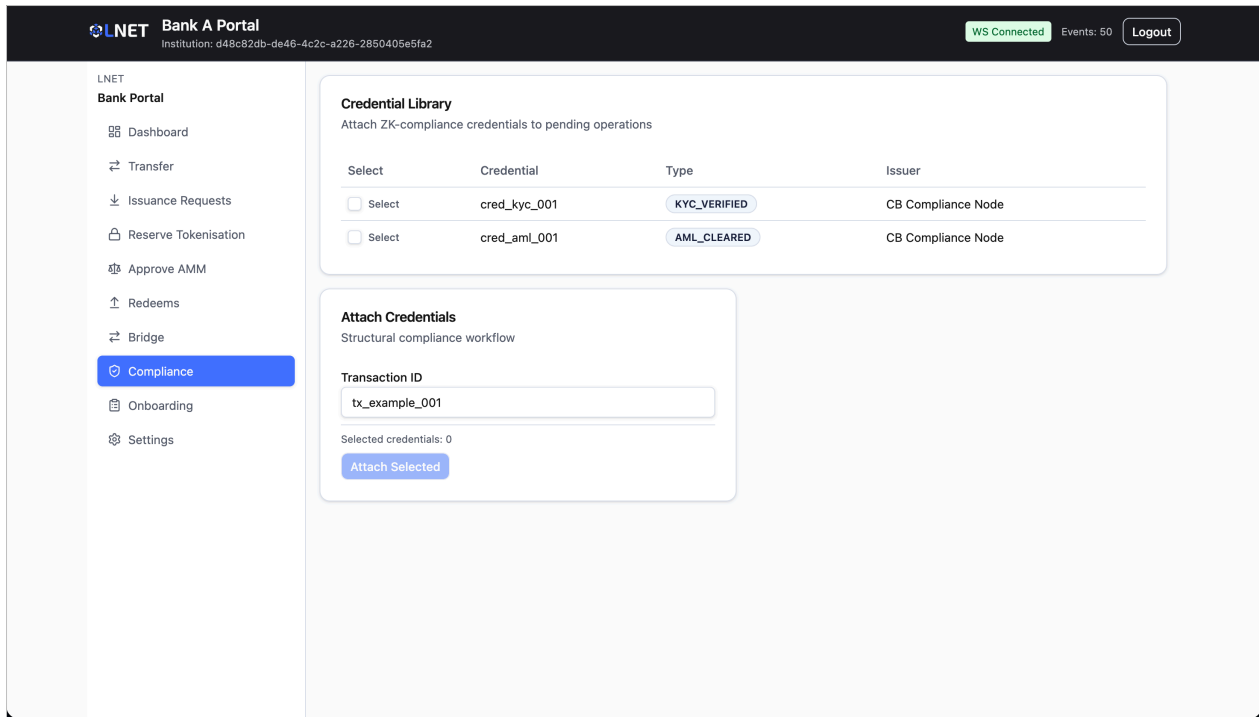


Figure 10: Compliance

The Compliance Center allows you to view your institution’s ZK-compliance credentials and attach them to pending on-chain operations. It is reachable at `/compliance` but is not listed in the Scenario B sidebar.

Credential Library

A table listing all credentials associated with your institution:

Column	Description
Select	Checkbox to include this credential in an attachment.
Credential	Unique credential identifier.
Type	Credential type badge (e.g. KYC, AML).
Issuer	The entity that issued this credential.

Attach Credentials

1. Select one or more credentials using the checkboxes.
2. Enter the **Transaction ID** of the pending operation you want to attach them to.
3. Click **Attach Selected**.

Credentials are issued by the Central Bank after KYC approval. Contact your Central Bank if the credential library is empty.

5. Typical Workflows

5.1 Getting started: obtain tCeBM

Before you can send a cross-currency payment you need tCeBM in your wallet. Follow these steps in order:

1. **Onboard** (if not already done): navigate to **Onboarding** and complete all four wizard stages.
2. **Request issuance**: go to **Issuance Requests**, enter a fiat amount, and confirm. Wait for the Central Bank to approve the request (status changes from **PENDING** to **APPROVED**).
3. **Tokenise the reserve**: go to **Reserve Tokenisation**, select the approved Deposit ID, enter an amount, and confirm. Wait for Central Bank approval (status changes to **APPROVED**). Your tCeBM balance increases.

5.2 Send a cross-currency payment (Bridge)

Prerequisites: tCeBM balance available; an **ACTIVE** pool that includes your sovereign currency.

1. Navigate to **Bridge**.
2. In **Step 1 — Select Pool**, choose a liquidity pool. The direction is fixed: you spend your sovereign currency and the beneficiary receives the counterpart. Confirm the pool banner shows the pool is **ACTIVE**; if it shows another state, contact your Central Bank.
3. In **Step 2 — Get Quote**, enter the **Amount Out** (the amount the recipient should receive) and click **Get Quote**. Review the effective rate and Amount In in the Quote Details card.
4. Check the TTL — if it is low, proceed quickly or re-quote.
5. In **Step 3 — Execute Bridge**, click **Use Suggested Max** to fill the Max Amount In field (or enter a custom ceiling).
6. Enter the **Beneficiary Bank ID**.
7. Click **Execute Bridge**.
8. Monitor the four-step progress indicator until the status reaches **COMPLETED**.
9. Note the Bridge Tx Hash and Correlation ID for your records. Click **New Bridge** to prepare for the next operation.
10. To review past operations, open **Bridge History**.

5.3 Redeem tCeBM for fiat

1. Navigate to **Redeems**.
2. Confirm your tCeBM balance is sufficient.
3. Enter the tCeBM amount to redeem.
4. Click **Review Redeem Request**, then **Confirm Request**.
5. The request appears as **PENDING**. The Central Bank approves it from the Governance Portal; status changes to **APPROVED** once the fiat reserve is released.

6. Status Reference

Issuance (Deposit) request statuses

Status	Meaning
PENDING	Submitted; awaiting Central Bank approval.
APPROVED	Approved by the Central Bank; tCeBM has been minted.
REJECTED	Declined by the Central Bank; check the Rejection Reason column.
MINT_FAILED	Approved but the on-chain mint failed — contact your Central Bank.

Tokenisation (Escrow) request statuses

Status	Meaning
PENDING	Submitted; awaiting Central Bank approval.
APPROVED	Approved; fCeBM burned and tCeBM minted on-chain.
REJECTED	Declined by the Central Bank.

Redeem request statuses

Status	Meaning
PENDING	Submitted; awaiting Central Bank approval.
APPROVED	Approved; fiat reserve released.
REJECTED	Declined by the Central Bank; check the Rejection Reason column.
MINT_FAILED	Error during on-chain processing — contact your Central Bank.

Bridge progress states (Bridge page)

Shown in the four-step progress indicator on the Bridge page.

State	Meaning
BRIDGE_IN_PROGRESS	Your tCeBM is being locked on your spoke.
SWAP_IN_PROGRESS	The AMM swap is executing on the Hub.
BRIDGE_OUT_PROGRESS	Swapped tokens are being delivered on the beneficiary's spoke.
COMPLETED	The bridge fully settled.

Bridge History statuses

Shown as badges in the Bridge History table.

Status	Displayed as	Meaning
QUOTING	Quoting	A quote is being prepared.
BRIDGE_IN_PROGRESS	Bridging in	Funds are being locked on the source spoke.
SWAP_IN_PROGRESS	Swapping	The AMM swap is executing on the Hub.
BRIDGE_OUT_PROGRESS	Bridging out	Swapped tokens are being delivered on the beneficiary's spoke.
COMPLETED	Completed	Operation settled end-to-end.
FAILED	Failed	Operation errored (hover the badge for the failure reason).

Pool and circuit-breaker states (Liquidity Pools page)

Pool status	Displayed as	Meaning
ACTIVE	Active	Pool is live and swappable.
PENDING_COUNTERPART	Awaiting counterpart	The counterpart Central Bank has not yet confirmed the corridor.
EMPTY	Empty	Pool has no liquidity.

Circuit-breaker state	Displayed as	Meaning
LIVE	Operational	Swaps are open on this pool.
HALTED	Paused	A Central Bank triggered the circuit breaker (1-of-N pause). Swaps are suspended.
RESUME_PENDING	Resume pending	A resume proposal is awaiting the 2-of-N Central Bank quorum.

Onboarding statuses

Status	Meaning
NONE	No onboarding request exists. Start the wizard.
PENDING	Onboarding request submitted; awaiting Central Bank KYC review.
CREDENTIAL_REQUESTED	System is internally requesting the PKI credential (shows as PENDING to you).
KYC_APPROVED	Central Bank approved KYC; final on-chain activation in progress.
ACTIVE	Institution fully onboarded; all operational screens are accessible.
REJECTED	Central Bank rejected the onboarding request. Contact governance support.
REVOKED	Institution access was revoked after activation.
FROZEN	Institution account was frozen.

7. Troubleshooting

Sign-in fails with “unauthorized”

- Verify your **Username** and **Password** are correct.
- Confirm the spoke backend services are running and reachable (check with your infrastructure team).
- Confirm your institution’s status is **ACTIVE** — if it is not, complete onboarding first.

Operational screens are inaccessible after login

- Your institution is not yet **ACTIVE**. Navigate to **Onboarding** to check the wizard state and continue from where it left off.

Issuance, tokenisation, or redeem request is stuck in PENDING

- The Central Bank must approve it from the Governance Portal. Contact your Central Bank governance operator.
- Click **Refresh** on the relevant page to check for status updates.

Pool is not ACTIVE — Execute Bridge disabled

- The selected pool is not yet activated for your corridor. Select another **ACTIVE** pool from the Step 1 drop-down, or contact your Central Bank to activate the pool. You can also review pool availability on the **Liquidity Pools** page.

Circuit breaker paused — swaps disabled

- Cross-border swaps on the pool have been suspended by a Central Bank (1-of-N pause governance). Wait for governance to resume (2-of-N resume). No action is required from the bank operator. The **Liquidity Pools** page shows the current breaker state per pool.

Quote expired — Execute Bridge disabled

- Click **Get Quote** again to obtain a fresh quote. If the quote was auto-refreshed, review the new figures before confirming. If the pool is paused or inactive, swaps are not possible until it is operational again.

A bridge operation failed

- The Bridge page shows a **Bridge Failed** card with a plain-language reason and a **Try Again** button.
- A “bridge in failed” or “swap failed” message means the operation was rolled back and no funds were moved — you may retry safely.
- For a slippage error, re-quote and increase your Max Amount In within tolerance.
- If the message indicates a daily transfer limit, contact your Central Bank to adjust the cap.

Onboarding wizard appears stuck at Step 3 (KYC Pending)

- Click **Refresh status** on the Step 3 screen to force an immediate status poll.
- If your institution already shows **ACTIVE** in the Governance Portal but the wizard still shows Step 3, log out and log back in, then navigate to Onboarding again.

Compliance credential library is empty

- Credentials are issued by the Central Bank as part of the onboarding process. If the library is empty after your institution is **ACTIVE**, contact your Central Bank governance operator.

Governance Portal — User Manual

(Scenario B)

Audience: Central bank governance operators. **Scenario:** Scenario B — International Hub (FX-Agreement + AMM + Cacti relay). **Data source:** Live backend via API Gateway. No mock data toggle is present in Scenario B build.

1. Overview

The Governance Portal is the sovereign control surface for a central bank participating in **Scenario B — International Hub**. In Scenario B, the hub is a shared blockchain network with an Automated Market Maker (AMM) at its centre. Cross-border payments are executed through AMM swaps rather than bilateral HTLC channels.

This portal gives the central bank governance team authority over:

- **Participant admission** — KYC approval for commercial banks requesting access to the network.
- **Account intervention** — emergency freeze of a participant account.
- **AMM oversight** — read-only visibility into in-progress and historical swaps.
- **Circuit breaker** — pause and resume AMM swaps per currency pair using a 1-of-N pause / 2-of-N resume multi-signature mechanism.
- **Transfer limits** — set daily transfer limits per participant in the spoke's sovereign currency.
- **Oversight and disclosure** — open and co-sign AML/CFT disclosure requests for cross-border transactions.
- **Audit trail** — immutable record of every governance action.

Important — Scenario B vs Scenario A. The Governance Portal serves both scenarios from the same codebase. When `VITE_SCENARIO=b`, a Scenario B-specific route set is activated. In Scenario B, the Deposits Approval, Escrows Approval, Redeems Approval, HTLC Monitor, and Parameters screens are **not available** in this portal — they belong to other portals (Treasury, Bank) in the Scenario B architecture. Do not expect those screens to be accessible.

Actor	Portal	Role
Central Bank Governance Operator	Governance Portal	Admits participants, freezes and unfreezes accounts, monitors swaps, operates the circuit breaker, sets transfer limits, manages AML disclosures

2. Who uses it

Central bank governance and compliance officers. Access is provisioned by the network administrator. A Keycloak institutional account with the `ROLE_GOVERNANCE` role is required — the login page explicitly states that this is the highest RBAC tier and is restricted to governance operators.

3. Access and login

Open the Governance Portal URL for your central bank entity. The URL is defined in the **Portal Ports** table of `scenario-b/frontend/README.md` for the local deployment, or provided by the network administrator for production deployments.

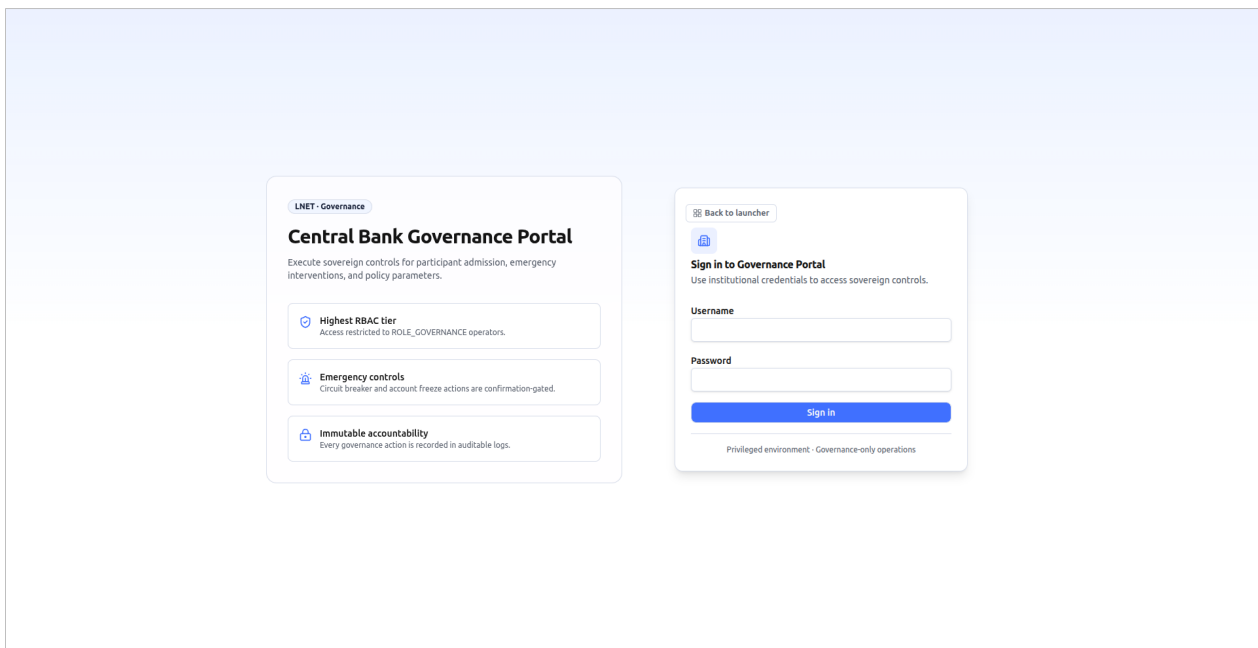


Figure 11: Login

The login form requires two fields:

Field	Description
Username	Institutional service account identifier (minimum 3 characters).
Password	Institutional service account secret (minimum 6 characters).

After submitting valid credentials, the portal authenticates via Keycloak OIDC and verifies the `ROLE_GOVERNANCE` role. If the role check passes, the operator is redirected to the Dashboard with a “Welcome to the Governance Portal” confirmation. If the credentials are invalid or the role is missing, an error message is displayed inline on the login form.

Every governance action performed after login is recorded in the immutable audit log. Operators should treat their credentials as privileged secrets.

4. Navigation

In Scenario B, the sidebar contains exactly the following items in order:

Sidebar label	Route	Purpose
Dashboard	<code>/</code>	Network health summary and alert cards
Registry	<code>/registry</code>	Participant compliance registry and KYC approvals
Accounts	<code>/accounts</code>	Account freeze for emergency intervention
Swap Monitor	<code>/swap-monitor</code>	Read-only visibility into AMM swap operations
Circuit Breaker	<code>/circuit-breaker</code>	Pause / resume AMM swaps per pair (multi-sig)
Transfer Limits	<code>/transfer-limits</code>	Daily transfer limits per participant
Oversight	<code>/oversight</code>	AML/CFT disclosure requests
Audit	<code>/audit</code>	Full governance action audit trail
Settings	<code>/settings</code>	Session and display preferences

Unknown routes redirect to `/` automatically.

5. Screens

5.1 Dashboard

Route: `/`

The Dashboard is the first screen after login. It provides a real-time summary of the most important network health indicators.

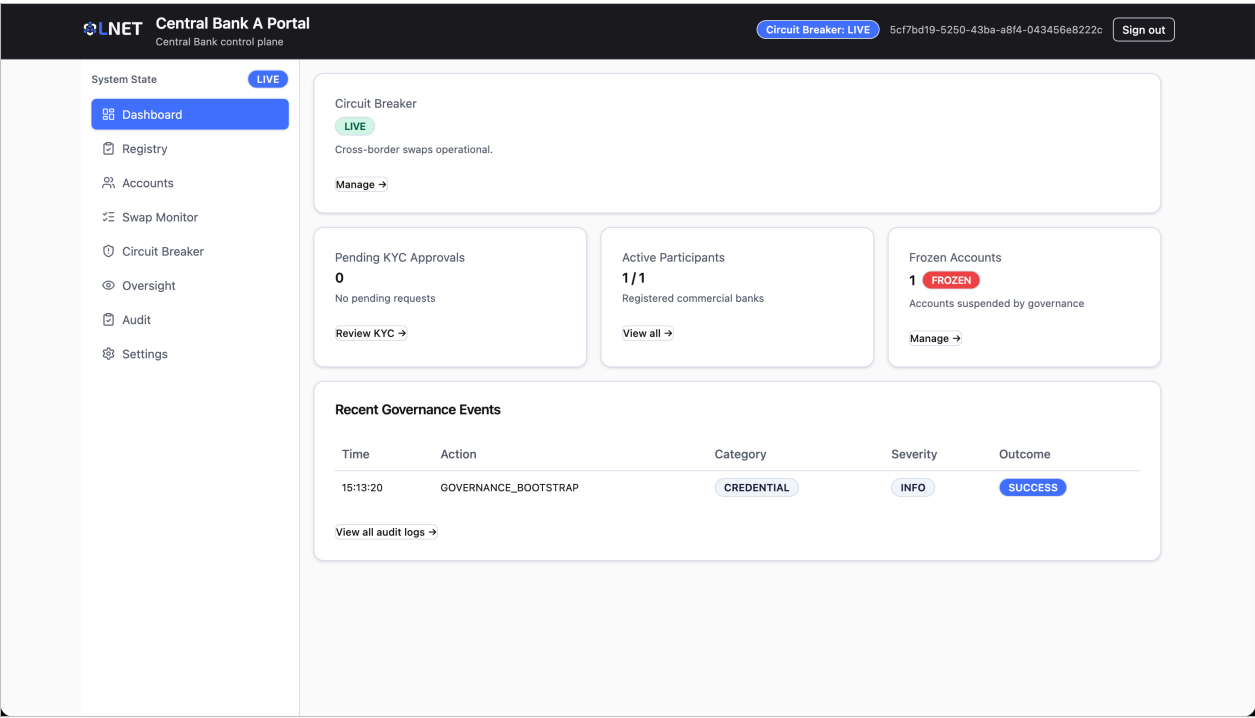


Figure 12: Dashboard

Circuit Breaker status card

In Scenario B, the top of the Dashboard shows a dedicated Circuit Breaker card. When the breaker is in the HALTED state, the card border turns red and the status badge is highlighted in destructive (red) colour. The card reads “Cross-border swaps are paused network-wide.” A “Manage →” link navigates directly to the Circuit Breaker screen.

Summary cards

Three summary cards are displayed side by side:

Card	Description
Pending KYC Approvals	Count of commercial banks waiting for Central Bank validation. A yellow border and “NEEDS ATTENTION” badge appear when the count is above zero. Links to Registry.
Active Participants	Count of participants with ACTIVE status out of total registered institutions. Links to Accounts.
Frozen Accounts	Count of accounts currently frozen by governance. A red border and “FROZEN” badge appear when the count is above zero. Links to Accounts.

Recent Governance Events

A table showing the five most recent audit log entries with columns: Time, Action, Category, Severity, and Outcome. Severity badges are colour-coded: CRITICAL is red, WARNING is amber, and INFO is grey. Outcome badges are green for SUCCESS and red for any failure. A “View all audit logs →” link navigates to the Audit screen.

5.2 Registry

Route: /registry

The Registry is the compliance registry of all institutions known to the network. It has two sections: the full participant list and a live queue of pending KYC approvals.

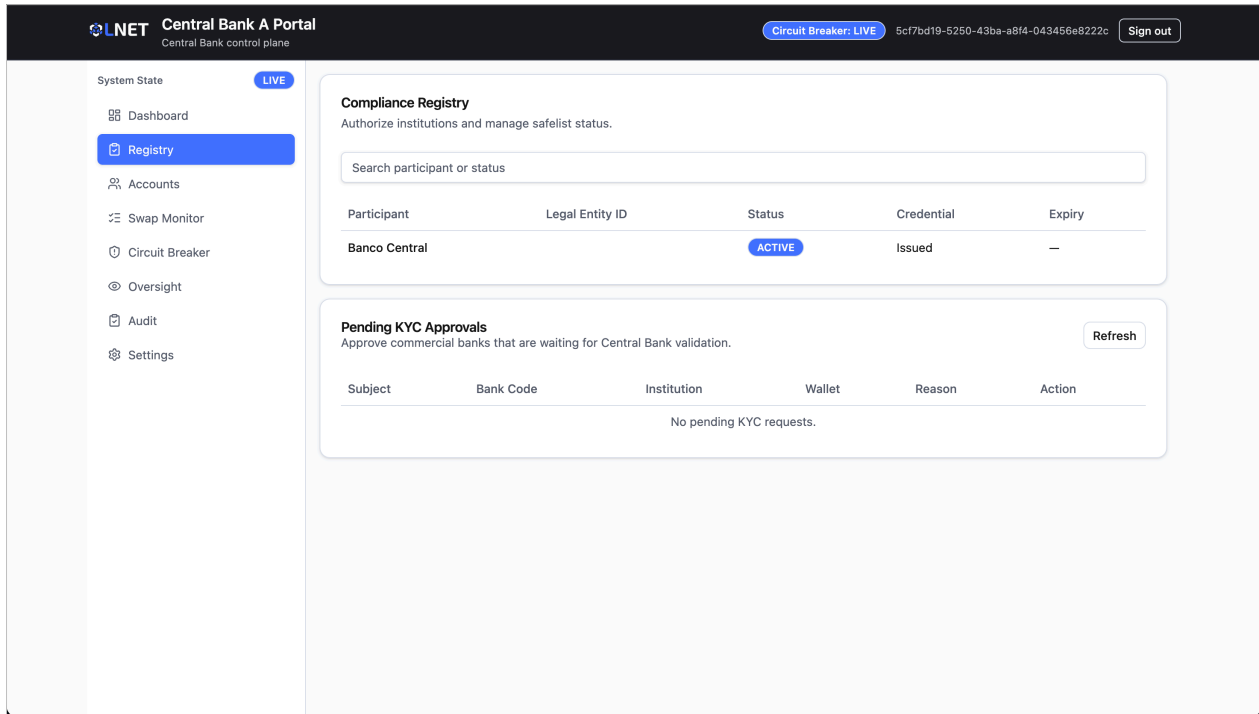


Figure 13: Registry

Compliance Registry table

A searchable table of all registered participants. The search field filters by participant name or status.

Column	Description
Participant	Institution display name.
Legal Entity ID	The institution’s legal entity identifier.
Status	Current compliance status (see Status reference).
Credential	Credential identifier if one has been issued, otherwise “—”.
Expiry	Credential expiry date, or “—” if not applicable.

Pending KYC Approvals table

This section refreshes automatically every 15 seconds. It shows commercial banks that have completed the onboarding flow and are waiting for the central bank to grant network access.

Column	Description
Subject	Keycloak subject identifier of the bank’s service account.
Bank Code	Short bank code, if available.

Column	Description
Institution	Full institution name.
Wallet	On-chain wallet address registered for the bank.
Reason	Text input where the governance operator enters an approval reason (minimum 10 characters).
Action	“Approve KYC” button.

To approve a pending bank:

1. Enter an approval reason of at least 10 characters in the Reason field for that row.
2. Click **Approve KYC**.
3. A success notification confirms the approval. The entry disappears from the pending list on the next refresh.

The “Issue Credential” form visible in earlier versions of this portal is currently commented out in the source and is not available.

5.3 Accounts

Route: /accounts

The Accounts screen (titled “Account Intervention”) provides emergency account intervention: governance operators can freeze and unfreeze participant accounts. Every action must be accompanied by a written reason of at least 10 characters.

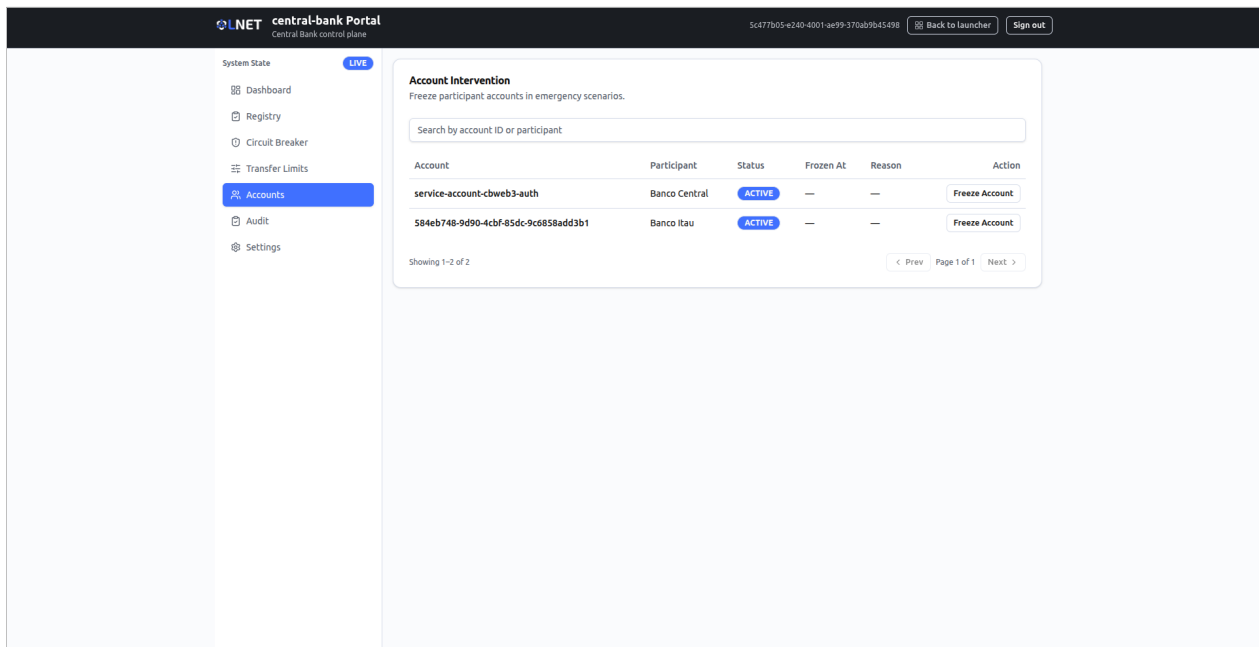


Figure 14: Accounts

Account list

A searchable, paginated table of all participant accounts. The search field filters by account ID or participant name. The table shows 10 accounts per page, with Prev / Next controls and a “Showing X–Y of N” indicator below it.

Column	Description
Account	Account identifier.
Participant	Name of the institution that holds the account. Central Bank accounts are tagged with an “Admin” badge.
Status	ACTIVE (green) or FROZEN (red).
Frozen At	Timestamp when the account was frozen, or “—” if active.
Reason	The reason recorded at freeze time, or “—” if active.
Action	“Freeze Account” for active accounts, “Unfreeze Account” for frozen accounts. Central Bank accounts show “Central Bank — protected” and cannot be frozen or unfrozen.

Freezing or unfreezing an account

1. Locate the account in the list using the search field if needed.
2. Click **Freeze Account** (or **Unfreeze Account** for a frozen account) on the target row. A confirmation panel appears below the list, titled “Confirm Freeze” or “Confirm Unfreeze”.
3. Enter a reason of at least 10 characters in the text area.
4. Click **Confirm Freeze** (or **Confirm Unfreeze**). The action is submitted and a success notification appears.
5. Click **Cancel** to abandon the action.

Freeze and unfreeze actions are recorded permanently in the audit log under the **FREEZE** category. The Central Bank is the governance administrator and can never freeze or unfreeze its own account — its row is protected.

5.4 Swap Monitor

Route: /swap-monitor

The Swap Monitor provides read-only operational visibility into the Scenario B payment pipeline. It shows live counts of pending approval queues and allows governance operators to track individual swaps by swap ID during a session.

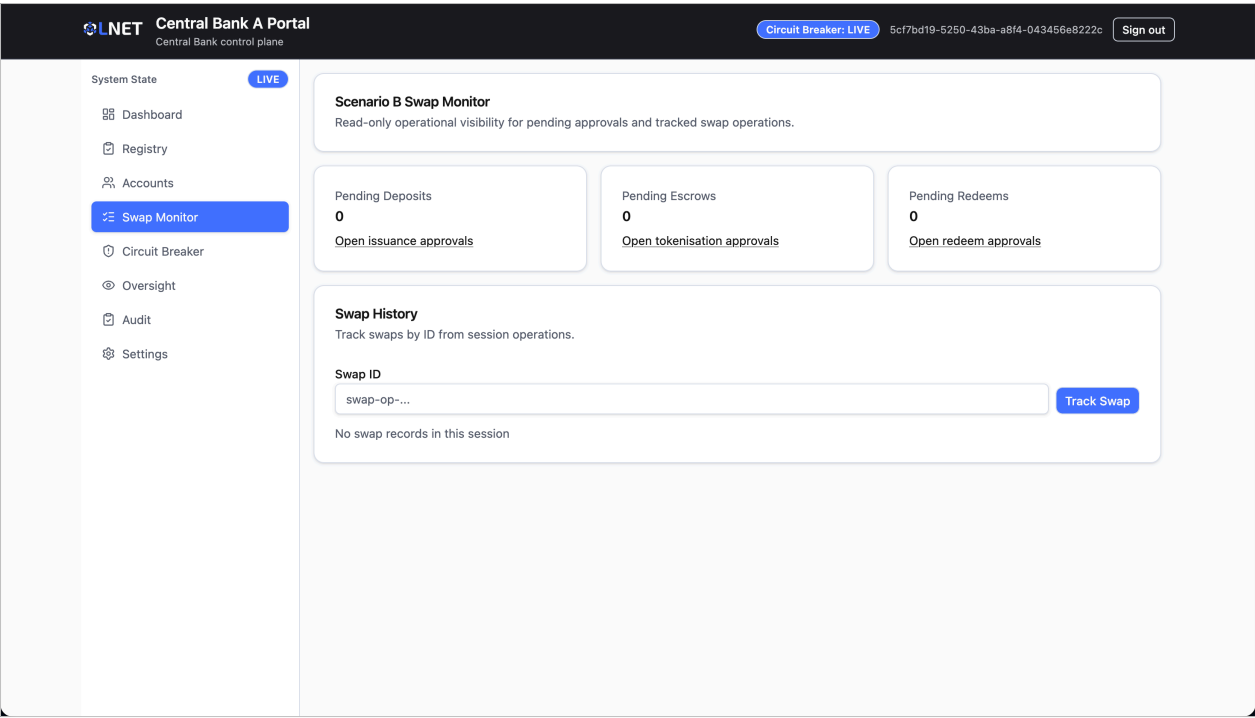


Figure 15: Swap Monitor

Pending approval summary

Three summary cards at the top of the screen:

Card	What it counts
Pending Deposits	Issuance requests awaiting approval.
Pending Escrows	Tokenisation requests awaiting approval.
Pending Redeems	Redemption requests awaiting approval.

These counts are informational only. Each card carries a link (Open issuance approvals / Open tokenisation approvals / Open redeem approvals), but the approval screens for deposits, escrows, and redeems are part of other portals in the Scenario B architecture and are not present in the Governance Portal navigation. The cards show the queue depth so governance operators can monitor pipeline health.

Swap History

Governance operators can track individual swap operations by ID. Enter a swap ID in the **Swap ID** field and click **Track Swap**. The swap record is fetched and displayed as a card with the following fields:

Field	Description
Status	Current swap status (see swap status table in Status reference).
Payer Bank	Bank ID of the institution initiating the payment.
Beneficiary Bank	Bank ID of the receiving institution.
Bridge In Position	Position ID for the incoming bridge leg, or “—”.

Field	Description
Swap Tx Ref	On-chain transaction reference for the AMM swap, or “—”.
Bridge Out Position	Position ID for the outgoing bridge leg, or “—”.
Created At	Timestamp when the swap was initiated.
Updated At	Timestamp of the most recent status change.

Multiple swaps can be tracked in a single session; each appears as a separate card. Tracked swap IDs are retained for the duration of the browser session only.

5.5 Circuit Breaker

Route: /circuit-breaker

The Circuit Breaker is the emergency control for AMM swap operations on the hub. The breaker is applied **per currency pair** — one AMM instance exists per pair — so all actions on this screen apply to the pair currently selected. Governance operators use this screen to halt or resume swap processing for a pair.

In Scenario B, the circuit breaker uses a **multi-signature governance model** across Central Banks:

- **Pausing** the breaker is a unilateral action: any single Central Bank can halt the selected pair immediately (1-of-N).
- **Resuming** the breaker is a two-step process that requires two different Central Banks (2-of-N). One Central Bank proposes the resume; a second Central Bank signs it before swaps are re-enabled.

The acting institution identity is derived from the operator session (falling back to the configured institution name) and is **never typed manually**. The institutional signature is generated server-side from the Central Bank PKI key — operators do not paste a signature. The page polls the current breaker state automatically every 15 seconds and reads the state directly from the blockchain, so every Central Bank sees the same state.

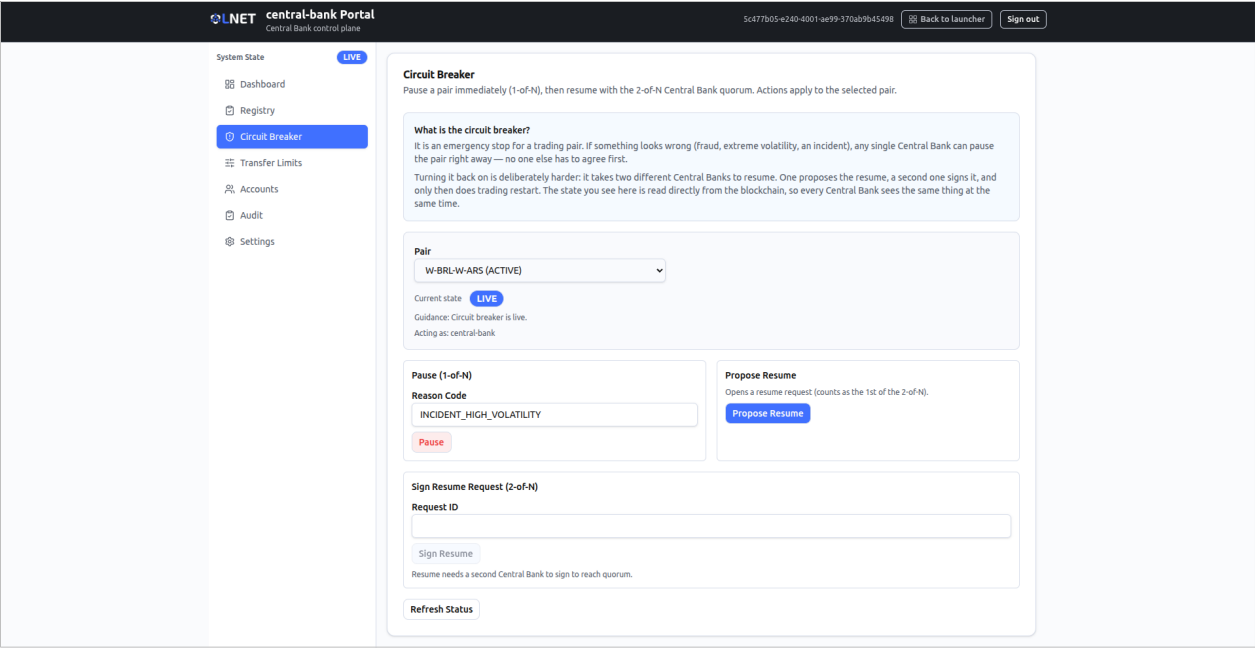


Figure 16: Circuit Breaker

At the top of the screen, an explanatory panel titled “What is the circuit breaker?” describes the control in plain language.

Pair selector and current state

Element	Description
Pair	A dropdown of the AMM pairs created on the hub. Each option shows the pair identifier and its status, for example BRL–USD (ACTIVE). If no pairs exist yet, the selector reads “No pairs created yet”.
Current state badge	LIVE (default), HALTED (red), or RESUME_PENDING (amber).
Resume proposal	When a resume proposal is in flight, a line shows the truncated resume request ID and signature progress, for example X/Y signatures.
Guidance	Operational guidance text derived from the current state and freshness.
Stale indicator	If the most recent status refresh failed, an amber warning reads “Showing last known state. Latest status refresh failed.”
Acting as	The Central Bank identity that pause and resume actions are recorded under.

Pause (1-of-N)

Use this panel to halt the selected pair immediately.

Field	Description
Reason Code	Machine-readable reason code, pre-filled with <code>INCIDENT_HIGH_VOLATILITY</code> .

Click **Pause** to submit (disabled until a pair is selected). The state updates to **HALTED** on success.

Pausing stops swaps for the selected pair immediately and does not require a second operator. Choose the reason code carefully — it is recorded permanently.

Propose Resume

Use this panel to open the resume process after a pause. It counts as the first of the 2-of-N signatures.

Click **Propose Resume** (disabled until a pair is selected). A resume request ID is created and displayed below the button as “Created request: ...”. A second Central Bank must then sign it.

Sign Resume Request (2-of-N)

Use this panel to add the second Central Bank signature to a pending resume proposal.

Field	Description
Request ID	The resume request ID. This field is auto-filled with the resume request discovered on-chain, so a Central Bank that did not propose can co-sign in one click.

Click **Sign Resume** (enabled only when a pair is selected and a Request ID is present). Once the 2-of-N quorum is reached, swaps are re-enabled. Click **Refresh Status** at any time to confirm the current state.

Operational protocol. When a pause is engaged, communicate the event to all participant banks promptly — their transfer operations for that pair will fail until the breaker is resumed. Resuming requires two different Central Banks. Signatures are produced server-side from each Central Bank’s PKI key; the quorum cannot be satisfied by a single institution.

5.6 Transfer Limits

Route: `/transfer-limits`

The Transfer Limits screen lets a Central Bank set daily transfer limits for participants on its spoke. Limits always apply in the spoke’s own sovereign currency; a Central Bank cannot set a limit in another currency.

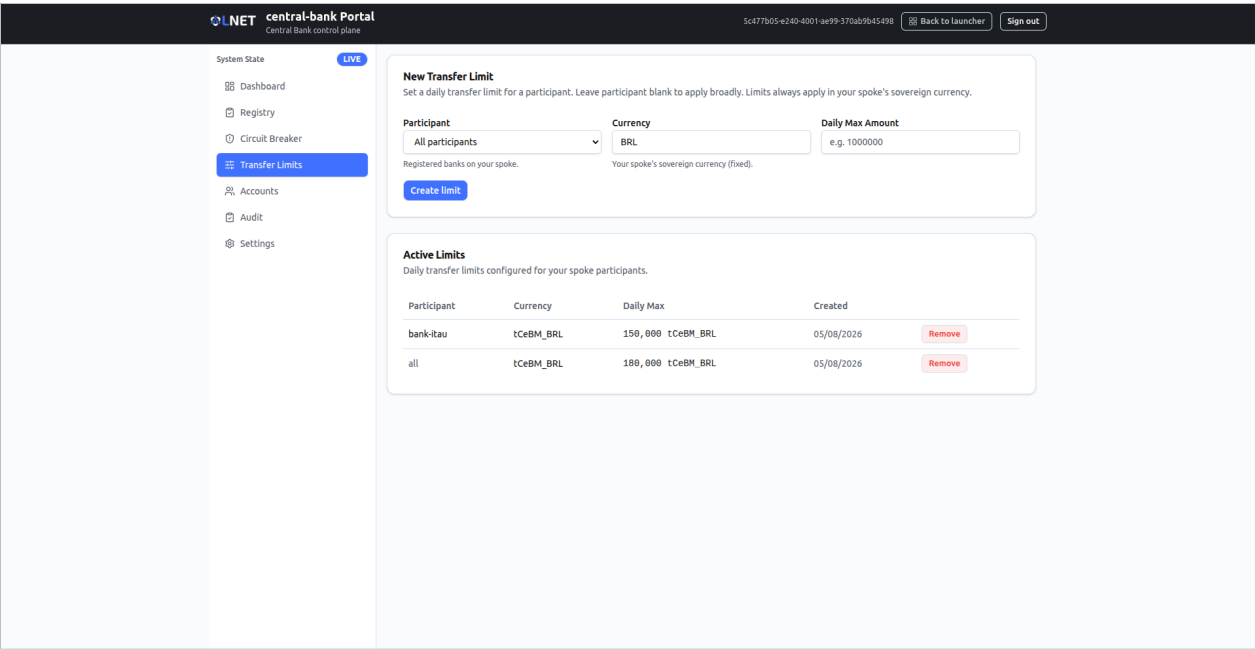


Figure 17: Transfer Limits

New Transfer Limit

Field	Description
Participant	Dropdown of registered banks on the spoke. The default option, “All participants”, applies the limit broadly.
Currency	Read-only. The spoke’s sovereign currency (the human code, for example COP, derived from the reserve token such as tCeBM_COP). The backend fixes this value.
Daily Max Amount	The daily maximum transfer amount, for example 1000000. Required.

Click **Create limit** to submit. A success notification confirms the new limit.

Active Limits

A table of the daily transfer limits configured for the spoke. When no limits exist, the panel reads “No limits configured.”

Column	Description
Participant	The participant bank code, or “all” if the limit applies broadly.
Currency	The currency of the limit, or “all”.
Daily Max	The formatted daily maximum amount with its currency.
Created (action)	Date the limit was created. “Remove” button that deletes the limit.

5.7 Oversight

Route: /oversight

The Oversight screen is for AML/CFT disclosure requests. A governance operator can open a formal disclosure request tied to a cross-border transaction reference, and a second operator from another institution can co-sign the request to reach quorum.

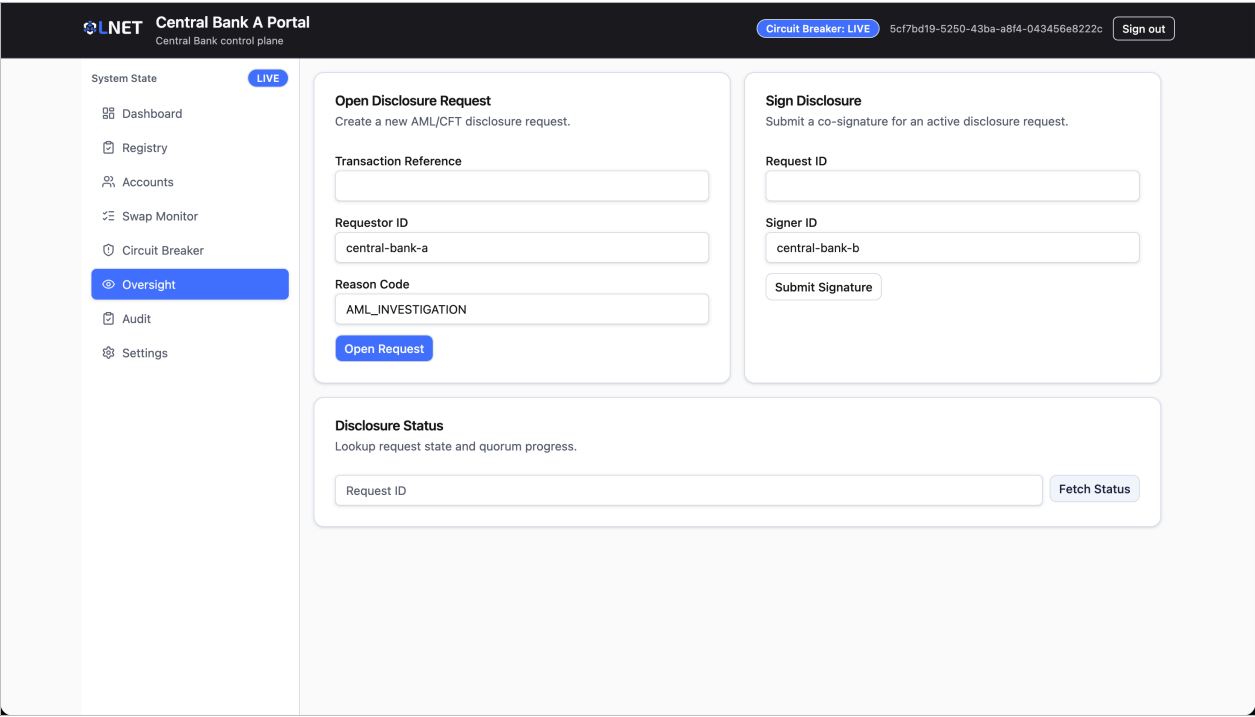


Figure 18: Oversight

Open Disclosure Request

Field	Description
Transaction Reference	The on-chain or backend transaction reference under investigation.
Requestor ID	Institutional identifier of the requesting central bank (e.g. <code>central-bank-a</code>).
Reason Code	Machine-readable reason code (e.g. <code>AML_INVESTIGATION</code>).

Click **Open Request** to submit. The new request appears in the Disclosure Status panel.

Sign Disclosure

Field	Description
Request ID	ID of the disclosure request to co-sign.
Signer ID	Institutional identifier of the co-signing central bank.

Click **Submit Signature** to record the co-signature. An error message is shown if this institution has already signed or the request is closed.

Disclosure Status

Enter a Request ID and click **Fetch Status** to retrieve the current state of any disclosure request. The result panel shows:

Field	Description
State	Current disclosure state (see Status reference).
Request ID	Identifier of the disclosure request.
Quorum	Current signatures collected vs required signatures.
Expires At	Timestamp after which the request automatically expires.
Closed At	Timestamp when the request was closed, or “—” if still open.

A full JSON view of the disclosure record is shown below the summary for detailed inspection.

5.8 Audit

Route: /audit

The Audit screen (titled “Governance Audit Trail”) provides the immutable governance action trail. Every action taken through this portal — credential approvals, account freezes, parameter changes, circuit breaker operations — is recorded here.

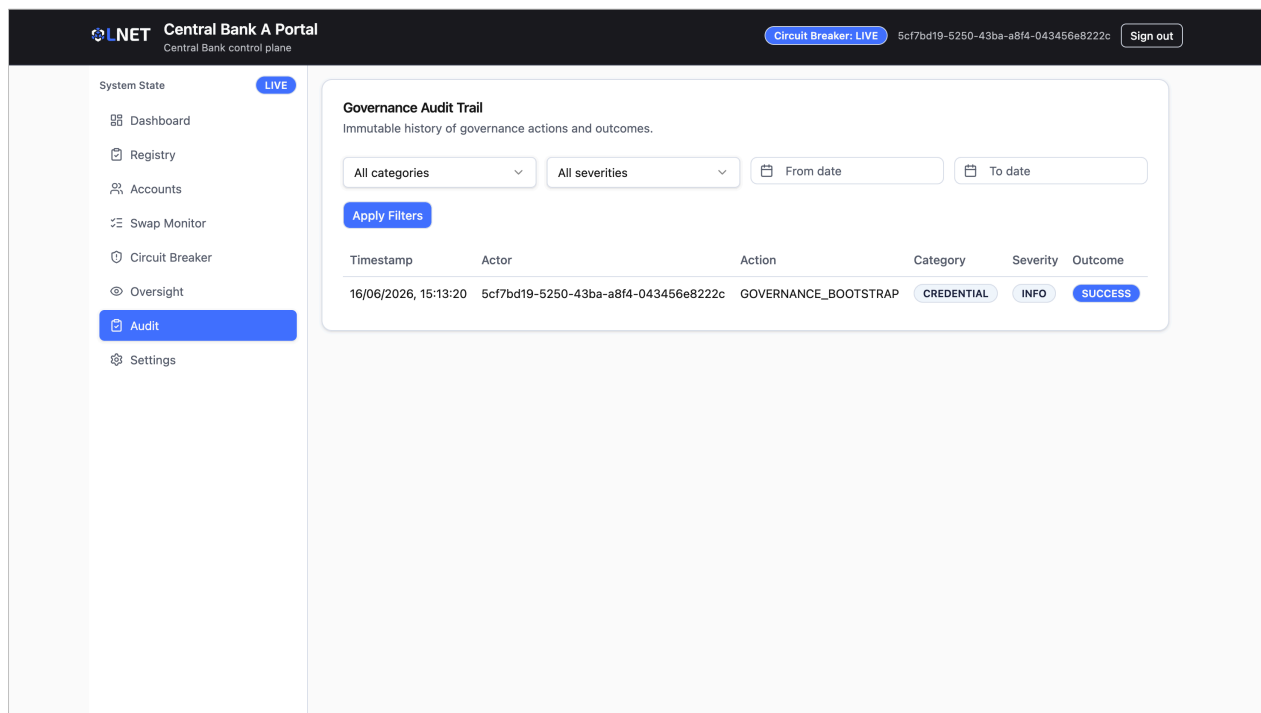


Figure 19: Audit

Filters

Four filters can be applied independently or in combination:

Filter	Options
Category	All, Credential, Circuit Breaker, Freeze, Parameter
Severity	All, Info, Warning, Critical
From date	Date picker — earliest timestamp to include
To date	Date picker — latest timestamp to include

Click **Apply Filters** to execute the filtered query. The filter state is not persisted across page loads.

Audit log table

Column	Description
Timestamp	ISO-8601 timestamp of the action.
Actor	Operator or service account that performed the action.
Action	Human-readable description of the governance event.
Category	CREDENTIAL, CIRCUIT_BREAKER, FREEZE, or PARAMETER.
Severity	INFO, WARNING, or CRITICAL.
Outcome	SUCCESS (green) or failure status (red).

5.9 Settings

Route: /settings

The Settings screen holds operator session information and local display preferences.

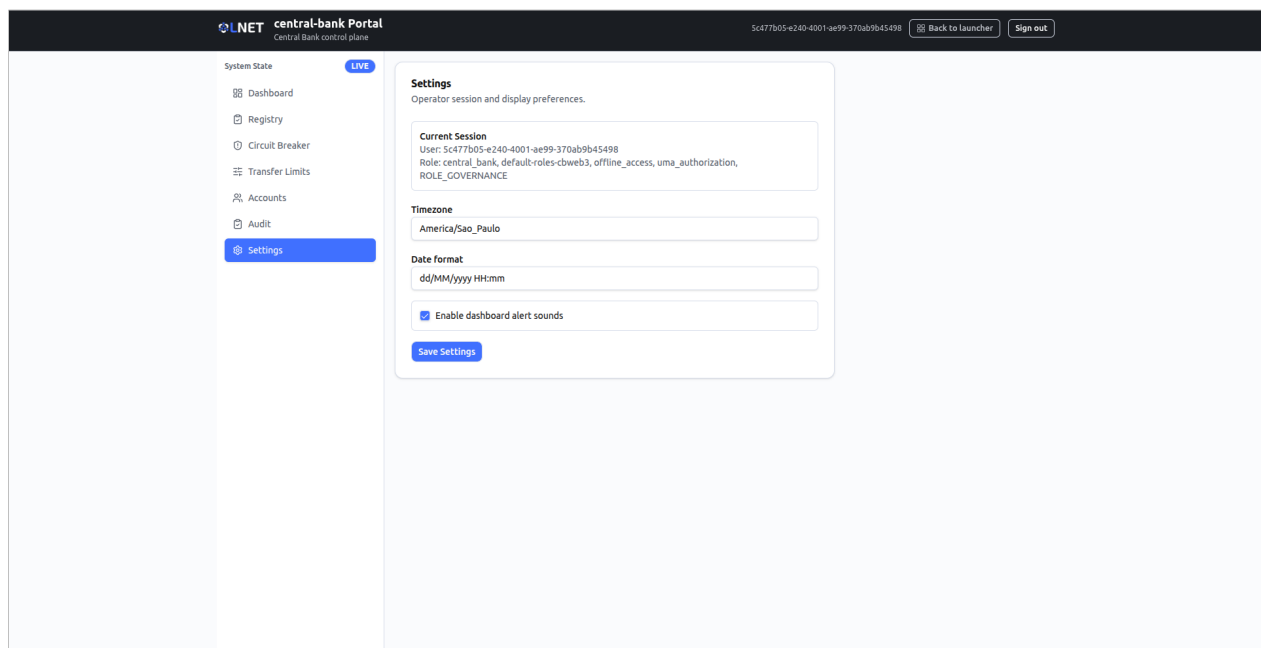


Figure 20: Settings

Current Session

A read-only panel showing the signed-in operator:

Field	Description
User	The operator's subject identifier from the session.
Role	The roles carried by the session, joined by commas.

Display preferences

Control	Description
Timezone	Free-text timezone, default <code>America/Sao_Paulo</code> .
Date format	Free-text date format, default <code>dd/MM/yyyy HH:mm</code> .
Enable dashboard alert sounds	Checkbox, enabled by default.

Click **Save Settings** to confirm. These preferences are display-only client-side settings.

6. Typical workflows

Onboarding a new commercial bank

1. The bank completes the onboarding wizard (see the Onboarding Portal documentation).
2. Navigate to **Registry** (`/registry`).
3. Scroll to the **Pending KYC Approvals** table. The bank's entry appears here automatically — the table refreshes every 15 seconds.
4. Review the bank's subject ID, bank code, institution name, and wallet address.
5. Enter an approval reason (at least 10 characters) in the Reason field on the bank's row.
6. Click **Approve KYC**. A success notification confirms the approval.
7. The bank's entry moves from the pending table to the Compliance Registry with **ACTIVE** status.

Freezing an account in an emergency

1. Navigate to **Accounts** (`/accounts`).
2. Use the search field to locate the account by participant name or account ID.
3. Click **Freeze Account** on the target row.
4. In the confirmation panel that appears, enter a reason of at least 10 characters.
5. Click **Confirm Freeze**.
6. Notify the relevant parties that the account has been frozen. The freeze reason is visible in the Accounts table and in the Audit log.

Halting AMM swaps (emergency pause)

1. Navigate to **Circuit Breaker** (`/circuit-breaker`).
2. Select the affected pair in the **Pair** dropdown and confirm the current state is **LIVE**.
3. In the **Pause (1-of-N)** panel, adjust the reason code if required.
4. Click **Pause**.
5. The state updates to **HALTED**. Communicate the pause to all participant banks immediately.
6. The pause is recorded in the Audit log under the **CIRCUIT_BREAKER** category.

Resuming AMM swaps (two-signature process)

1. **First Central Bank:** Navigate to **Circuit Breaker** (/circuit-breaker). Select the pair, then click **Propose Resume** in the **Propose Resume** panel. Note the resume request ID that appears as “Created request: ...”.
2. A second Central Bank opens the **Circuit Breaker** screen and selects the same pair. The resume request ID is auto-filled in the **Sign Resume Request (2-of-N)** panel from the on-chain state; if needed, the request ID from step 1 can be entered manually.
3. **Second Central Bank:** Click **Sign Resume**.
4. The state updates to **LIVE** once the 2-of-N quorum is reached. Verify with **Refresh Status**.
5. Both signature events are recorded in the Audit log. Signatures are produced server-side from each Central Bank’s PKI key.

Investigating a suspicious transaction (AML disclosure)

1. Obtain the transaction reference (on-chain tx hash or backend transaction ID) from the NOC or compliance team.
2. Navigate to **Oversight** (/oversight).
3. In the **Open Disclosure Request** panel, enter the transaction reference, your institution’s requestor ID, and a reason code such as **AML_INVESTIGATION**. Click **Open Request**.
4. Note the request ID from the Disclosure Status panel or the API response.
5. Share the request ID with a second central bank operator for co-signature.
6. The second operator navigates to **Oversight**, enters the request ID and their signer ID in the **Sign Disclosure** panel, and clicks **Submit Signature**.
7. Monitor quorum progress in the **Disclosure Status** panel by entering the request ID and clicking **Fetch Status**.

Monitoring swap pipeline health

1. Navigate to **Swap Monitor** (/swap-monitor).
2. Review the three pending count cards (Deposits, Escrows, Redeems). Elevated counts indicate a processing backlog in the payment pipeline.
3. To investigate a specific swap, obtain the swap ID from the payment team or NOC, enter it in the **Swap ID** field, and click **Track Swap**.
4. Review the swap record card for status, parties, and transaction references.
5. If the swap shows a failure status (e.g. **BRIDGE_OUT_FAILED**), escalate to the NOC team — do not attempt to intervene from this portal.

Setting a daily transfer limit

1. Navigate to **Transfer Limits** (/transfer-limits).
2. In the **New Transfer Limit** form, choose a participant, or leave “All participants” selected to apply the limit broadly.
3. Confirm the read-only **Currency** — it is fixed to your spoke’s sovereign currency.
4. Enter the **Daily Max Amount** and click **Create limit**. A success notification confirms the new limit.
5. The limit appears in the **Active Limits** table. Use its **Remove** button to delete it if needed.

7. Status reference

Participant / registry status

Status	Meaning
ACTIVE	Onboarded and cleared to transact.
APPROVED	Credential approved; may be transitioning to ACTIVE.
PENDING	Onboarding or credential issuance in progress.
CREDENTIAL_REQUESTED	Awaiting credential approval in the Registry.
KYC_APPROVED	KYC step completed; awaiting full activation.
REVOKED	Credential has been revoked; institution cannot transact.
REJECTED	Onboarding or credential request was rejected.
FROZEN	Account suspended by governance action.

Circuit breaker state

State	Meaning
LIVE	AMM swaps for the pair are operational. Normal operating state and the default shown when no state has been retrieved.
HALTED	AMM swaps for the pair are stopped. Cross-border transfers on that pair fail closed until resumed.
RESUME_PENDING	A resume proposal is open and collecting signatures. Swaps remain halted until the 2-of-N quorum is reached.
UNKNOWN	Shown on the Dashboard circuit breaker card when the state could not be retrieved from the backend; check connectivity.

Swap status

Status	Meaning
COMPLETED	Swap fully settled on both sides.
PENDING	Swap initiated, awaiting processing.
*_PROGRESS	Any status ending in _PROGRESS — swap is actively processing (bridge or AMM step in flight).
BRIDGE_OUT_FAILED	Settlement failed on the outgoing bridge leg; manual investigation required.

Audit event severity

Severity	Meaning
INFO	Routine governance action.
WARNING	Action that warrants attention; no immediate incident.
CRITICAL	High-priority event requiring immediate review.

Disclosure request state

State	Meaning
PENDING	Request is open and collecting signatures.
QUORUM_REACHED	Sufficient signatures collected; disclosure is complete.
EXPIRED	The request expired before quorum was reached.
REJECTED	The request was closed without quorum.

8. Troubleshooting

The Circuit Breaker state panel shows “Showing last known state. Latest status refresh failed.”

The portal could not reach the backend API on the most recent 15-second poll. The displayed state is the last known value and may be stale. Check network connectivity and confirm the API Gateway service is running. Click **Refresh Status** manually to retry.

Banks report that cross-border transfers are failing

1. Navigate to **Circuit Breaker** (/circuit-breaker). If the state is **HALTED**, swaps are intentionally blocked. Initiate the resume process if the halt is no longer required (see [Resuming AMM swaps](#)).
2. If the state is **LIVE**, navigate to **Swap Monitor** (/swap-monitor) and check whether pending queue counts are elevated or individual swaps show **BRIDGE_OUT_FAILED**. Escalate to the NOC team for infrastructure investigation.

A pending KYC entry does not appear in the Registry

The Pending KYC table refreshes every 15 seconds. Click **Refresh** on the Pending KYC panel header to trigger an immediate reload. If the entry still does not appear, confirm that the bank completed the full onboarding wizard and that the auth service is reachable.

Resume does not re-enable the AMM

The resume process requires signatures from two different institutions. Confirm:

1. A resume proposal was submitted by one institution and a valid request ID was created.
2. A second institution submitted a co-signature against that exact request ID.
3. Neither signature came from the same institution.

If both conditions are met and the state remains **HALTED**, click **Refresh Status** and check the API Gateway logs for errors.

KYC approval fails with “Provide an approval reason with at least 10 characters”

The reason field on the KYC approval row requires a minimum of 10 characters. Enter a descriptive reason before clicking **Approve KYC**.

Account freeze fails with “Reason must contain at least 10 characters”

The freeze confirmation panel requires a reason of at least 10 characters. Enter a more detailed reason and retry.

The Oversight “Sign Disclosure” panel returns “Already signed or invalid request”

This error appears when the signer ID has already submitted a signature for this request, or when the request is closed (expired or quorum already reached). Fetch the disclosure status by request ID to confirm its current state.

Audit log is empty after applying filters

Filters are applied as a server-side query. If no results appear, try expanding the date range or selecting “All” for category and severity. If the log remains empty with no filters applied, confirm the audit service is running and the API Gateway is accessible.

Treasury Portal — User Manual

(Scenario B)

Audience: Central bank treasury officers and issuance operators. **Portal:** Treasury Portal (Scenario B — International Hub) **Data source:** Live backend. All data shown is real unless an individual section states otherwise.

Important — Scope relative to Scenario A

In Scenario B the responsibilities of the treasury team are narrower than in Scenario A. Cross-border governance actions — circuit-breaker activation, FX-agreement management, multi-party approvals — are handled in the **Governance Portal**. The Treasury Portal focuses on the domestic token lifecycle (issuance, tokenisation, redemption), liquidity pool monitoring, cooperative liquidity provisioning, and audit.

1. Overview

The Treasury Portal is the central bank’s operational workbench for managing the tCeBM (tokenised Central Bank Money) lifecycle on a spoke network within the International Hub architecture. It covers three approval queues, read-only liquidity pool monitoring, cooperative liquidity provisioning, and a structured audit log.

The Governance Portal handles cross-cutting decisions that affect multiple central banks or that require multi-party signatures. The table below summarises the division of responsibilities.

Function	Treasury Portal	Governance Portal
Approve / reject fiat-backed issuance requests	Yes	No
Approve / reject tCeBM tokenisation (lock-to-mint)	Yes	No
Approve / reject tCeBM redemption (burn-to-release)	Yes	No
Provision liquidity pools (propose / confirm pairs, seed reserves)	Yes	No
Monitor hub pool and circuit-breaker state	Yes (read-only)	Full controls

Function	Treasury Portal	Governance Portal
Circuit-breaker (halt / resume swaps)	No	Yes
FX-agreement approval	No	Yes
Identity / compliance governance	No	Yes

2. Access and Login

Open the Treasury Portal URL provided by your platform administrator and sign in with your institutional client credentials.

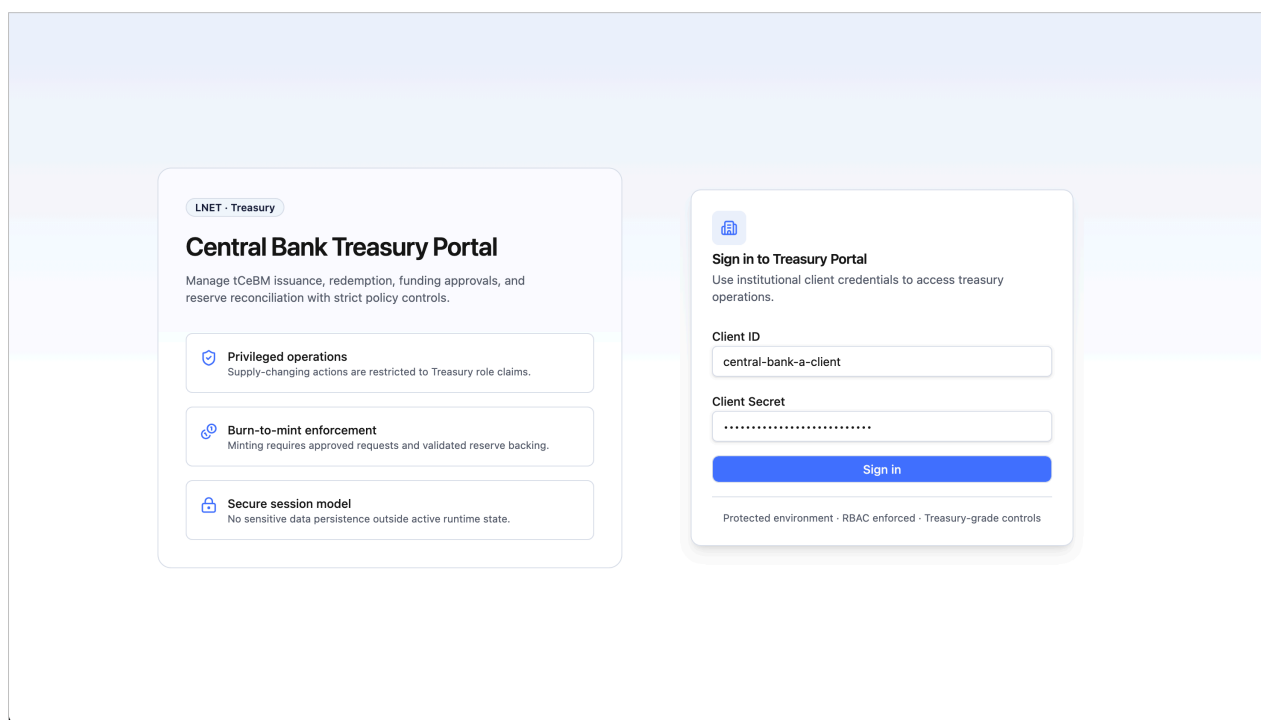


Figure 21: Login

The login form requires two fields:

Field	Description
Username	The Keycloak client ID assigned to this treasury institution (e.g. <code>central-bank-a-treasury</code>). Minimum three characters.
Password	The corresponding client secret. Minimum six characters.

Credentials are validated against Keycloak. A successful authentication redirects to the Dashboard. Access to every route and every approval action requires the **TREASURY** role claim; requests without that claim are blocked at the API gateway.

Credential policy. Mint and burn operations additionally require the `authorizedIssuer` attribute on the authenticated identity. If your account is an authorised issuer, the Dashboard displays a green **Authorised issuer** badge beside your institution name.

Session security. No authentication tokens or sensitive payloads are written to local storage or session storage. Closing the browser tab terminates the active session.

3. Navigation

After login, the application shell presents a sidebar with the following items:

Sidebar label	Route	Description
Dashboard	/	Overview of balance, pending queues, recent operations, and a seven-day activity chart
Issuance Approvals	/deposits-approval	Issuance request queue
Tokenisation Approvals	/escrows-approval	Tokenisation request queue
Redeem Approvals	/redeems-approval	Redemption request queue
Liquidity Management	/liquidity	Read-only monitoring of the pools this central bank provisions
Liquidity Provisioning	/liquidity-provisioning	Currency registry, pair proposal / confirmation, and sovereign pool seeding
Audit	/audit	Structured audit log
Settings	/settings	Security profile reference

The sidebar links appear in the order shown above.

Unauthenticated requests to any protected route are redirected to `/login`. All other unrecognised paths redirect to the Dashboard.

4. Screens

4.1 Dashboard

Route: /

The Dashboard provides a real-time overview of the treasury position and pending workload. It is the first screen seen after login and the primary situational-awareness view.

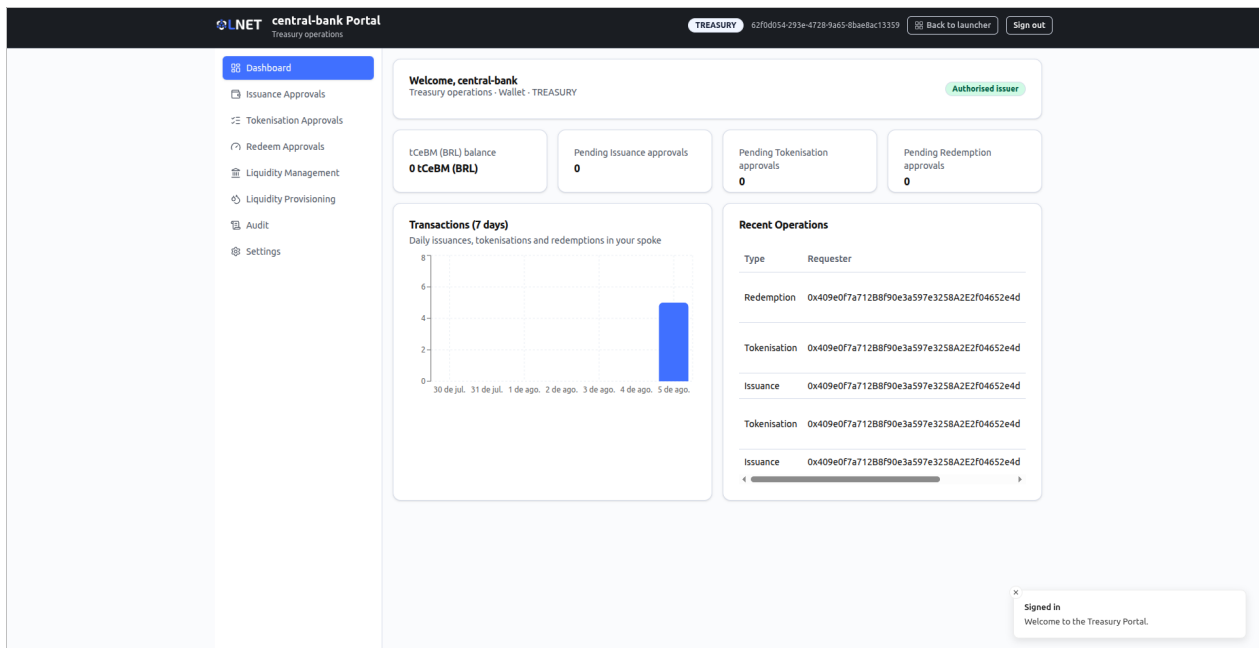


Figure 22: Dashboard

Identity card. The top card greets the authenticated institution (“Welcome, ...”, sourced from `VITE_INSTITUTION_NAME`, the Keycloak institution ID, or the user name), and shows the truncated wallet address, the user role, and an **Authorised issuer** badge when applicable.

KPI row. Four summary cards show:

- tCeBM balance of the treasury wallet.
- Number of pending Issuance approvals.
- Number of pending Tokenisation approvals.
- Number of pending Redemption approvals.

Transactions (7 days). A bar chart of daily activity over the last seven days, counting this spoke’s own operations (issuances, tokenisations, and redemptions) by their creation date.

Recent Operations table. Shows the eight most recent operations across all three queues (Issuance, Tokenisation, Redemption), sorted by creation time descending. Columns: Type, Requester, Amount, Status. When there is no activity, the table shows “No operations yet.”

4.2 Issuance Approvals

Sidebar label: Issuance Approvals **Route:** `/deposits-approval`

This screen manages requests from commercial bank participants to obtain fiat-backed tCeBM (the “Issuance” or “Deposit” flow). When a participant deposits fiat reserves and requests a corresponding tCeBM mint, the request appears here for treasury approval.

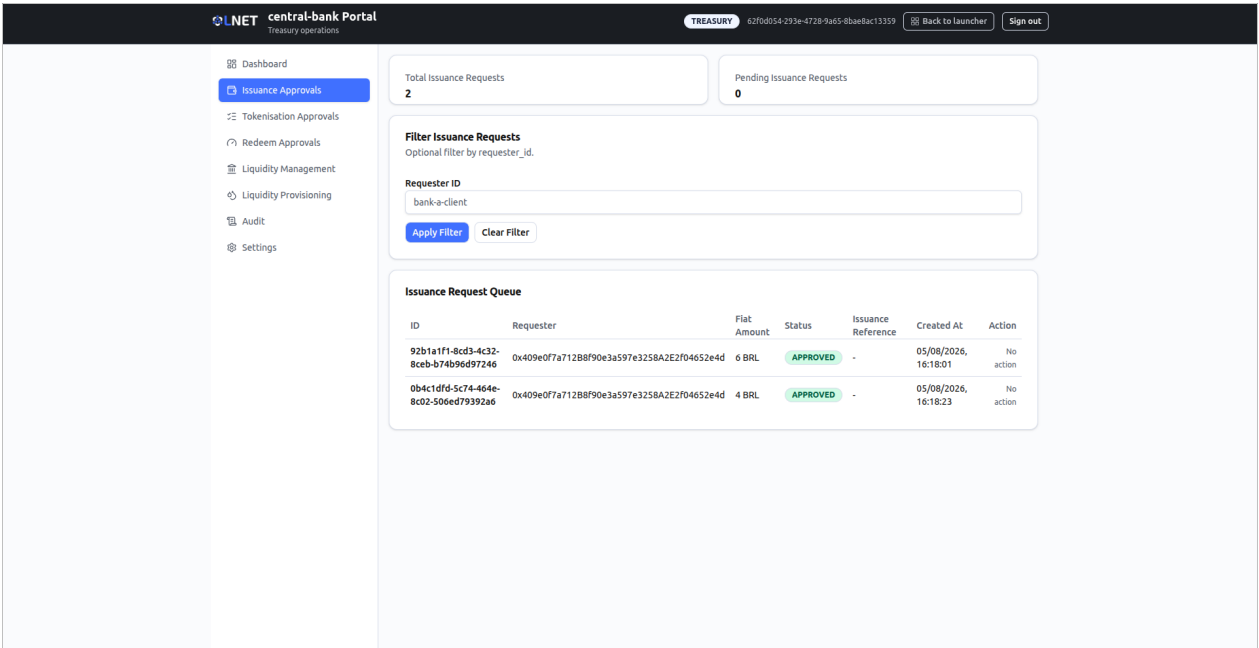


Figure 23: Issuance Approvals

Summary cards. Show the total number of issuance requests and the number currently in **PENDING** status.

Filter. An optional requester ID filter narrows the queue to requests from a specific participant. Enter a participant identifier (e.g. **bank-a-client**) and click **Apply Filter**. Click **Clear Filter** to return to the full list.

Issuance Request Queue. Each row represents one request:

Column	Description
ID	Internal request identifier.
Requester	ID of the participant who submitted the request.
Fiat Amount	The fiat value deposited, expressed in the spoke’s domestic currency.
Status	Current lifecycle status (see Section 6).
Issuance Reference	Truncated on-chain mint transaction hash (hover for full hash).
Created At	Timestamp of request submission.
Action	Approve or Reject buttons, visible only for PENDING requests.

Approving a request.

1. Locate the request in the queue (use the filter if the queue is long).
2. Click **Approve Issuance**. A confirmation card appears at the bottom of the page showing the request ID.
3. Click **Confirm Action** to submit. A success notification confirms the approval and the on-chain mint transaction is triggered. The row status updates to **APPROVED**.

Rejecting a request.

1. Click **Reject** on the target row. A rejection card appears.

2. Enter a reason (minimum three characters — the field will not submit without one).
3. Click **Confirm Reject**. The status updates to **REJECTED** and the reason is recorded.

Only one confirmation card (approve or reject) is visible at a time. Clicking **Cancel** in either card dismisses it without taking action.

4.3 Tokenisation Approvals

Sidebar label: Tokenisation Approvals **Route:** /escrows-approval

This screen handles tokenisation requests — the cross-currency flow where a participant burns tokens on one spoke network (Redemption leg) and mints tokens on a destination spoke (Issuance leg) via the hub. Both legs are coordinated atomically; approving an escrow triggers both the on-chain burn (Redemption tx hash) and the on-chain mint (Issuance ref).

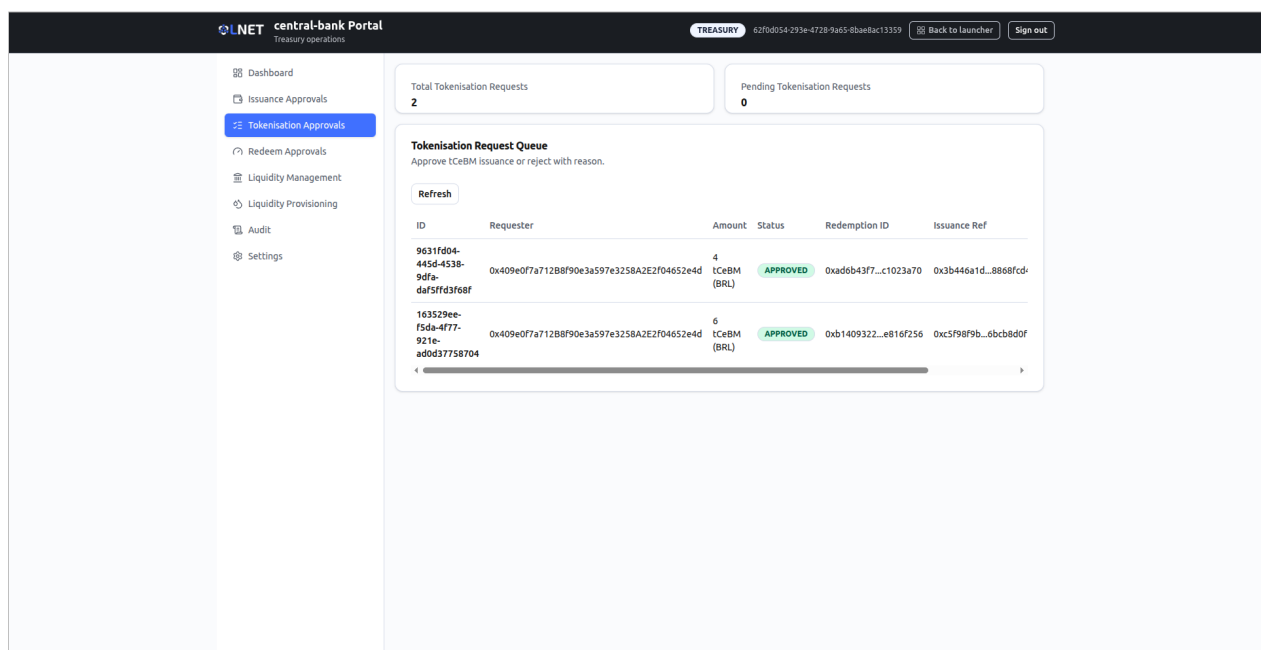


Figure 24: Tokenisation Approvals

Summary cards. Total and pending tokenisation request counts.

Tokenisation Request Queue. Columns:

Column	Description
ID	Internal escrow request identifier.
Requester	ID of the originating participant.
Amount	tCeBM amount, labelled with the spoke's domestic currency code.
Status	Current lifecycle status (see Section 6).
Redemption ID	Truncated burn transaction hash (hover for full hash).
Issuance Ref	Truncated mint transaction hash (hover for full hash).
Created At	Timestamp of request submission.

Column	Description
Action	Approve or Reject, visible only for PENDING requests.

Approving a tokenisation request.

1. Click **Approve** on the target row.
2. In the confirmation card, click **Confirm Approve**.
3. On success, a notification shows truncated hashes for both the burn and the mint transactions. Both legs complete atomically.

Rejecting a tokenisation request.

1. Click **Reject** on the target row.
2. Enter a reason (minimum three characters).
3. Click **Confirm Reject**.

Use the **Refresh** button to reload the queue from the backend at any time.

4.4 Redeem Approvals

Sidebar label: Redeem Approvals **Route:** /redeems-approval

Redemption requests represent a participant returning tCeBM to the central bank in exchange for fiat reserves. Approving a redeem burns the participant's tCeBM via a Zeto privacy transfer and releases the corresponding fiat reserve.

The screenshot displays the 'Treasury operations' section of the 'LNET central-bank Portal'. The sidebar on the left contains links to Dashboard, Issuance Approvals, Tokenisation Approvals, Redeem Approvals (highlighted), Liquidity Management, Liquidity Provisioning, Audit, and Settings. The main content area features two summary cards: 'Total Redeems' with a count of 1, and 'Pending Redeems' with a count of 0. Below these is the 'Redeem Queue' section, which includes a 'Refresh' button and a table for redeem requests. The table has the following columns: ID, Requester, Amount, Status, Zeto Transfer Tx Hash, Redemption ID, Created At, and Action. A single row is shown with the following data: ID '924a1c8e-ea79-447b-9419-da5fa739408d', Requester '0x409e0f7a712b8f90e3a597e3258a2e2f04652e4d', Amount '1 tCeBM (BRL)', Status 'APPROVED', Zeto Transfer Tx Hash '-', Redemption ID '-', Created At '05/08/2026, 16:18:29', and Action 'No action'.

Figure 25: Redeem Approvals

Summary cards. Total and pending redeem counts.

Redeem Queue. Columns:

Column	Description
ID	Internal redeem request identifier.
Requester	ID of the originating participant.
Amount	tCeBM amount to be burned.
Status	Current lifecycle status (see Section 6).
Zeto Transfer Tx Hash	Truncated hash of the privacy-preserving burn transaction (hover for full hash).
Redemption ID	Truncated fiat mint transaction hash representing the fiat reserve release (hover for full hash).
Created At	Timestamp of request submission.
Action	Approve or Reject, visible only for PENDING requests.

Approving a redemption.

1. Click **Approve** on the target row.
2. In the confirmation card, click **Confirm Approve**.
3. On success, a notification shows the truncated fiat mint transaction hash.

Rejecting a redemption.

1. Click **Reject** on the target row.
2. Enter a reason (minimum three characters).
3. Click **Confirm Reject**.

Use the **Refresh** button to reload the queue at any time.

4.5 Liquidity Management

Sidebar label: Liquidity Management **Route:** /liquidity

This screen is a read-only monitor of the cross-currency pools that this central bank provisions. It lists only the pools whose pair includes the institution's own national currency (the corridors it sovereignly provisions). Provisioning actions — proposing and confirming pairs and seeding reserves — are performed on the separate **Liquidity Provisioning** screen (Section 4.6).

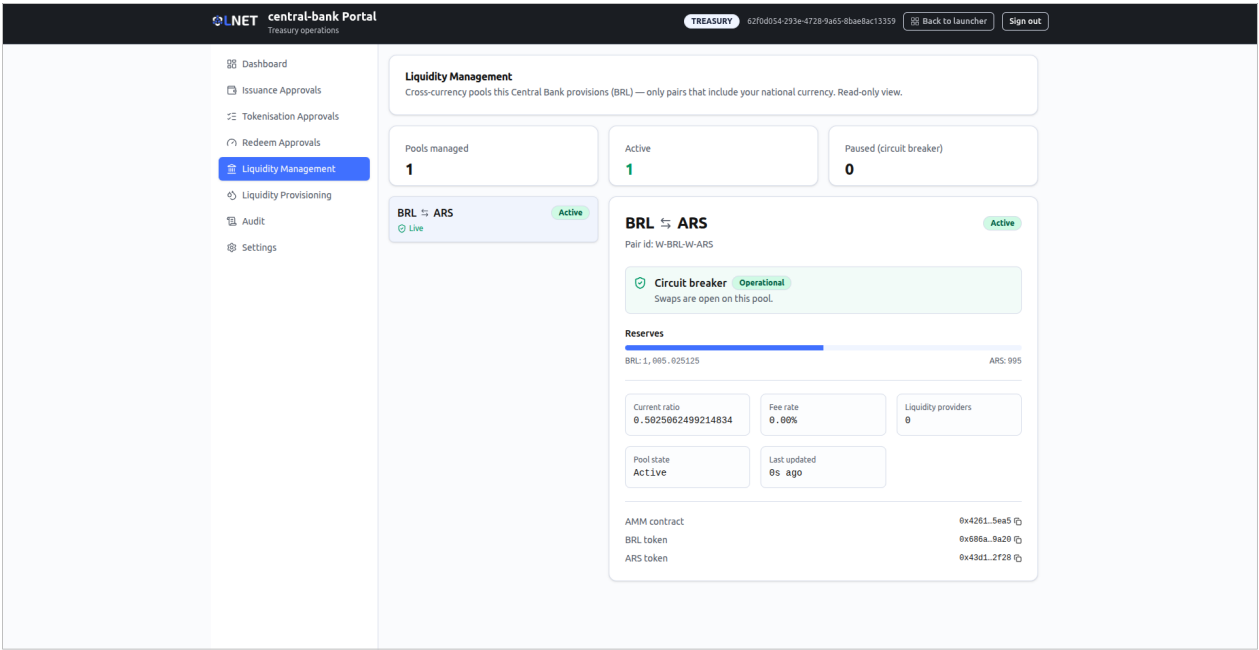


Figure 26: Liquidity Management

Live pool and circuit-breaker data refresh automatically about every 15 seconds. If the national currency cannot be determined (for example, the balance lookup fails), the screen falls back to showing all pools rather than hiding everything.

Summary cards. Three cards at the top:

Card	Description
Pools managed	Number of pools that include this central bank’s national currency.
Active	Number of those pools whose status is ACTIVE .
Paused (circuit breaker)	Number of those pools whose circuit breaker is HALTED .

Pool list. A selectable list of the managed pools. Each entry shows the currency pair (e.g. BRL ↔ COP), a status badge (Active, Awaiting counterpart, Empty, or Unknown), and the circuit-breaker state (Live or Paused). Selecting a pool opens its detail panel.

Pool detail panel. For the selected pool:

- **Circuit breaker banner.** Shows the current breaker state for that pool. This is read-only in the Treasury Portal; pausing and resuming are performed in the Governance Portal.

Breaker state	Badge	Meaning
LIVE	Operational	Swaps are open on this pool.
HALTED	Paused	A central bank triggered the circuit breaker; swaps are suspended. The pause reason and initiator are shown when available.

Breaker state	Badge	Meaning
RESUME_PENDING	Resume pending	A resume proposal is awaiting the 2-of-N central bank quorum.
(other)	Unknown	Circuit-breaker state is not available.

- **Reserves.** A balance bar with each side's reserve amount. An **Imbalanced** badge appears when the backend flags a reserve imbalance.
- **Metrics grid.** Current ratio, fee rate, liquidity provider count, pool state, and last-updated time.
- **On-chain references.** The AMM contract address, both token addresses (click to copy), and the proposer and confirmer central bank identifiers when present.

4.6 Liquidity Provisioning

Sidebar label: Liquidity Provisioning **Route:** /liquidity-provisioning

This screen is where a central bank provisions cross-currency liquidity: it registers pairs on the hub and seeds pool reserves. Signing is performed by this central bank's own gateway; each institution acts only on its own side (sovereign provisioning).

Registered Currencies
National currencies registered on the Hub currency registry. [Refresh](#)

Symbol	Country	Token Address	Proposer CB
W-tCeBM_BRL	Sovereign BRL	0x886afd...949a29	spoke-brl
W-tCeBM_ARS	Sovereign ARS	0x43d1f9...b22f28	spoke-ars

Propose Pair
Pick two registered currencies to propose a new AMM pair. Token addresses and the proposer CB are derived automatically; signing is done by this central bank's gateway.

Currency A:

Currency B:

Pair ID:

Token A: -
Token B: -
Proposer CB: -

[Propose Pair](#)

Confirm Pair
Select a proposed pair to confirm it. The confirmer CB is derived from the pair's token B currency.

Proposed Pair:

Confirmer CB (derived):

[Confirm Pair](#)

Figure 27: Liquidity Provisioning

Registered Currencies. A table of the national currencies registered on the hub currency registry. Columns: Symbol, Country, Token Address, Proposer CB. A **Refresh** button reloads the list.

Propose Pair. Creates a new AMM pair from two registered currencies.

1. Select **Currency A** and **Currency B** from the registered currencies.
2. The **Pair ID**, both token addresses, and the **Proposer CB** are derived automatically (the proposer CB is taken from Currency A). These derived values are shown for reference.

3. Click **Propose Pair**. The gateway deploys a dedicated per-pair AMM bound to the two tokens; there is no operator-supplied AMM address. A confirmation shows the resulting pair ID, status, and the AMM address when returned.

Confirm Pair. Confirms a pair that is still awaiting confirmation.

1. Select a proposed pair from the drop-down. Only pairs in **PROPOSED** or **PENDING** status appear.
2. The **Confirmer CB** is derived from the pair’s token B currency and shown for reference.
3. Click **Confirm Pair**. A confirmation shows the pair ID, status, and transaction hash.

Seed Liquidity (sovereign). Funds a pool’s reserves through an escrow-and-finalize flow. Each central bank deposits only its own currency; its side is auto-detected from the pair.

1. Select a **Pool Pair** and enter an **Amount** in whole tokens of your currency (converted to 18-decimal base units on submit).
2. Click **Deposit My Side** to escrow your side. The escrow status panel shows whether Side A and Side B are deposited and whether the pool is finalized.
3. When both sides are escrowed, anyone may click **Finalize Pool** to fund the pool reserves atomically (shares for both sides are reported on success).
4. Before finalize, you may click **Reclaim My Side** to withdraw your own pending deposit.

4.7 Audit

Sidebar label: Audit **Route:** /audit

The Audit screen (titled “Audit Logs”) provides a filterable log of treasury-relevant events recorded by the backend. This is the primary tool for regulatory review and incident investigation.

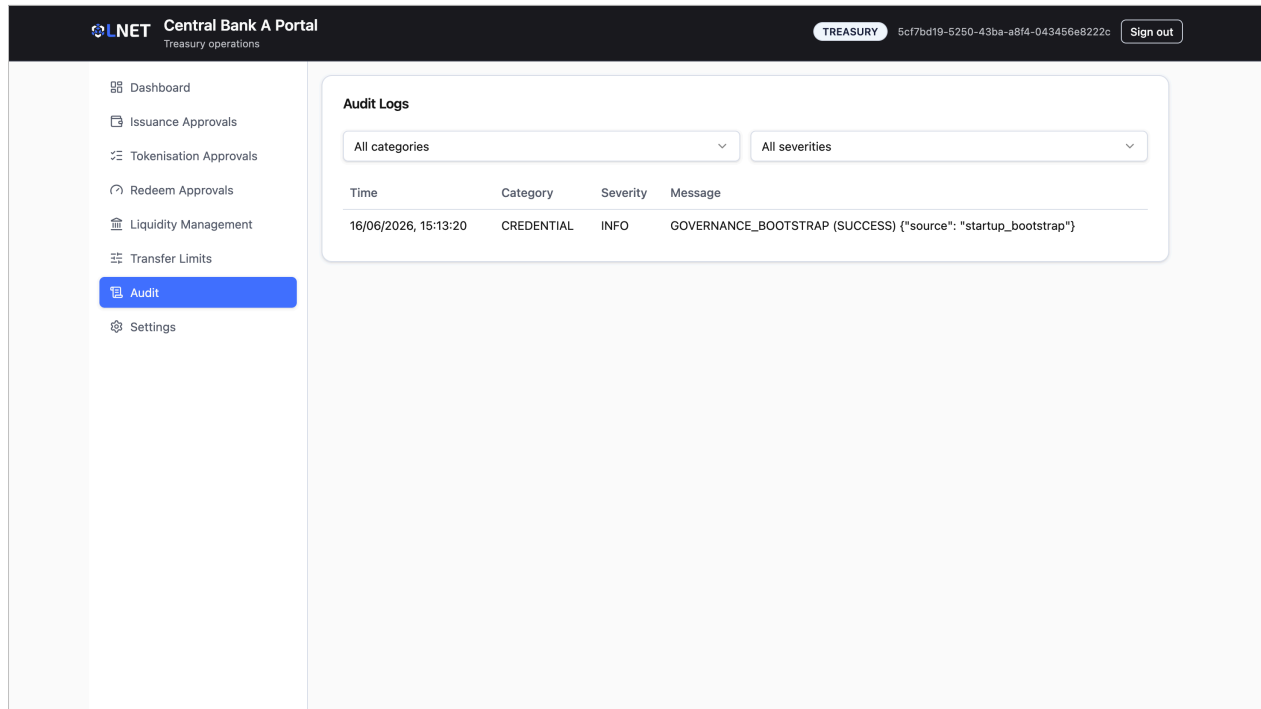


Figure 28: Audit

Filters. Two dropdown filters can be combined independently:

Filter	Options
Category	ALL, AUTH, FUNDING, TREASURY, KYC, RECONCILIATION
Severity	ALL, INFO, WARNING, CRITICAL

Selecting a filter applies it immediately to the displayed results.

Log table. Columns:

Column	Description
Time	Local timestamp of the audit event.
Category	Functional area that generated the event.
Severity	INFO (routine), WARNING (requires attention), or CRITICAL (immediate action needed).
Message	Human-readable description of the event.

The log is loaded from the backend on page open. A loading indicator appears while the request is in flight. Any backend error is displayed below the table.

The audit log is append-only on the backend. Records cannot be edited or deleted from this interface.

4.8 Settings

Sidebar label: Settings **Route:** /settings

The Settings screen (titled “Security Settings”) is a read-only reference to the portal’s operational security profile. It has no editable fields. Three sections are shown:

Section	Content
RBAC	Route and action guards require the TREASURY role claim.
Credential Policy	Mint operations require an approved funding request and valid issuer credentials.
Session Handling	No local storage or session storage is used for authentication tokens or sensitive payloads.

5. Typical Workflows

5.1 Processing a fiat deposit and issuing tCeBM

1. Navigate to **Issuance Approvals**.
2. Review the pending issuance request: verify the requester identity and fiat amount.
3. If the deposit is valid: click **Approve Issuance**, then **Confirm Action**. The payment orchestrator triggers the on-chain mint; the status moves to **APPROVED**.
4. If the deposit is invalid or policy-non-compliant: click **Reject**, enter the reason, click **Confirm Reject**. The status moves to **REJECTED** and the reason is persisted.

5.2 Processing a cross-border tokenisation (escrow)

1. Navigate to **Tokenisation Approvals**.
2. Review the pending tokenisation request: verify the requester, the tCeBM amount, and the associated Redemption and Issuance transaction references.
3. If the request is valid: click **Approve**, then **Confirm Approve**. Both the burn (on the source spoke) and the mint (on the destination spoke) execute atomically. The notification confirms both transaction hashes.
4. If the request should be declined: click **Reject**, enter the reason, click **Confirm Reject**.

5.3 Processing a tCeBM redemption

1. Navigate to **Redeem Approvals**.
2. Review the pending redeem: verify the requester identity, the tCeBM amount, and the Zeto transfer transaction hash.
3. If the redemption is valid: click **Approve**, then **Confirm Approve**. The burn is confirmed on-chain and the fiat reserve is released. The fiat mint transaction hash appears in the success notification.
4. If the redemption should be declined: click **Reject**, enter the reason, click **Confirm Reject**.

5.4 Provisioning and seeding a cross-currency pool

1. Coordinate with the counterpart central bank on the currencies and contribution amounts, using off-system communication as needed.
2. Navigate to **Liquidity Provisioning**.
3. Confirm both currencies appear in the **Registered Currencies** table.
4. Under **Propose Pair**, select Currency A and Currency B; the pair ID, token addresses, and proposer CB are derived automatically. Click **Propose Pair** to deploy the per-pair AMM.
5. The counterparty central bank selects the proposed pair under **Confirm Pair** and clicks **Confirm Pair**.
6. Under **Seed Liquidity (sovereign)**, each central bank selects the pool pair, enters the amount in whole tokens of its own currency, and clicks **Deposit My Side**. Your side is auto-detected from the pair. Use **Reclaim My Side** to withdraw a pending deposit before finalize.
7. Once both sides show as deposited, click **Finalize Pool** to fund the reserves atomically.
8. Monitor the resulting pool on the **Liquidity Management** screen, where its status moves to **ACTIVE** and commercial swaps can proceed.

5.5 Investigating a suspicious event

1. Navigate to **Audit**.
2. Set the **Severity** filter to **CRITICAL** or **WARNING** to surface high-priority events.
3. Optionally set the **Category** filter to narrow to a functional area (e.g. **TREASURY**).
4. Review the message column for event descriptions.
5. If further investigation is required, use the event timestamp to correlate with backend service logs or the Supervisor Portal's disclosure workflow.

6. Status Reference

The following status values apply to Issuance (Deposit), Tokenisation (Escrow), and Redemption (Redeem) records throughout the portal.

Status	Display colour	Meaning
PENDING	Amber	The request has been received and is awaiting a treasury decision. Approve or Reject actions are available.
APPROVED	Green	The request was approved. The on-chain mint or burn completed successfully.
REJECTED	Red	The request was declined by treasury. The rejection reason is stored on the record.
MINT FAILED	Red	The request was approved but the on-chain operation failed. Contact the platform operator.
UNKNOWN	Outline	The status value returned by the backend could not be recognised. Report to the platform operator.

7. Troubleshooting

The queue shows no requests

Verify that the backend payment-orchestrator service is reachable. Check the browser developer console for network errors. If a filter is active, click **Clear Filter** and reload. Confirm with the platform administrator that requests have been submitted by participants.

Approve or Reject returns an error

An error notification appears at the top of the page with the backend error message. Common causes:

- The request is no longer in **PENDING** status (was acted on by another operator in a concurrent session).
- The authenticated identity does not have the **authorizedIssuer** attribute (for mint operations).
- The backend payment-orchestrator or ledger-gateway is temporarily unavailable.

Refresh the queue and retry. If the error persists, escalate to the platform operator with the error text and the request ID.

A pool circuit breaker shows Paused (HALTED)

On the **Liquidity Management** screen, the selected pool's circuit-breaker banner shows "Paused" when swaps are suspended by a governance action. This is not a treasury action — contact the operator responsible for the **Governance Portal** to review the pause reason and initiate the resume process. The circuit-breaker state is read-only in the Treasury Portal.

Pool shows "Imbalanced"

The AMM reserves have diverged from the expected ratio. Open the pool on the **Liquidity Management** screen for current reserve levels. If the imbalance is caused by a recent large swap, it may

self-correct as subsequent swaps trade in the opposite direction. Additional reserves can be seeded on the **Liquidity Provisioning** screen.

A proposed pair does not appear under Confirm Pair

The **Confirm Pair** drop-down lists only pairs still in **PROPOSED** or **PENDING** status. If a pair is missing, click **Refresh** on the Registered Currencies card to reload registry data, and confirm the proposal transaction succeeded on the proposing side.

Finalize Pool is disabled

The **Finalize Pool** button is enabled only once both sides of the pool have been escrowed. Confirm the escrow status panel shows Side A and Side B as deposited, and that the pool is not already finalized.

Login fails

Verify that the Username and Password are correct for this institution. The username (Keycloak client ID) must be at least three characters; the password (client secret) at least six. If credentials are correct but login still fails, the Keycloak service may be temporarily unavailable — contact the platform administrator.

Audit log does not load

The audit backend may be temporarily unavailable. An error message appears below the log table when this occurs. Retry after a short interval. If the problem persists, escalate to the platform operator.

Supervisor Portal — User Manual

(Scenario B: International Hub)

Audience: Supervisors and regulators.

IMPORTANT — Preview Status

The Supervisor Portal is currently a **UI preview**. The portal's data sources depend on the backend services being reachable at the configured API base URL. When those services return data, the portal displays it; when they are unavailable (as is the case in most local or demo environments), each screen will show empty tables or zero counters.

Additionally, the authentication flow uses a username and password form that talks to the platform's OIDC provider. In environments where Keycloak is not running, sign-in will fail.

- **Do not** rely on any figure, pool status, audit entry, or participant record shown in a demo or offline environment for a real supervisory or regulatory decision.
 - This manual documents the screens as implemented and calls out every area where live backend connectivity is required.
-

1. Overview

The Supervisor Portal is the **read-only oversight surface** for supervisors and regulators in **Scenario B — International Hub**. Scenario B uses an AMM-based (Automated Market Maker) hub architecture for cross-border settlement, with privacy-preserving ZKP tokens (Zeto) on the spoke networks. The portal's surveillance responsibilities therefore differ substantially from Scenario A:

- **AMM liquidity health.** Pool reserves and imbalance thresholds are the primary stability signal; there are no HTLC expiry clocks to monitor.
- **Circuit-breaker state.** A governance circuit breaker can pause the entire hub. CRITICAL alerts are raised when the circuit breaker is tripped.
- **ZK commitment verification.** Because Zeto ZKP shielding means transaction amounts and parties are never revealed on-chain, the portal provides a ZK Pointer tool for confirming that a shielded transfer commitment was validly issued without decrypting its contents.
- **AML/CFT disclosure workflow.** Multi-party 2-of-N disclosure requests allow authorised supervisors to co-sign an investigation into a specific shielded transaction reference.

Actor	Portal	Intended role
Supervisor / Regulator	Supervisor Portal	Read-only monitoring, compliance oversight, audit review, ZK verification, AML/CFT investigation initiation

2. Who uses this portal

Regional supervisors and regulators with institutional SSO credentials issued by a participating central bank. The portal is a privileged environment: every route is protected and requires an authenticated session. There is no guest or read-only anonymous access.

3. Access and login

Route: /login

Navigate to the Supervisor Portal URL. The login page presents an institutional SSO/OIDC sign-in form.

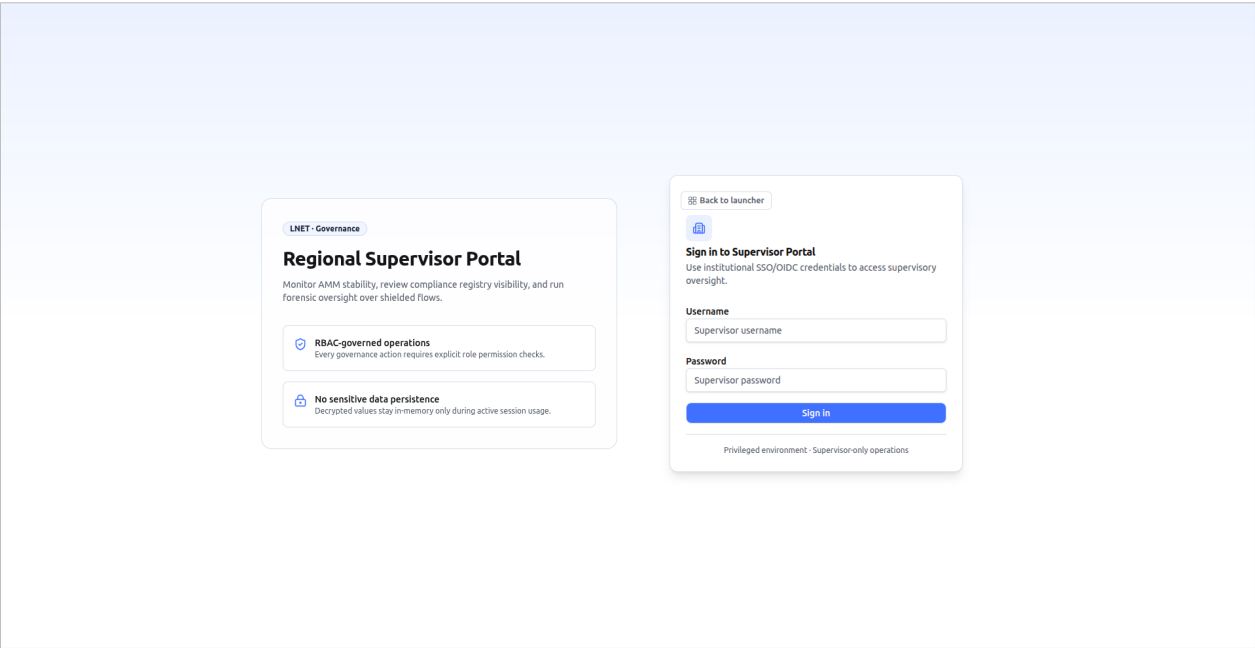


Figure 29: Login

Fields:

Field	Description
Username	The supervisor username issued to your institution (minimum 3 characters).
Password	The corresponding supervisor password (minimum 6 characters).

How authentication works:

The form submits the username and password to the platform’s Keycloak OIDC provider via the backend auth service (POST /api/v1/auth/login). On success, the portal establishes the session, displays a confirmation toast, and redirects to the Dashboard. A short-lived access token is renewed automatically for the duration of the session; closing the tab ends the session.

Privileged environment. This login is connected to the platform’s production identity provider. Credentials are institution-issued and supervised. Do not share your credentials.

Session security features (configured in Settings):

- Strict session management requires periodic re-authentication for privileged operations.
- Decrypted values are held in memory only during active session usage; they are never persisted to local storage or disk.

Unauthenticated requests to any protected route are redirected to /login automatically.

4. Navigation

After login, the left sidebar provides access to all portal screens. On smaller viewports the sidebar collapses to a top navigation bar.

Sidebar label	Route	Description
Dashboard	/	Network-level overview of AMM health and recent audit events
Liquidity Monitor	/liquidity	Detailed pool-by-pool AMM surveillance
Compliance Registry	/participants	Searchable registry of participating institutions
Audit Vault	/audit	Immutable compliance audit log with ZK pointer verification
Stability Insights	/stability	AMM risk posture summary and circuit-breaker awareness
Investigation	/investigation	AML/CFT disclosure request workflow
Settings	/settings	Session security and operational preferences

All routes require an authenticated session. Visiting an unknown path redirects to the Dashboard.

5. Screens

5.1 Dashboard

Route: / · Sidebar label: Dashboard

The Dashboard provides a real-time network-level overview by combining data from the network overview API and the stability (AMM pool) API. It is the recommended starting point for any supervisory session.

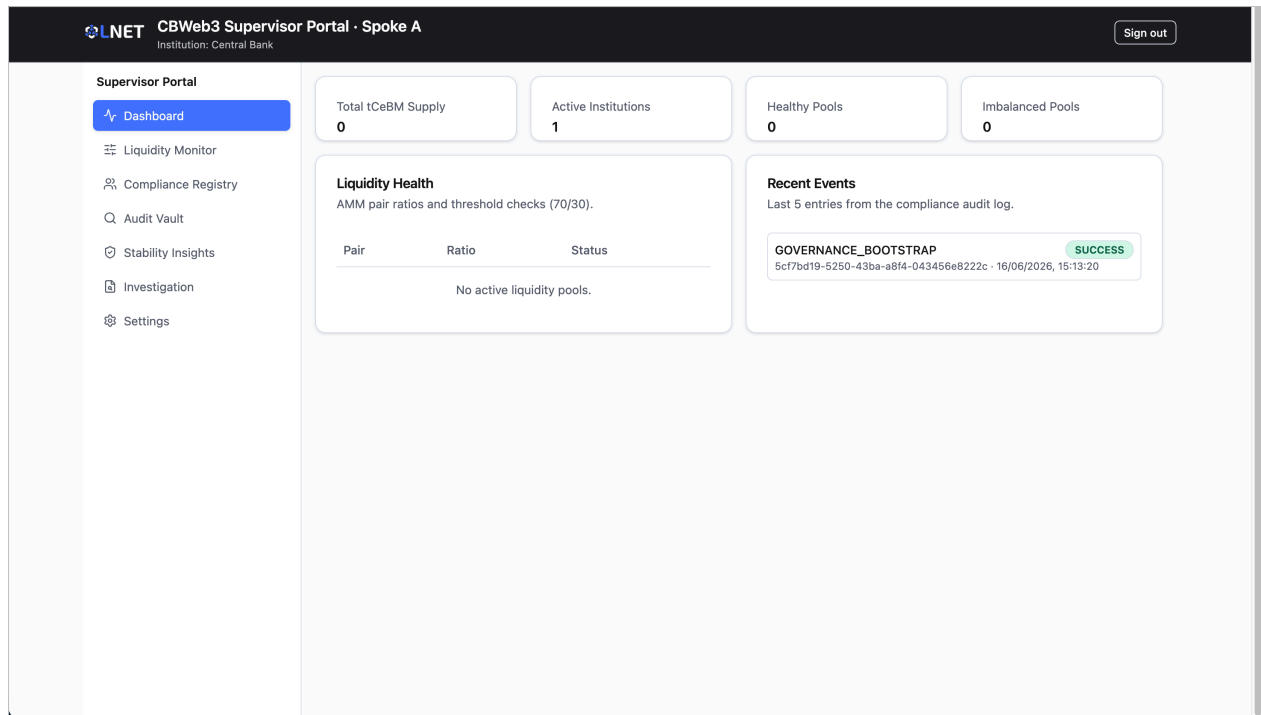


Figure 30: Dashboard

Summary cards (top row):

Card	Source	Description
Total tCeBM Supply	Network overview API	Aggregate reserve-layer tCeBM token supply across the hub.
Active Institutions	Network overview API	Count of institutions with an ACTIVE credential in the compliance registry.
Healthy Pools	Network overview API	Number of AMM pools whose reserve ratio is within the acceptable threshold.
Imbalanced Pools	Network overview API	Number of AMM pools flagged as imbalanced (ratio outside the 70/30 threshold).

Live data required. These counters are populated from the backend network overview API. In an offline environment they will display as dashes (-).

Liquidity Health table (lower left):

Displays AMM pair ratios from the stability API, using the same threshold logic as the Liquidity Monitor page. Each row shows:

- **Pair** — the currency pair label (for example, W-BRL-ARS).
- **Ratio** — the current reserve ratio expressed as `ratioA / ratioB`.
- **Status** — HEALTHY (green) or IMBALANCED (red) based on the pool's `imbalance_flag`.

The imbalance threshold is **70/30**: a pool whose dominant reserve exceeds 70% of the total is flagged as imbalanced.

Live data required. Pool ratios come from the AMM pool status API (GET /api/v2/amm/pool/{pair}/status). Empty tables indicate the backend is unreachable.

Recent Events panel (lower right):

Shows the five most recent entries from the compliance audit log. Each entry displays:

- The action name and outcome badge (SUCCESS or FAILURE).
- Actor identity, optional target subject, and timestamp.

Live data required. Entries come from the audit log API (GET /api/v1/compliance/audit/logs).

5.2 Liquidity Monitor

Route: /liquidity · **Sidebar label:** Liquidity Monitor

This screen is specific to Scenario B and provides detailed, pool-level AMM surveillance. It is the primary tool for assessing hub liquidity health during normal operations or a market stress event.

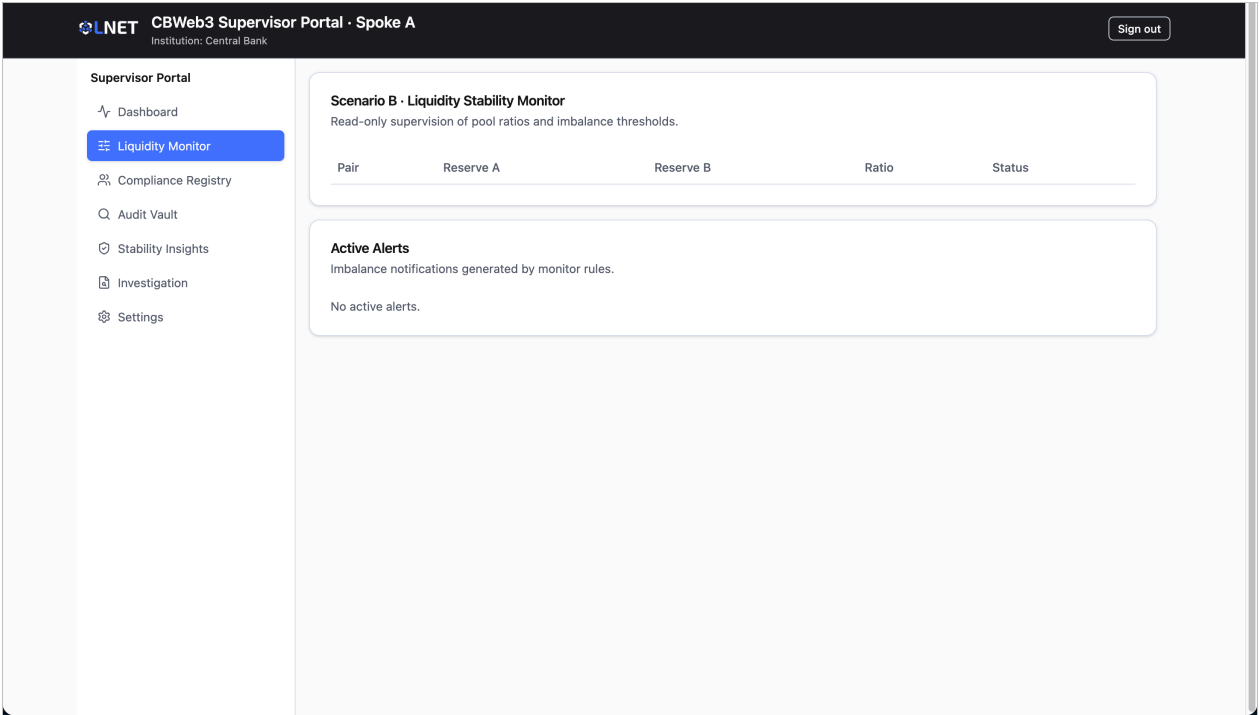


Figure 31: Liquidity Monitor

Scenario B specific. This page does not exist in the Scenario A Supervisor Portal. It reflects the AMM architecture of the International Hub.

Pool Status table:

One row per known AMM pair (currently W-BRL-ARS). Columns:

Column	Description
Pair	Currency pair identifier.

Column	Description
Reserve A	Raw reserve of token A in the pool (displayed in human-readable units, converted from wei).
Reserve B	Raw reserve of token B in the pool (in human-readable units).
Ratio	Current ratio expressed as <code>current_ratio</code> / 1.
Status	HEALTHY (green) or IMBALANCED (red).

Pools with `pool_status` = `EMPTY` are excluded from the table; they indicate the pool has not yet been funded.

Active Alerts panel:

Alert cards are generated in real time from two sources:

1. **Circuit-breaker alerts.** If the governance circuit breaker is in `PAUSED` state, a **CRITICAL** severity alert labelled **ALL** is displayed with the pause reason.
2. **Pool imbalance alerts.** For each active pool with `imbalance_flag` = `true`, a **HIGH** severity alert is shown.

Alert severity	Badge colour	Trigger
CRITICAL	Red	Circuit breaker is <code>PAUSED</code>
HIGH	Amber	A specific pool is imbalanced

When no alerts are active the panel displays “No active alerts.”

Live data required. Both pool status and circuit-breaker status are fetched from live backend APIs. In an offline environment the table will be empty and no alerts will be shown.

5.3 Compliance Registry

Route: `/participants` · **Sidebar label:** Compliance Registry

A searchable, read-only view of all institutions registered in the hub compliance system. Use this screen to verify participant credentials and jurisdictional standing.

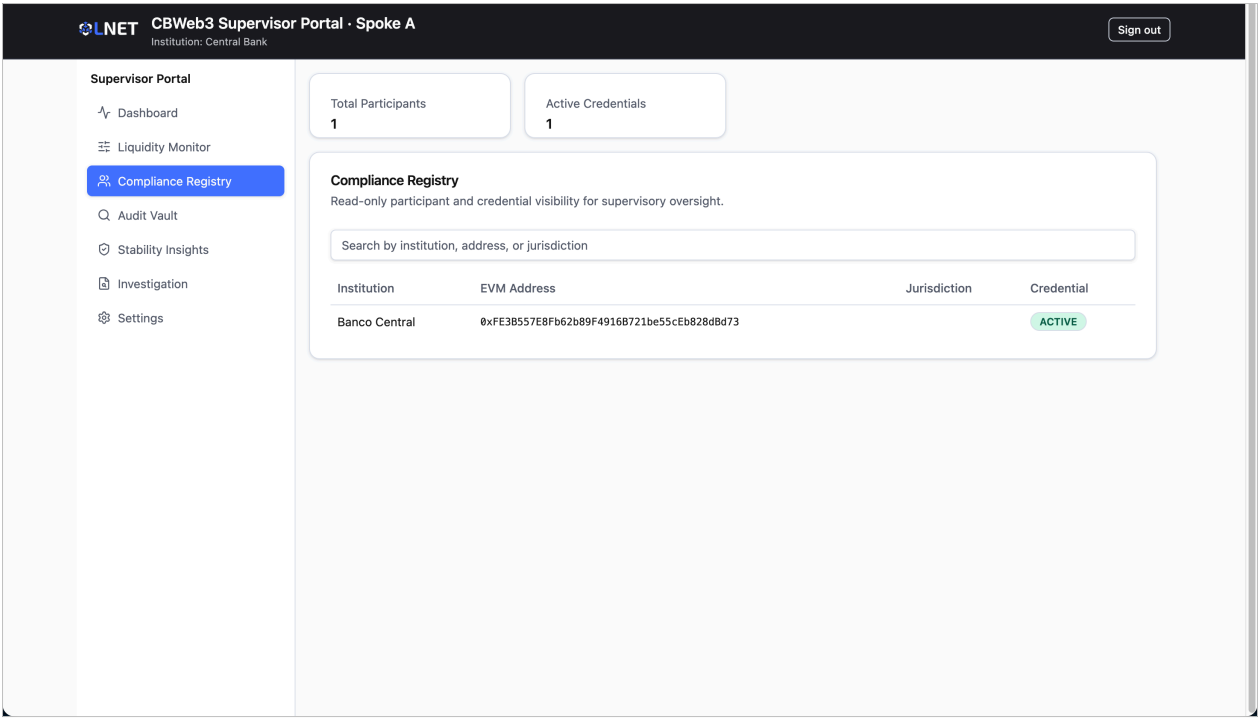


Figure 32: Compliance Registry

Summary cards:

Card	Description
Total Participants	Count of all records in the registry.
Active Credentials	Count of participants whose <code>credentialStatus</code> is <code>ACTIVE</code> .

Registry table:

Column	Description
Institution	Human-readable name of the institution.
EVM Address	On-chain address (displayed in monospace).
Jurisdiction	ISO jurisdiction code.
Credential	<code>ACTIVE</code> (green) or <code>REVOKED</code> / <code>PENDING</code> (red).

Search / filter:

The search box filters the visible rows in real time against institution name, EVM address, and jurisdiction code. No server round-trip is required for filtering; all records are loaded on page entry.

Live data required. The registry is populated from the compliance service participant API. In an offline environment the table will be empty.

5.4 Audit Vault

Route: `/audit` · Sidebar label: Audit Vault

The Audit Vault has two sections: a ZK Pointer verification tool and the immutable compliance audit log. This arrangement reflects Scenario B’s privacy model: because Zeto ZKP shielding means that amounts and parties are never revealed on-chain, classical “decrypt with view key” functionality does not apply. The ZK Pointer tool provides the appropriate alternative.

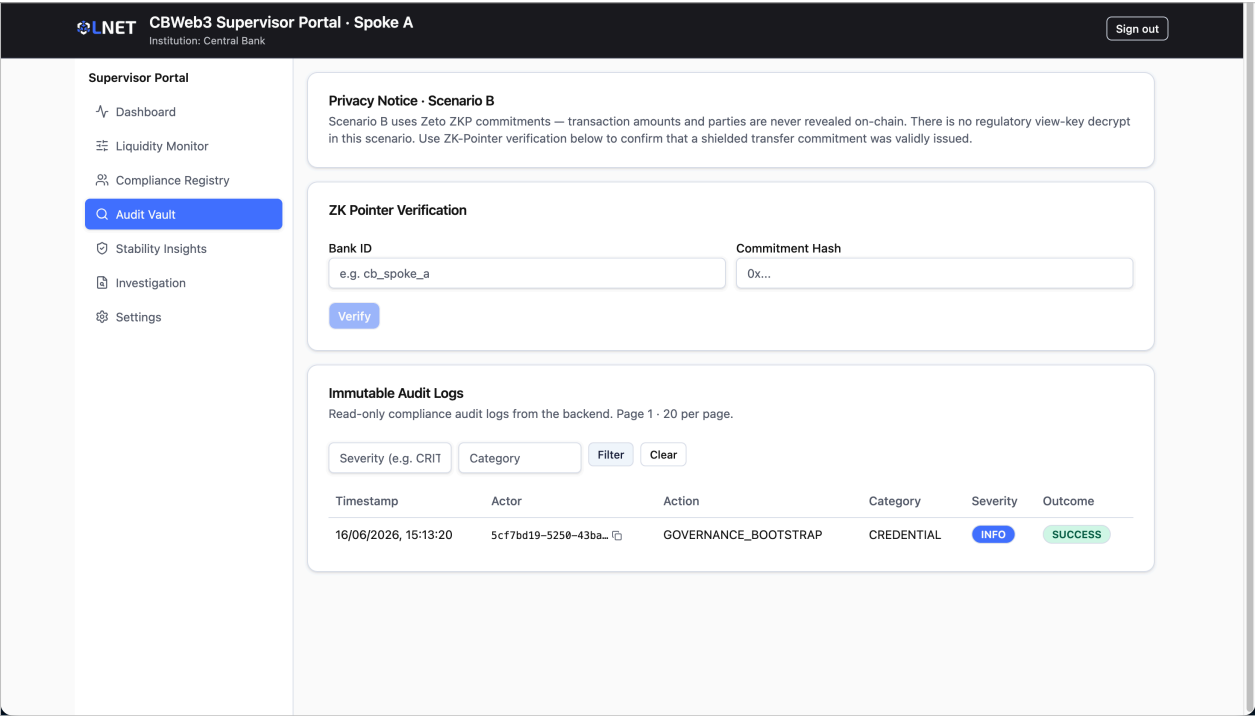


Figure 33: Audit Vault

Privacy notice (top card):

A non-dismissable notice reminding the user that Scenario B uses Zeto ZKP commitments. Transaction amounts and parties are not accessible even to supervisors via the portal. The ZK Pointer mechanism is the supported method for confirming the validity of a shielded transfer.

ZK Pointer Verification panel:

Allows a supervisor to confirm that a shielded transfer commitment was validly issued by a specific bank without revealing its contents.

Field	Description
Bank ID	Identifier of the issuing institution (example: <code>cb_spoke_a</code>).
Commitment Hash	The 0x-prefixed commitment hash of the shielded transfer.

Click **Verify** to submit. The panel displays a result card with:

Result field	Description
State	VALID (green), EXPIRED (amber), or INVALID (red).
Pointer ID	Internal pointer identifier (displayed if present).

Result field	Description
Commitment	The commitment hash that was verified (truncated for display).
Expires At	Expiry timestamp of the pointer (if applicable).

Live data required. Verification calls the ZK Pointer API (GET /api/v1/compliance/zk-pointer/verify). If the backend is unreachable or the combination of bank ID and commitment is not found (HTTP 404), the state is shown as **INVALID**.

Immutable Audit Log table:

Read-only compliance audit log sourced from the compliance backend.

Filters:

Filter	Description
Severity	Filter by severity keyword (e.g. CRITICAL , HIGH , MEDIUM , LOW , INFO).
Category	Filter by event category string.

Click **Filter** to apply. Click **Clear** to reset all filters and reload the unfiltered log. Filters are applied server-side.

Table columns:

Column	Description
Timestamp	ISO 8601 timestamp localised to the browser timezone.
Actor	Identity of the actor who triggered the event (truncated with copy button).
Action	Event type name.
Category	Compliance category of the event.
Severity	CRITICAL / HIGH (red), MEDIUM (amber), LOW / INFO (grey).
Outcome	SUCCESS (green) or FAILURE (red).

Live data required. Log entries are fetched from the audit log API (GET /api/v1/compliance/audit/logs). In an offline environment the table shows “No audit logs found.”

5.5 Stability Insights

Route: /stability · **Sidebar label:** Stability Insights

A summary dashboard of the AMM network’s current risk posture. This screen is the Scenario B equivalent of a stability controls page; it is intentionally read-only and does not expose any intervention buttons.

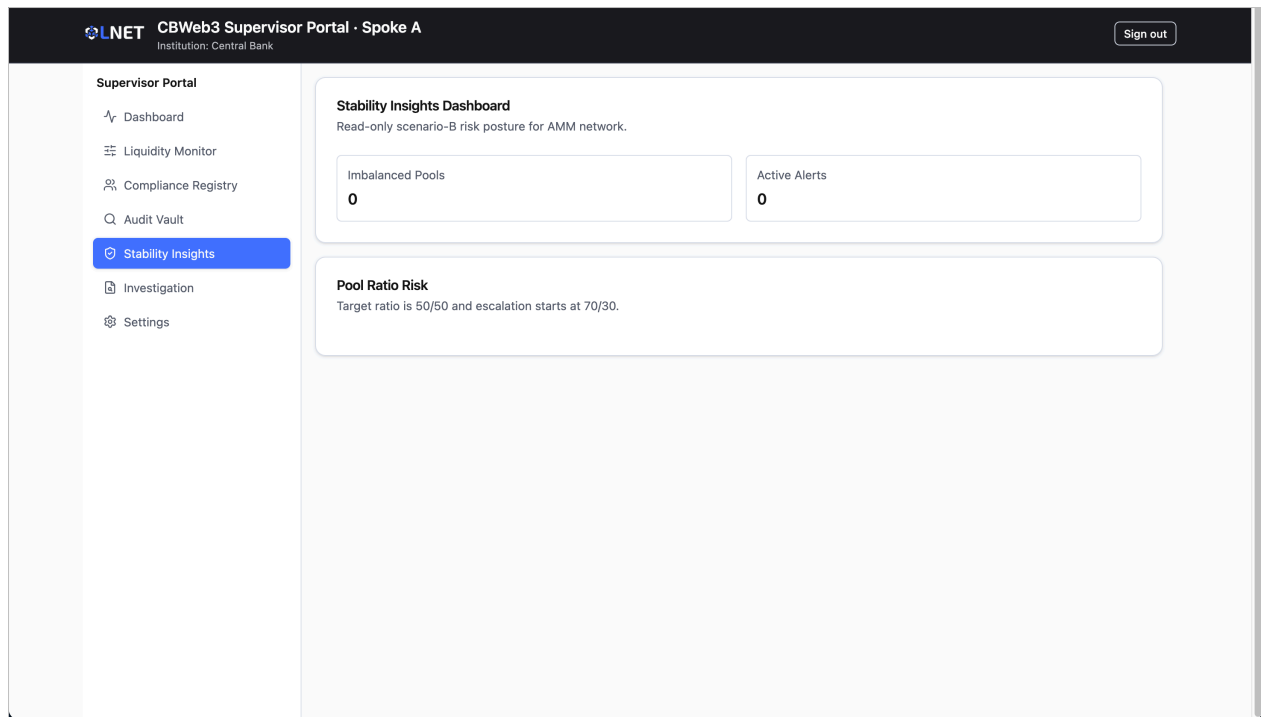


Figure 34: Stability Insights

Scenario B specific. This page focuses entirely on AMM pool risk. It does not display HTLC expiry or escrow risk indicators, which are Scenario A concerns.

Summary counters:

Counter	Description
Imbalanced Pools	Count of AMM pools currently outside the 70/30 ratio threshold.
Active Alerts	Count of open imbalance or circuit-breaker alerts.

Pool Ratio Risk section:

For each active pool, a progress bar visualises how far the dominant reserve has drifted from the 50/50 target. The bar represents `ratioA` as a percentage (0–100).

- **Target:** 50/50.
- **Escalation threshold:** 70/30. Pools that cross this boundary are flagged IMBALANCED on the Liquidity Monitor page and generate a HIGH alert.

Each pool row shows the pair label and the current ratio alongside the bar.

Live data required. Uses the same stability store and AMM pool APIs as the Liquidity Monitor. In an offline environment all pools are absent and both counters read zero.

5.6 Investigation

Route: `/investigation` · **Sidebar label:** Investigation

The Investigation screen supports the AML/CFT disclosure workflow. It enables authorised supervisors to open a multi-party disclosure request against a specific shielded transaction reference and

to co-sign requests opened by other supervisors.

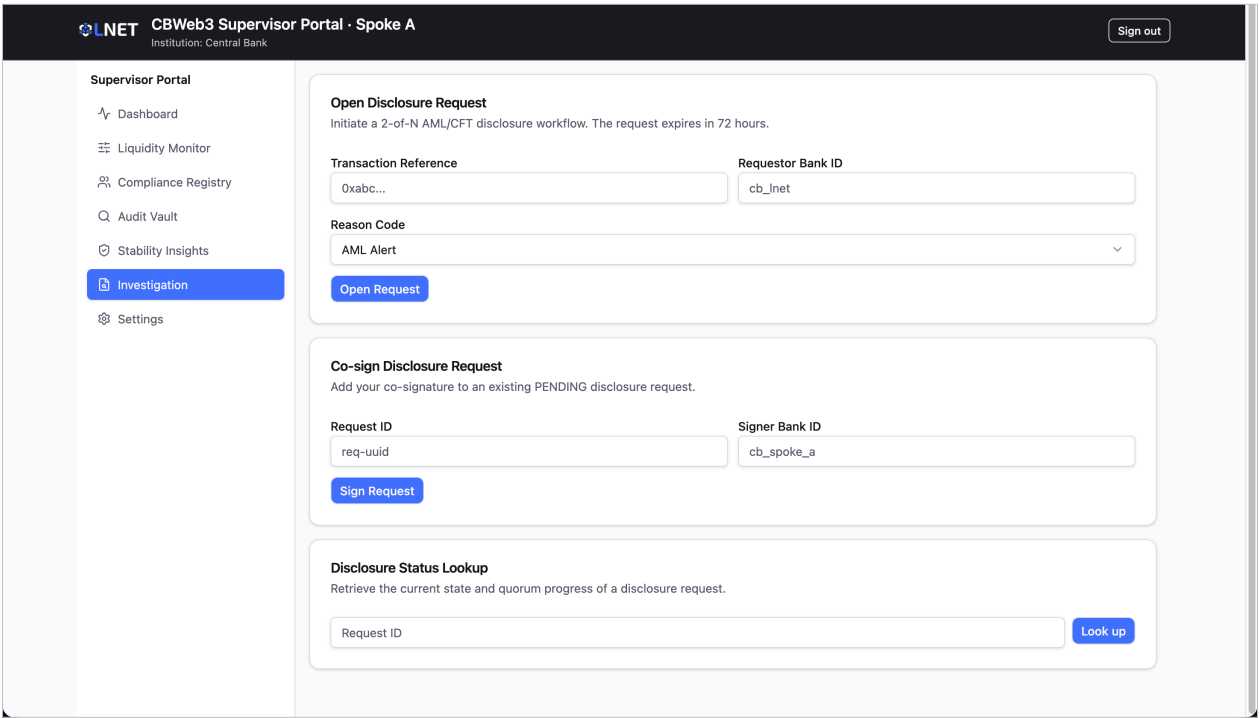


Figure 35: Investigation

Active write operations. Unlike the other screens, this page makes write API calls when opening or signing a disclosure request. Operations require a valid authenticated session and appropriate role permissions on the backend.

Workflow overview:

Disclosure requests use a **2-of-N quorum** model: a minimum number of supervisors must co-sign before a request reaches `QUORUM_REACHED` status. Requests expire 72 hours after creation.

Section 1 — Open Disclosure Request:

Opens a new disclosure request for a specific transaction.

Field	Required	Description
Transaction Reference	Yes	The on-chain transaction reference or hash to be investigated (e.g. <code>0xabc...</code>).
Requestor Bank ID	Yes	Identifier of the requesting institution (e.g. <code>cb_lnet</code>).
Reason Code	Yes	Regulatory reason for the request (see table below).

Reason codes:

Code	Label
<code>AML_ALERT</code>	AML Alert
<code>CFT_INVESTIGATION</code>	CFT Investigation
<code>COURT_ORDER</code>	Court Order
<code>REGULATORY_EXAM</code>	Regulatory Exam

Code	Label
------	-------

On success, the new disclosure request card is displayed inline. Record the **Request ID** — it is required for co-signing and status lookups.

Section 2 — Co-sign Disclosure Request:

Adds a co-signature to an existing PENDING request.

Field	Required	Description
Request ID	Yes	UUID of the disclosure request to sign.
Signer Bank ID	Yes	Identifier of the signing institution (e.g. <code>cb_spoke_a</code>).

Once the quorum threshold is met, the request transitions to QUORUM_REACHED.

Section 3 — Disclosure Status Lookup:

Retrieves the current state and quorum progress of any disclosure request by its Request ID.

Field	Description
Request ID	Input the UUID and click Look up .

The result card shows:

Field	Description
Request ID	Unique identifier.
State	PENDING, QUORUM_REACHED, or EXPIRED (see Status reference).
Target Transaction	Transaction reference under investigation.
Reason	Reason code applied.
Quorum	signatures received / signatures required.
Expires	Expiry timestamp.

Live data required. All three sections call the oversight API (`/api/v2/oversight/...`). Backend availability and correct role permissions are required for operations to succeed.

5.7 Settings

Route: `/settings` · **Sidebar label:** Settings

The Settings screen holds session security and operational preferences. All options are toggles that apply to the current browser session only; they are not persisted to the backend.

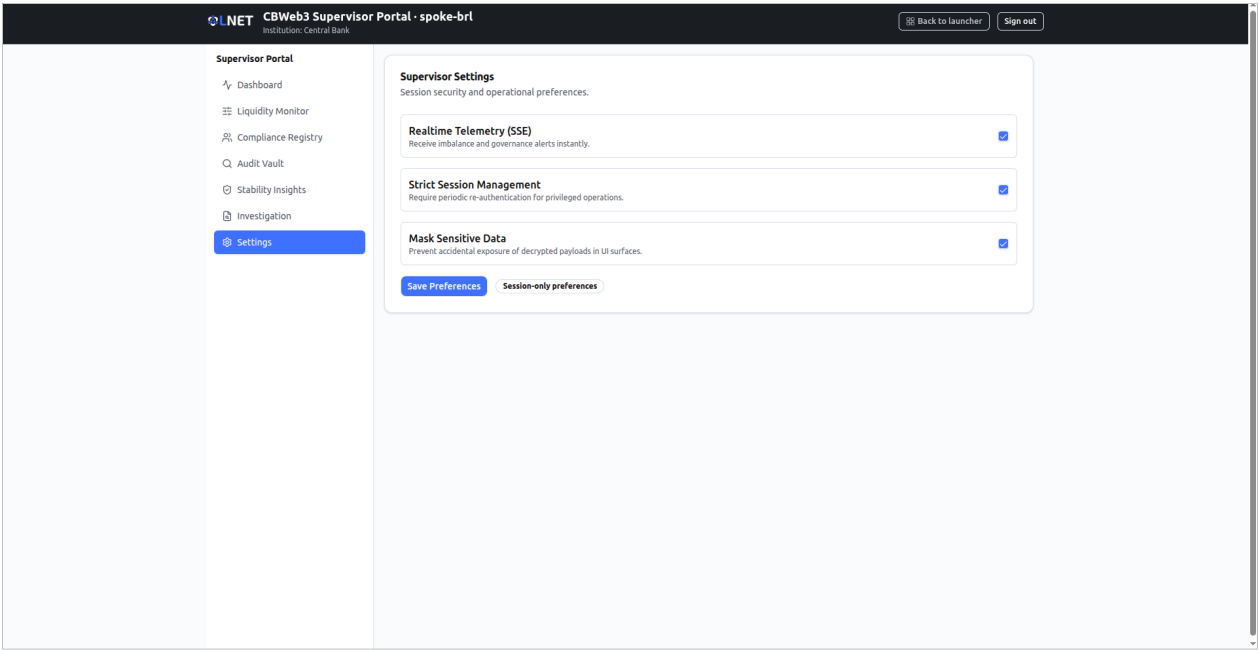


Figure 36: Settings

Preferences:

Preference	Default	Description
Realtime Telemetry (SSE)	On	Receive imbalance and governance alerts instantly over server-sent events.
Strict Session Management	On	Require periodic re-authentication for privileged operations.
Mask Sensitive Data	On	Prevent accidental exposure of decrypted payloads in UI surfaces.

Click **Save Preferences** to apply the current selections. A confirmation toast is shown and a **Session-only preferences** badge indicates that the settings are not stored beyond the active session.

Session-only. Preferences reset when the session ends. Re-apply them at the start of each session if required.

6. Typical workflows

6.1 Morning liquidity check

1. Sign in with institutional credentials.
2. Review the **Dashboard** summary cards — note any non-zero Imbalanced Pools counter or IMBALANCED rows in the Liquidity Health table.
3. Navigate to **Liquidity Monitor** for pool-level detail. Check Reserve A, Reserve B, and the ratio for each active pair.

4. Review Active Alerts. A **CRITICAL** alert means the circuit breaker is currently **PAUSED**; escalate to the governance team immediately.
5. Navigate to **Stability Insights** to confirm the pool ratio progress bars are within acceptable range.

6.2 Investigating a suspicious shielded transfer

1. Obtain the transaction reference (on-chain hash or reference ID) and the issuing bank's Bank ID from the referring analyst.
2. Navigate to **Audit Vault**. Use the ZK Pointer Verification panel to confirm the commitment hash is **VALID** and has not **EXPIRED**. Record the Pointer ID.
3. If further investigation is warranted, navigate to **Investigation**.
4. In "Open Disclosure Request", enter the transaction reference, your institution's Requestor Bank ID, and the appropriate reason code. Click **Open Request**.
5. Record the returned Request ID. Share it with co-signing supervisors.
6. Each co-signing supervisor navigates to Investigation, enters the Request ID and their Signer Bank ID, and clicks **Sign Request**.
7. Use "Disclosure Status Lookup" to monitor quorum progress. Once **QUORUM_REACHED** status is confirmed, proceed with the investigation according to your institution's regulatory procedures.

6.3 Verifying participant credential status

1. Navigate to **Compliance Registry**.
2. Enter the institution name, EVM address, or jurisdiction code in the search box.
3. Confirm the Credential column shows **ACTIVE**. A **REVOKED** status means the institution is not cleared to transact.

6.4 Reviewing audit events for a specific actor

1. Navigate to **Audit Vault**.
2. In the filter row, enter a severity or category filter and click **Filter**.
3. Scroll the table to locate entries for the actor of interest. The Actor column supports copy-to-clipboard for precise matching.

7. Status reference

AMM pool status

Status	Badge colour	Meaning
HEALTHY	Green	Pool reserves are within the 70/30 threshold.
IMBALANCED	Red	Pool reserves have exceeded the 70/30 threshold and may require rebalancing.

Alert severity

Severity	Badge colour	Typical trigger
CRITICAL	Red	Governance circuit breaker is PAUSED.
HIGH	Amber	A specific AMM pool is imbalanced.
MEDIUM	Amber	Intermediate severity (reserved for future monitor rules).
LOW	Grey	Informational threshold breach.
INFO	Grey	Background informational event.

Participant credential status

Status	Badge colour	Meaning
ACTIVE	Green	Institution is cleared to transact in the hub.
PENDING	Red	Onboarding in progress; not yet cleared. Any non-ACTIVE status renders red.
REVOKED	Red	Credential has been revoked; institution is blocked.

Audit log outcome

Outcome	Badge colour	Meaning
SUCCESS	Green	The recorded operation completed without error.
FAILURE	Red	The recorded operation failed or was rejected.

Audit log severity

Severity	Badge colour	Meaning
CRITICAL	Red	Highest severity compliance event.
HIGH	Red	High-priority compliance event.
MEDIUM	Amber	Medium-priority event requiring review.
LOW	Grey	Low-priority informational event.
INFO	Grey	Routine informational record.

Disclosure request state

State	Badge	Meaning
PENDING	Grey	Request is open and collecting co-signatures.
QUORUM_REACHED	Default (blue)	The required number of co-signatures has been received; investigation may proceed.

State	Badge	Meaning
EXPIRED	Red	The 72-hour window closed before quorum was reached; a new request must be opened if still needed.

ZK Pointer verification state

State	Badge colour	Meaning
VALID	Green	The commitment hash was found and is within its validity window.
EXPIRED	Amber	The pointer was found but the validity window has passed.
INVALID	Red	No matching pointer was found for the given bank ID and commitment hash.

8. Troubleshooting

The Dashboard shows dashes or zero counts

The network overview API is unreachable. Verify that the backend services are running and that the portal is configured with the correct API base URL. This is expected in fully offline demo environments.

Pool tables are empty on Liquidity Monitor or Stability Insights

The AMM pool status API (`/api/v2/amm/pool/{pair}/status`) is not returning data. Check that the backend FX / AMM service is healthy. Also confirm that at least one pool has been funded (EMPTY pools are intentionally excluded from the display).

Sign-in fails with “Unable to login”

Verify that Keycloak is running and reachable from the browser. Confirm your username and password are correct and have not been rotated. If using a local development stack, check that the auth service container is up.

ZK Pointer returns INVALID for a known commitment

Either the bank ID does not match the issuing institution exactly, or the commitment hash has not been indexed yet. Allow a few seconds for the ZK Pointer service to index a newly created commitment and retry. If the hash is confirmed correct but the result remains INVALID, the transfer may not have been registered via the ZK Pointer API.

Disclosure request creation fails

Confirm that your session has the required RBAC role for disclosure requests. Also confirm that the transaction reference is a valid, existing reference known to the compliance service. Check the backend oversight service logs for a detailed error.

Co-sign returns an error

The most common causes are: the request has already expired (**EXPIRED** state), the request ID is incorrect, or your institution's signer ID is not on the eligible signers list for this hub. Use "Disclosure Status Lookup" to check the current state before retrying.

Audit log shows "No audit logs found"

The compliance audit service is unreachable or has no records matching the current filters. Click **Clear** to remove all filters and reload the unfiltered log. If the table remains empty, check backend compliance service health.

Preferences reset between sessions

This is expected. Settings are session-only and are not persisted to the backend. Re-apply your preferred settings at the start of each session.

When the portal is connected to a fully operational stack, this manual will be updated to include screenshots, precise API endpoint versions, and confirmation of which features are available to which RBAC roles.

NOC Portal — User Manual

(Scenario B)

Audience: Network operations / SRE engineers. **Scenario:** B — International Hub (hub-and-spoke, AMM-based FX settlement). **Data source:** The portal connects to a live NOC backend service. Component health, alerts, logs, and topology data reflect actual spoke and hub state when the stack is running. Every screen refreshes by HTTP polling — there is no telemetry push stream.

Read first — current status

The NOC Portal is wired to a real backend API (Keycloak auth + NOC backend service). Component health, alerts, and logs reflect actual spoke and hub state when the stack is running. There is **no WebSocket push stream** — the portal runs in HTTP polling mode on every screen, at the interval configured in [Settings](#). Authenticated sessions are stored as JWT tokens in `localStorage` and renewed automatically (see [Access and Login](#)).

Pool Stability requires the NOC backend to be configured with a reachable api-gateway (`spec.noc.ammGatewayURL`) and at least one active AMM pair. The page states which of the two is missing, so an empty table is never ambiguous.

1. Overview

The NOC (Network Operations Center) Portal is the monitoring surface for engineers responsible for the health of Scenario B's hub-and-spoke infrastructure. Scenario B uses a **Regional Hub** network connected to multiple **spoke networks** (one per central bank) via a Cacti-based relay. The AMM (AutomatedMarketMaker) handles FX settlement between spoke currencies.

The NOC Portal gives operations engineers a unified view of:

- Real-time health status for Hyperledger Besu nodes, Paladin privacy nodes, payment orchestrators, and CACTI relay containers across all spokes and the Hub.
- An active alert feed with acknowledge/dismiss workflow.
- Container log inspection for any registered component.
- AMM liquidity pool reserve ratios and 70/30 breach alerts.
- An interactive topology diagram showing the full hub-and-spoke layout.
- An immutable audit trail of operator actions.

Actor	Portal	Role
Network Operations Engineer / SRE	NOC Portal	Infrastructure health, alert triage, relay monitoring, pool stability, log review, topology

The portal is **read-only with two exceptions**: operators can acknowledge and dismiss alerts. No destructive infrastructure controls are exposed here.

How Scenario B differs from Scenario A

Dimension	Scenario A	Scenario B
Architecture	Peer-to-peer corridors (HTLC)	Hub-and-spoke (Regional Hub + spoke networks)
4th Dashboard KPI	Total Spokes	Pool Breaches (70/30)
Pool Stability page	Not present	Present — monitors AMM 70/30 reserve ratio
Topology node kinds	BESU, PALADIN, CACTI	BESU, PALADIN, CACTI, HUB
Log Viewer page	Present	Present
Privacy tokens	ZetoToken / NotoToken	ZetoToken / NotoToken via Paladin

2. Access and Login

URL: The NOC Portal URL provided by your system administrator (default local port: 5910).

Authentication: Keycloak OIDC. The portal authenticates against the `cbweb3` realm with client ID `noc-portal`. Only accounts carrying the `SYS_ADMIN` role are accepted.

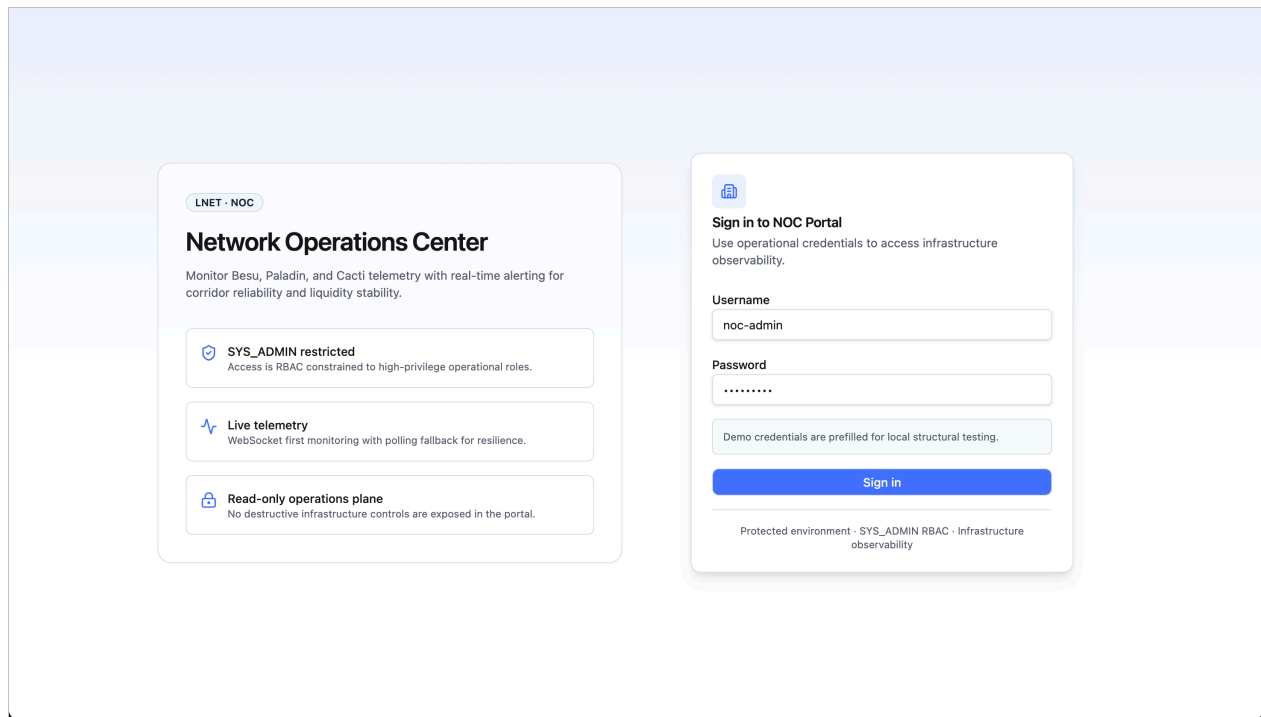


Figure 37: Login Page

Login steps

1. Open the NOC Portal URL in your browser.
2. Enter your `SYS_ADMIN` username and password.
3. Click **Sign in**.
4. On success, you are redirected to the Dashboard.

Login page layout

The login page has two panels on desktop:

- **Left panel** — feature summary (SYS_ADMIN restriction, continuous monitoring, read-only operations plane).
- **Right panel** — sign-in form (username, password, submit button).

Field	Notes
Username	Your <code>SYS_ADMIN</code> Keycloak username.
Password	Your <code>SYS_ADMIN</code> Keycloak password.

Standard bank-portal or governance-portal credentials cannot be used here. If login fails immediately, verify the account has the `SYS_ADMIN` role in Keycloak.

Demo credentials (local environments only): username `noc.admin`, password `NOCAdmin2026!`. These must not be used in any non-local deployment.

Session persistence: On successful login the portal stores `noc_access_token` and `noc_refresh_token` in `localStorage`. The access token is short-lived (5 minutes by realm default), so the portal renews it automatically — shortly before expiry and again if a request is still rejected — using the refresh token. A portal left open therefore keeps working without re-authentication up to the realm’s SSO session limit (10 hours by default), and a page reload within that window restores the session instead of returning to the login page.

When renewal is no longer possible (refresh token expired, session revoked), the portal drops the session and returns to the login page with “*Session expired. Sign in again.*”. **Sign out** clears both tokens and returns to the login page; use **Back to launcher** to leave the portal for the entity launcher.

3. Navigation

The portal uses a persistent left sidebar. All routes are protected — the login page is the only publicly accessible screen.

Sidebar label	Route	Description
Dashboard	/	Platform-wide health snapshot and alert feed.
Infrastructure	/infrastructure	Per-component health table (nodes, orchestrators).
Relays	/relays	CACTI relay container health per spoke.
Pool Stability	/pool-stability	AMM pool reserve ratios and 70/30 breach status.
Topology	/topology	Interactive hub-and-spoke network diagram.
Audit	/audit	Operator action history and resolved alerts.
Settings	/settings	Polling interval and notification preferences.

The **Log Viewer** (/logs/:componentId) has no sidebar entry. It is reached by clicking the **View Logs** button on the Infrastructure or Relays pages.

Unknown routes redirect to / (Dashboard).

Status bar

A thin bar above the header carries three badges, visible on every screen:

Badge	Meaning
POLLING <n>s	The active refresh interval, as configured in Settings .
LIVE DATA / STALE DATA	STALE DATA appears when no backend request has succeeded for three polling cycles (minimum 20 seconds) — the screen is showing the last data received. A banner with the same meaning appears above the page content.
CRITICAL ALERTS: <n>	Count of CRITICAL alerts currently loaded, turning red above zero.

STALE DATA points at the portal’s link to the NOC backend, not at the health of the monitored components — a spoke can be perfectly healthy while the browser cannot reach the backend, and vice versa.

4. Screens

4.1 Dashboard (/)

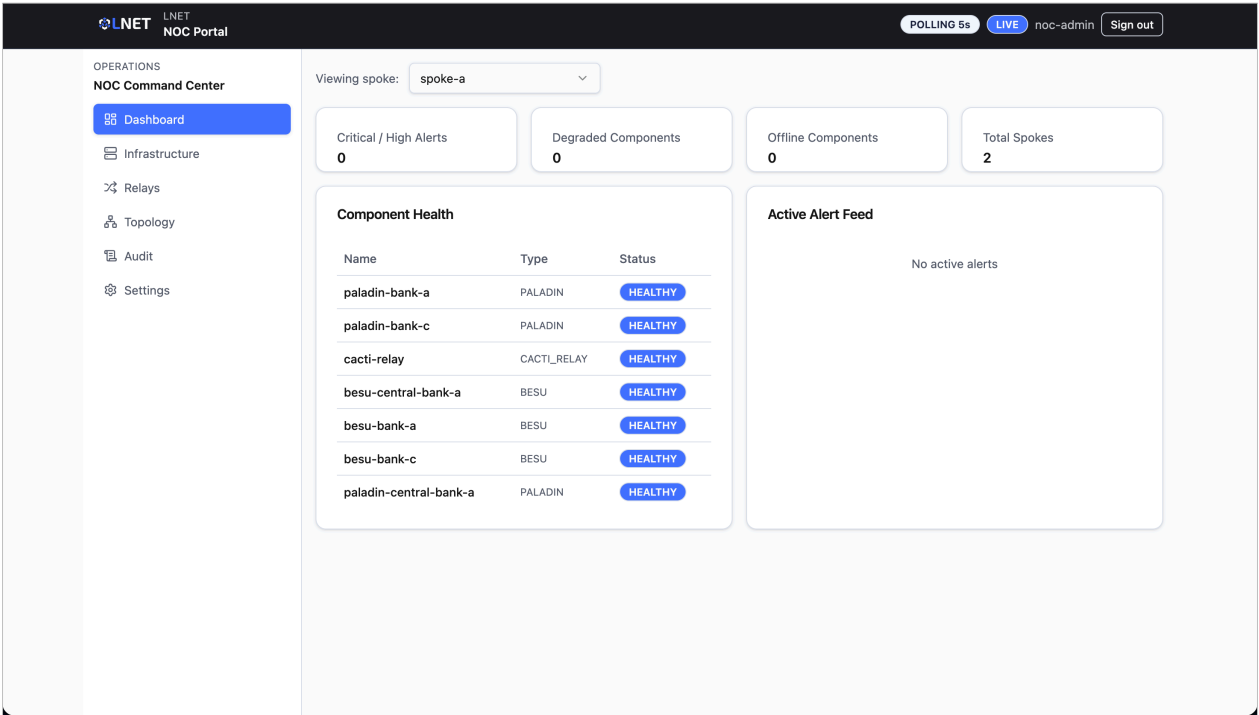


Figure 38: Dashboard

The Dashboard is the primary monitoring view. It aggregates platform-wide health data across all spokes and the Regional Hub (or a single selected spoke) and displays the active alert feed.

Spoke Selector

A dropdown at the top of the page. By default, the first available spoke is selected automatically. Select **a specific spoke** to filter both the Component Health table and the Alert Feed to that spoke only.

Select **All Spokes** for a platform-wide view: the Alert Feed then lists alerts from every registered spoke. Component health is only available per spoke, so in this mode the Component Health table shows “*Select a spoke to see component health*” and the **Degraded Components / Offline Components** cards show – instead of a count.

Summary Cards

Four cards provide an instant platform status snapshot:

Card	Colour when non-zero	Description
Critical / High Alerts	Red	Active alerts with severity HIGH or CRITICAL. Requires immediate attention.
Degraded Components	Orange	Components not in HEALTHY state (includes DEGRADED, OFFLINE, UNKNOWN).
Offline Components	Red	Components in OFFLINE state. Requires immediate attention.

Card	Colour when non-zero	Description
Pool Breaches (70/30)	Red	AMM pools whose reserve ratio has exceeded the 70/30 threshold.

Component Health Table

Shows the first 8 components for the selected scope. Columns: **Name**, **Type**, **Status** badge. For the full component list, navigate to [Infrastructure](#).

Active Alert Feed

Lists up to 8 active alerts. Each alert row shows:

Element	Description
Title	Short description of the alert condition.
Severity badge	INFO, WARNING, HIGH, or CRITICAL.
ACK badge	Shown when the alert has been acknowledged by any operator.
Dismiss button	X icon on the right — click to dismiss directly from the feed.

When **Critical Alerts Only** is enabled in [Settings](#), the feed lists only HIGH and CRITICAL alerts and a CRITICAL ONLY badge appears next to the feed title. The Summary Cards are not affected by that setting.

Click any alert row (not the dismiss button) to open the **Alert Detail Modal**.

Alert Detail Modal fields:

Field	Description
Title	Alert title.
Severity	Severity badge.
State	ACTIVE or RESOLVED.
Affected component	The component that raised the alert with its health status.
Created	Timestamp of alert creation (shown in local time).
Root cause sig	System-generated root cause pattern identifier.
Acknowledge button	Marks the alert as seen; keeps it in the feed.
Dismiss button	Removes the alert from the active feed.

Polling

The Dashboard refreshes automatically at the interval configured in [Settings](#) (default: 15 seconds, minimum: 5 seconds). The WebSocket stream is not yet active; the portal always uses HTTP polling.

4.2 Infrastructure (/infrastructure)

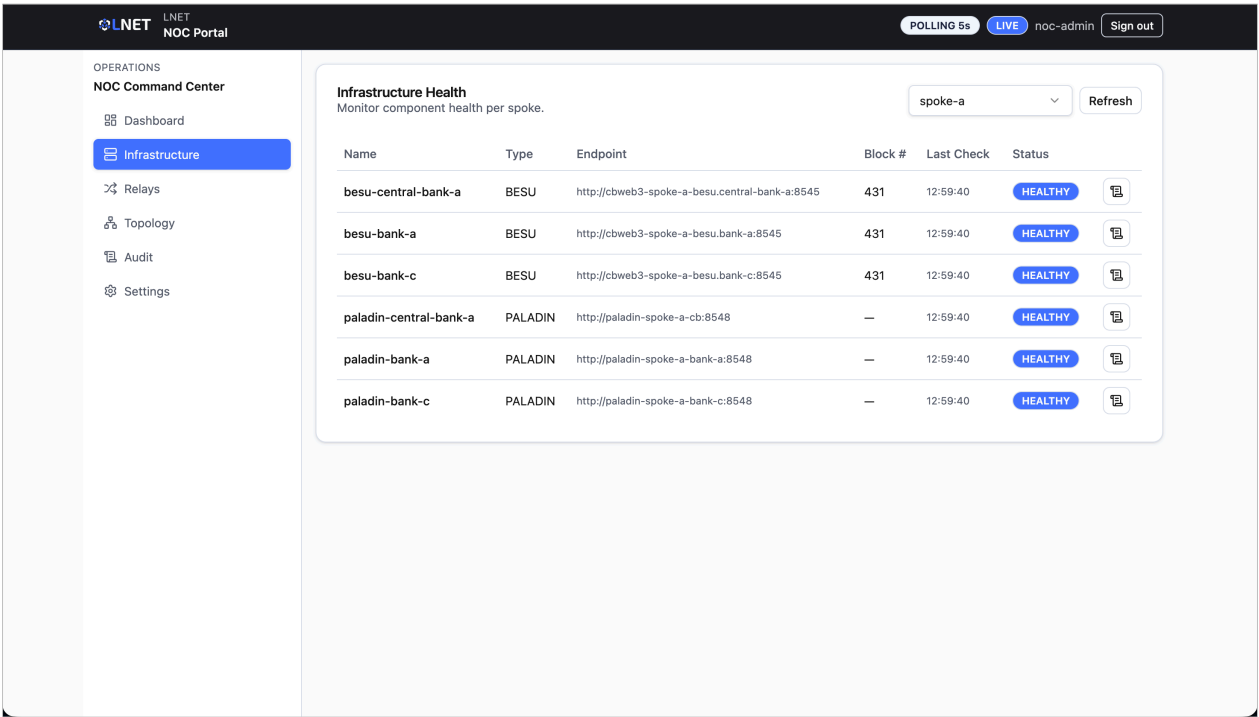


Figure 39: Infrastructure

Lists the health status of all non-relay infrastructure components for the selected spoke or the Regional Hub: Hyperledger Besu blockchain nodes, Paladin privacy nodes, and payment orchestrators. CACTI relay containers are excluded here — use [Relays](#) for those.

Controls

Control	Description
Spoke Selector	Select a spoke or the Regional Hub. The table is empty until a spoke is selected.
Refresh button	Force an immediate health poll; disabled while loading.

Component Health Table

Column	Description
Name	The component’s display name.
Type	Component category: BESU, PALADIN, or PAYMENT_ORCHESTRATOR.
Endpoint	The component’s API or RPC endpoint (truncated).
Block #	Latest block number — populated for BESU nodes only. – for other types.
Last Check	Timestamp of the most recent health poll (shown as local time).
Status	Health status badge. See Status Reference .

Column	Description
View Logs	Icon button — navigates to the Log Viewer for this component.

The table auto-refreshes at the configured polling interval whenever a spoke is selected.

4.3 Relay Status (/relays)

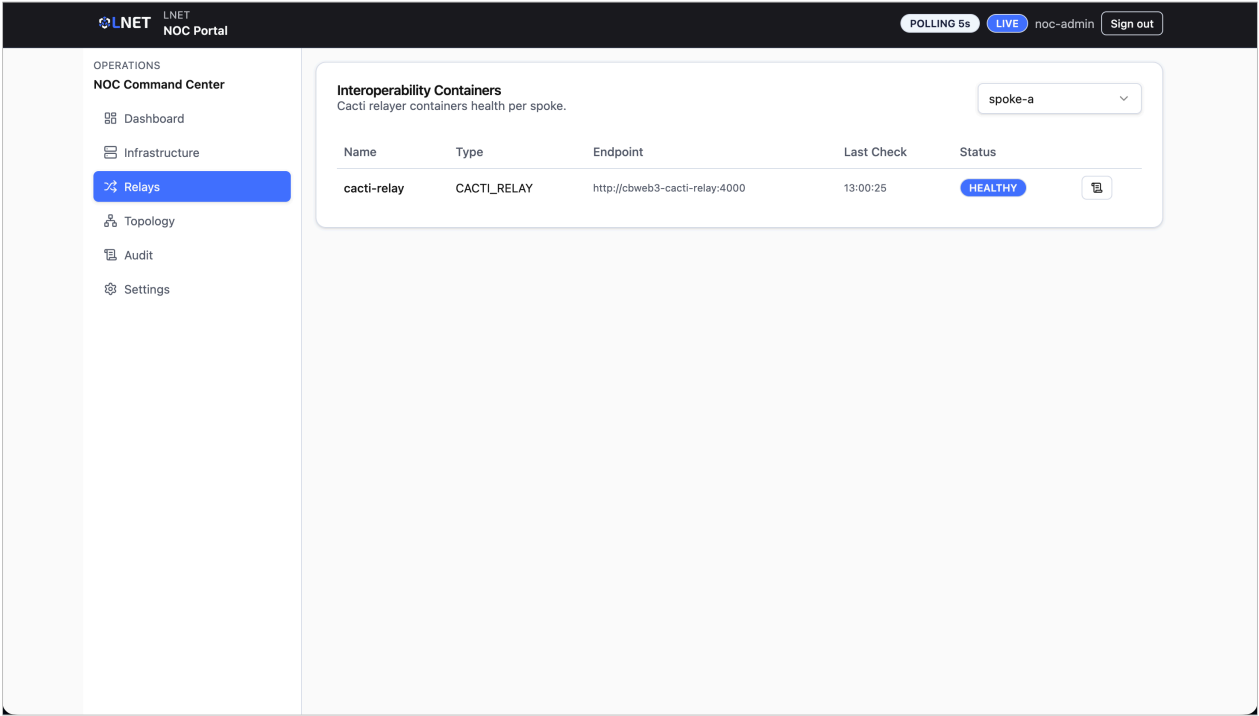


Figure 40: Relay Status

Shows the health of **CACTI interoperability relay containers** — the components responsible for bridging events between spoke networks and the Regional Hub. When a cross-spoke settlement appears stuck, this page is the first place to check. In the portal the page itself is titled **Interoperability Containers**; the sidebar entry is **Relays**.

Controls

Control	Description
Spoke Selector	You must select a specific spoke. The table is empty until a spoke is chosen.

Relay Components Table

Displays only CACTI_RELAY type components for the selected spoke.

Column	Description
Name	Relay container name.

Column	Description
Type	Always CACTI_RELAY.
Endpoint	The relay container’s API endpoint (truncated).
Last Check	Timestamp of the last health poll (local time).
Status	Health status badge. See Status Reference .
View Logs	Icon button — navigates to the Log Viewer for this relay.

If no relay components appear after selecting a spoke, verify that the CACTI relay containers for that spoke are registered in the NOC backend and that the backend service can reach them.

4.4 Pool Stability (/pool-stability)

This page is specific to Scenario B. It is not present in the Scenario A NOC Portal.

Displays the reserve ratios of each AMM (AutomatedMarketMaker) liquidity pool and flags pools that have breached the 70/30 threshold. A **Refresh** button triggers a manual data reload.

The NOC backend does not hold pool data of its own: it queries the api-gateway for each AMM pair it is configured with. The page therefore distinguishes three empty outcomes, and it is worth reading which one you are looking at:

What you see	What it means
<i>“No AMM pools are configured for this deployment.”</i>	The backend read the gateway successfully and there is nothing to show — no sovereign pair has been opened yet, or none is configured.
An amber warning listing pairs that could not be read	The gateway is unreachable or the pair does not exist there. This is a configuration or connectivity problem, not an empty AMM — the reason reported for each pair comes straight from the gateway.
A red banner: <i>“Could not reach the NOC backend.”</i>	The portal could not reach the NOC backend at all; nothing on this page is current.

Understanding the 70/30 rule

The AMM maintains two-sided reserve pools for each currency pair (e.g. BRL-tCeBM/ARS-tCeBM). The platform enforces a maximum reserve imbalance of 70% / 30% — if either side of the pool drops below 30% of total reserves, the circuit breaker may halt swaps for that corridor to prevent excessive price impact on settlement transactions.

Column reference

Column	Description
Pool	Currency pair identifier.
Reserve A	Total units held for token A.
Reserve B	Total units held for token B.
Split	Visual progress bar showing ratioA% / ratioB% — the current reserve split.

Column	Description
Status	STABLE (within 70/30) or BREACHED (threshold exceeded).
Updated At	Timestamp of the last pool snapshot.

Responding to a BREACHED pool

1. Note which pool pair is flagged and whether it is a single corridor or multiple.
2. Cross-reference with [Relay Status](#) — if the relay carrying that corridor is also degraded, the imbalance may be caused by undelivered settlement events rather than organic trading pressure.
3. Escalate to the liquidity operations team with the pool name, current ratios, and the time the breach appeared.
4. Do not attempt to rebalance the pool directly from this portal; the portal is read-only.

4.5 Topology (/topology)

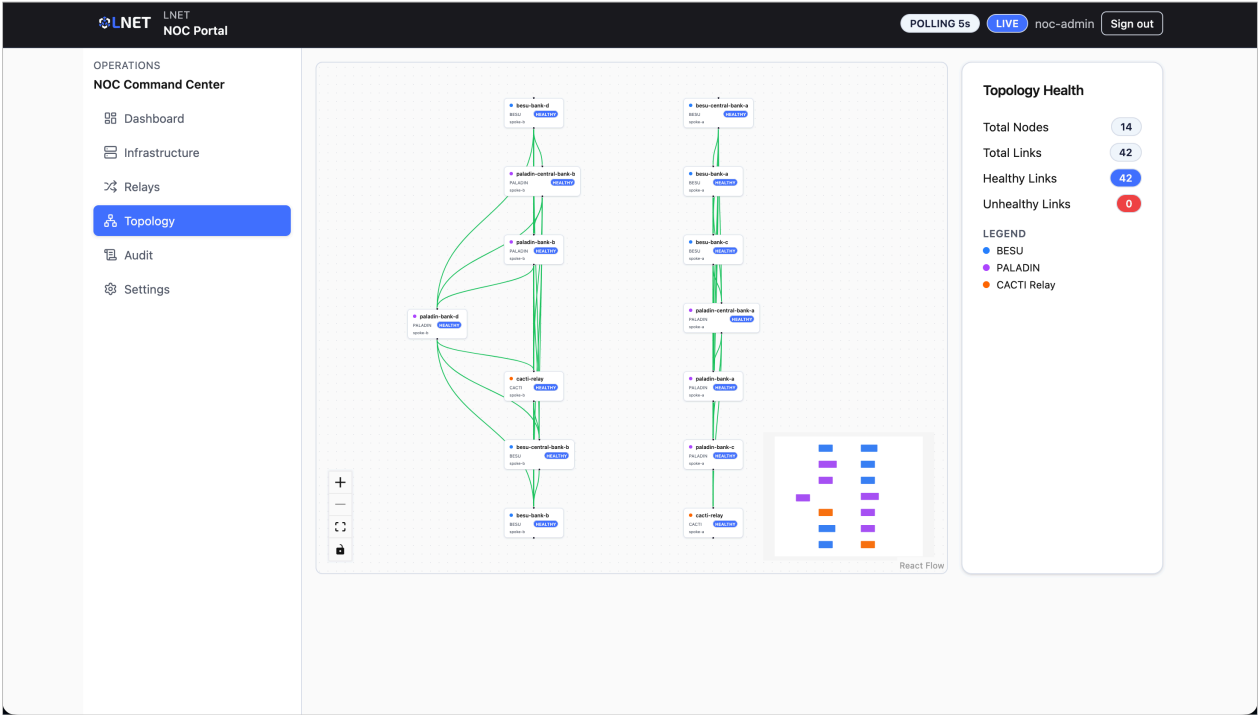


Figure 41: Topology

Renders an interactive network diagram of all nodes and their links across the hub-and-spoke topology. Useful for identifying where in the topology a fault is isolated and for verifying the Regional Hub’s connectivity to all spokes.

Diagram Elements

Element	Colour	Description
BESU node	Blue dot	A Hyperledger Besu blockchain node.

Element	Colour	Description
PALADIN node	Purple dot	A Paladin privacy node.
CACTI relay node	Orange dot	An interoperability relay container.
HUB node	Green dot	The Regional Hub network node.
Green edge	Green line	A healthy link between two nodes.
Red edge	Red line	An unhealthy link between two nodes.

Each node card shows: a colour-coded dot, the node name, its kind label, a health status badge, and the name of the spoke or hub it belongs to.

Nodes are grouped by spoke in columns. The Regional Hub nodes appear in their own column. The layout is generated automatically based on spoke membership.

Diagram Controls

Control	Action
Scroll wheel	Zoom in / out.
Click and drag (on the canvas)	Pan the diagram.
Mini-map (bottom-right corner)	Navigate quickly across a large topology.
Controls panel (bottom-left)	Zoom in, zoom out, fit view, lock layout buttons provided by ReactFlow.

Topology Health Summary Card

Located to the right of the diagram:

Metric	Description
Total Nodes	Number of nodes registered in the topology.
Total Links	Number of directed edges between nodes.
Healthy Links	Count of edges marked as healthy (green).
Unhealthy Links	Count of edges marked as unhealthy (red).

A colour legend below the metrics identifies each node type, including the HUB kind which is specific to Scenario B.



4.6 Log Viewer (/logs/:componentId)

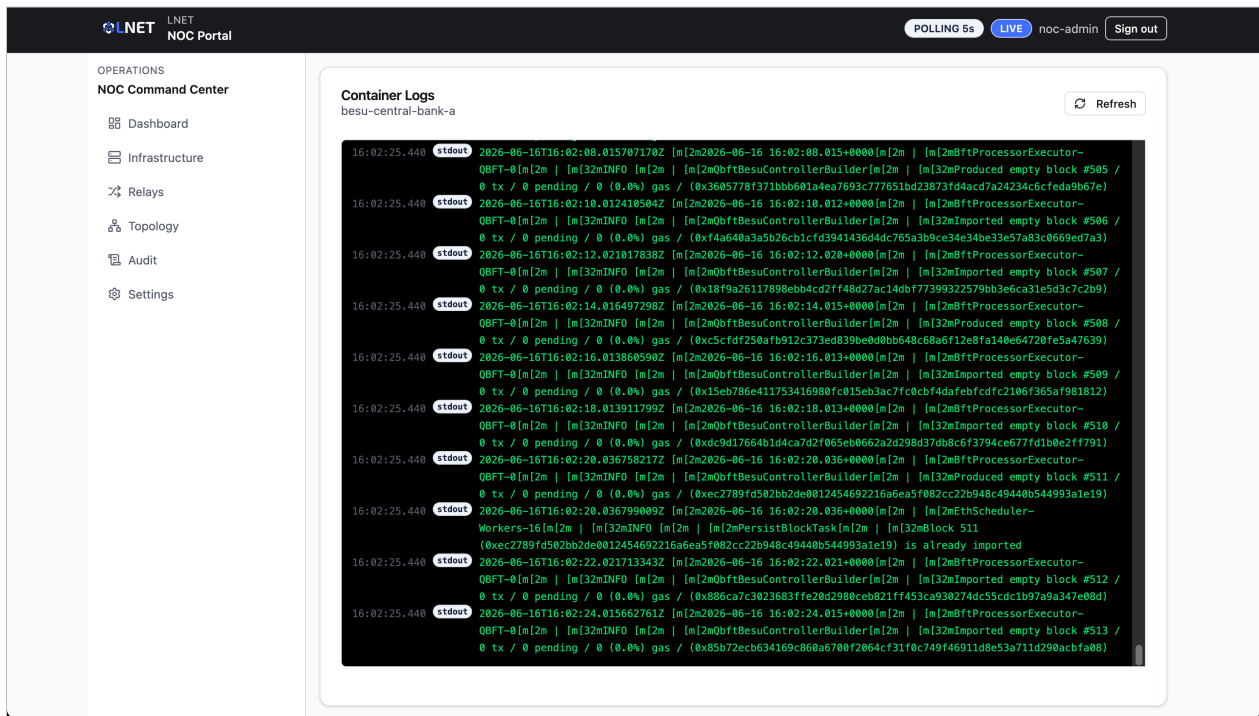


Figure 42: Log Viewer

Displays container logs for a specific infrastructure or relay component. There is no sidebar entry for this screen — navigate here by clicking the **View Logs** button from the [Infrastructure](#) or [Relays](#) pages.

The component name is shown in the page subtitle. The `componentId` in the URL is the internal component ID; the human-readable name is passed as a query parameter (`?name=...`).

Controls

Control	Description
Refresh button	Force an immediate log fetch. Spinner is shown while loading.
SNAPSHOT <n>s OLD badge	Appears when the newest line collected is more than 60 seconds old — see <i>Snapshots, not a live tail</i> below.

Log Panel

Terminal-style viewer (black background, green text).

Column	Description
Timestamp	Time portion of the log line's <code>occurred_at</code> timestamp (HH:MM:SS.mmm). This is when the agent collected the batch, so a whole batch shares one timestamp; each line's own time appears inside the log text.

Column	Description
Stream badge	<code>stdout</code> (normal output) or <code>stderr</code> (error output). <code>stderr</code> lines appear in red.
Log line	Raw log text from the container.

- Displays the **last 500 lines**.
- Auto-scrolls to the bottom on initial load and on each refresh.
- Refreshes automatically every **15 seconds**. This cycle is fixed and is not affected by the Settings polling interval.

Snapshots, not a live tail

Logs are collected by the spoke’s NOC agent and pushed to the backend; the viewer reads what the backend stored. If the container stops, or its agent stops collecting, the screen keeps showing the last snapshot with no other visible change — the **SNAPSHOT <n>s OLD** badge is what tells you the lines are not current.

Log collection requires the component to be registered with its container name. A component registered without one always reports “*No logs available.*” even while healthy — for relay components this depends on `spec.relay.containerName` being set in the entity’s deployment manifest.

4.7 Audit (/audit)

The screenshot displays the LNET NOC Portal interface. The top navigation bar includes the LNET logo, 'NOC Portal', a 'POLLING 5s' indicator, a 'LIVE' status, the user 'noc-admin', and a 'Sign out' button. The left sidebar lists navigation options: Dashboard, Infrastructure, Relays, Topology, Audit (highlighted in blue), and Settings. The main content area features a search bar labeled 'Search actor, action, target, detail...'. Below the search bar, there are two sections: 'Audit Trail' and 'Resolved Alerts'. The 'Audit Trail' section, titled 'Immutable log of operator actions.', shows a table with columns: Timestamp, Actor, Action, Target, and Detail. It currently displays 'No audit records'. The 'Resolved Alerts' section, titled 'Alerts that have been resolved and are no longer active.', shows a table with columns: Resolved At, Title, Severity, and Root Cause Sig. It lists two resolved alerts: one from 16/06/2026, 13:05:10 with title '[CRITICAL] besu-bank-a OFFLINE' and severity 'CRITICAL', and another from 16/06/2026, 13:04:55 with title '[CRITICAL] besu-bank-c OFFLINE' and severity 'CRITICAL'. Both alerts have a root cause signature starting with 'comp:'. The 'Severity' column for both alerts contains a red 'CRITICAL' badge.

Figure 43: Audit

Provides an immutable record of all operator actions taken within the NOC Portal and a history of resolved alerts.

A **search box** at the top filters both tables simultaneously. The filter matches against actor, action, target ID, and detail text in the Audit Trail table; and against title, severity, and root cause

signature in the Resolved Alerts table.

Audit Trail Table

Every alert acknowledgement and dismissal performed by any NOC operator is recorded here.

Column	Description
Timestamp	When the action was performed (local time).
Actor	Username of the NOC operator who performed the action, taken from the preferred_username claim of the token used. Deployments whose token carries no username fall back to the account's internal ID, and a request with no readable identity is recorded as dev-user .
Action	The action type: Acknowledge (from ACKNOWLEDGE_ALERT) or Dismiss (from DISMISS_ALERT).
Target	The internal ID of the alert that was acted upon.
Detail	Any additional context recorded at the time of the action.

Resolved Alerts Table

A searchable history of alerts that have been resolved (automatically or via dismissal).

Column	Description
Resolved At	Timestamp when the alert transitioned to resolved (local time). - if not yet recorded.
Title	The alert title.
Severity	Severity badge at the time of resolution.
Root Cause Sig	System-generated identifier for the root cause pattern (monospaced, truncated).

4.8 Settings (/settings)

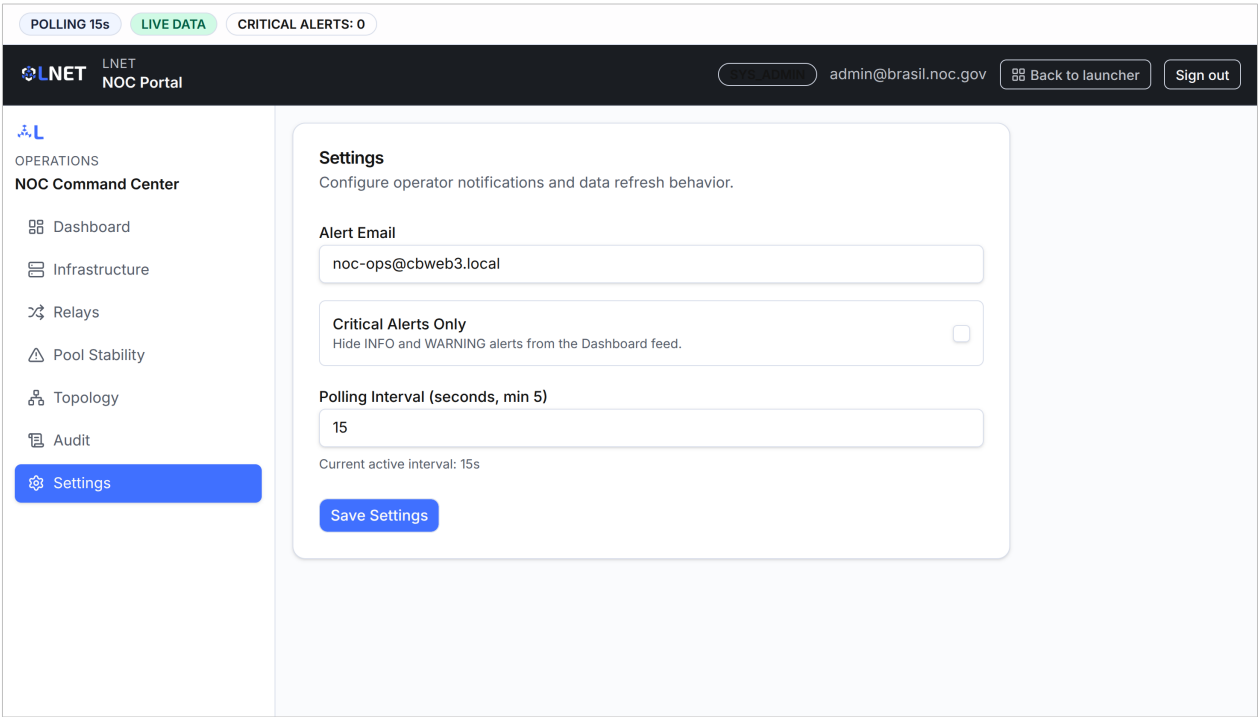


Figure 44: Settings

Configures operator-level monitoring preferences. Settings are persisted to `localStorage` (key: `noc-ui-settings`) and survive page reloads within the same browser. They reset when the browser storage is cleared.

Setting	Default	Description
Alert Email	<code>noc-ops@cbweb3.local</code>	Email address for alert notifications. Stored locally — not sent to the backend. Applied on Save Settings .
Critical Alerts Only	Off	When checked, hides <code>INFO</code> and <code>WARNING</code> severity alerts from the Dashboard’s Active Alert Feed. Only <code>HIGH</code> and <code>CRITICAL</code> alerts appear, and a <code>CRITICAL ONLY</code> badge is shown on the feed header. Applied immediately on toggle.
Polling Interval	15 seconds	How frequently the auto-refreshing pages reload data. Minimum: 5 seconds. Values below 5 are rejected. Applied on Save Settings .

Click **Save Settings** to apply the Alert Email and Polling Interval. A toast confirms success, and the active polling interval shown below the input updates immediately. If the interval is below 5 seconds or not a number, an error toast appears and **nothing is saved** — including the Alert

Email. The **Critical Alerts Only** checkbox does not require Save; it takes effect as soon as it is toggled.

The Polling Interval governs every auto-refreshing page in the portal — [Dashboard](#), [Infrastructure](#) and [Relay Status](#) — not the Dashboard alone. The Container Logs refresh cycle is fixed at 15 seconds and is not affected by this setting.

Note that the **Critical / High Alerts** counter in the Dashboard KPI row always counts **HIGH** and **CRITICAL** active alerts regardless of this setting.

5. Alert Management

NOC operators can take two actions on active alerts: **Acknowledge** and **Dismiss**. Both actions are logged in the [Audit](#) page.

Acknowledge vs. Dismiss

Action	Meaning	Effect
Acknowledge	An operator has seen the alert and is actively investigating.	Alert remains in the active feed. An ACK badge appears on the row. Action logged in Audit.
Dismiss	The alert is not actionable, is a false positive, or has been resolved externally.	Alert is removed from the active feed and moved to Resolved Alerts on the Audit page. Action logged in Audit.

How to Acknowledge

1. On the Dashboard, locate the alert in the **Active Alert Feed**.
2. Click the alert row to open the **Alert Detail Modal**.
3. Review the alert title, severity, affected component, and description.
4. Click **Acknowledge**.

The ACK badge appears on the alert row. The action appears in the Audit Trail.

How to Dismiss

Option A — from the alert feed directly:

1. Click the **X (dismiss) button** on the right side of the alert row.
2. The alert is removed immediately.

Option B — from the Alert Detail Modal:

1. Click the alert row to open the modal.
2. Click **Dismiss** in the modal.

In both cases the alert disappears from the feed and is logged in the Audit Trail. Dismissed alerts cannot be re-activated from the portal.

Dismissing does not silence an ongoing fault. If the component is still unhealthy, the next agent report raises a new alert for it — dismissal clears the row you acted on, not the condition behind it. The alert stops coming back once the component reports **HEALTHY**, which also resolves it

automatically. Use **Acknowledge** for a fault you are actively working on: it keeps the alert in the feed and marks it as taken.

6. Typical Workflows

6.1 Morning health check

1. Log in to the NOC Portal.
2. On the **Dashboard**, check the four summary cards. Any non-zero values in **Critical / High Alerts**, **Offline Components**, or **Pool Breaches** require immediate action.
3. If critical/high alerts are present, click each alert row to open the Alert Detail Modal and review the affected component.
4. Navigate to **Infrastructure** and select each spoke and the Regional Hub in turn. Confirm no unexpected **DEGRADED** or **OFFLINE** components.
5. Navigate to **Relays** and select each spoke. Confirm all CACTI relay containers show **HEALTHY**.
6. Open **Pool Stability** and confirm all pools are **STABLE**.
7. Navigate to **Topology** and verify that all edges are green (healthy links), including the Hub-to-spoke connections.

6.2 Investigating a stuck cross-spoke transfer

1. Navigate to **Relays** and select the spoke involved in the settlement.
2. Identify any relay in **DEGRADED** or **OFFLINE** state.
3. Click **View Logs** for the suspect relay to open the Log Viewer.
4. Review **stderr** lines (highlighted in red) for error messages — connection failures, timeout errors, or panics.
5. If the relay log shows no recent activity within the expected settlement window, the relay may be stalled — not the contract.
6. Open **Pool Stability** for the affected currency pair. A **BREACHED** pool may cause settlement to fail on price impact rather than the relay being the bottleneck.
7. Check the **Topology** panel for unhealthy links on the Hub-to-spoke corridor.
8. Escalate with: spoke name, relay component ID, log excerpt, pool status, and last-checked timestamp.

6.3 Acknowledging and tracking an active alert

1. On the **Dashboard**, click the alert row in the Active Alert Feed.
2. Read the Alert Detail Modal fully: title, severity, affected component, created time, and root cause signature.
3. If you are taking ownership, click **Acknowledge**. The ACK badge appears.
4. Investigate using the **Log Viewer** for the affected component.
5. Once resolved externally, return to the Dashboard and **Dismiss** the alert.
6. Verify the action appears in **Audit** > Audit Trail.

6.4 Checking container logs for a specific component

1. Navigate to **Infrastructure** (for nodes/orchestrators) or **Relays** (for CACTI containers).
2. Select the appropriate spoke from the Spoke Selector.
3. Locate the component in the table and click the **View Logs** icon button.
4. The Log Viewer opens showing the last 500 lines.
5. Look for **stderr** lines (red) — these typically contain error or warning messages.
6. Click **Refresh** to force an immediate fetch if the 15-second auto-refresh has not triggered yet.

6.5 Monitoring pool stability before a high-volume settlement window

1. Open **Pool Stability** and click **Refresh**.
2. Review the **Split** column for each pool that will be used in the settlement window.
3. Alert the liquidity operations team if any pool's ratio is approaching 70/30 (e.g. A ratio above 65% or below 35%) even if not yet flagged **BREACHED**.
4. Monitor the **Dashboard** "Pool Breaches (70/30)" card during the window for any breach that occurs mid-flight.

6.6 Adjusting the polling interval during an incident

1. Navigate to **Settings**.
2. Lower the **Polling Interval** to 5 seconds (the minimum) for faster alert and health updates during active incident investigation.
3. Click **Save Settings**.
4. Return the interval to the normal value (15–30 seconds) after the incident is resolved to reduce backend load.

7. Alert Reference

Severity Levels

Severity	Badge colour	When to act
INFO	Grey	Informational — no action required. Review periodically.
WARNING	Yellow/orange	Potential issue — monitor. Investigate if condition persists or worsens.
HIGH	Red	Significant problem — investigate as soon as possible.
CRITICAL	Red	Severe failure — requires immediate action.

Recommended Response by Severity

Severity	Recommended response
INFO	Acknowledge to mark as seen; dismiss if it is a known benign event.
WARNING	Acknowledge and investigate root cause. Check component logs. Dismiss once confirmed non-critical.
HIGH	Acknowledge immediately. Investigate component logs and relay status. Escalate if not resolved within SLA.
CRITICAL	Page on-call engineer. Acknowledge immediately. Isolate affected spoke if necessary. Escalate to lead.

8. Status Reference

Component Health Statuses

Status	Badge colour	Meaning
HEALTHY	Green (default)	Component is responding correctly to health checks.
DEGRADED	Yellow/orange (warning)	Component is responding but with errors or elevated latency. Investigate promptly.
OFFLINE	Red (destructive)	Component is not responding to health checks. Requires immediate attention.
UNKNOWN	Grey (secondary)	Health check has not been performed yet, or the last result could not be determined.

Relay-Specific Interpretation

Status	Relay meaning
HEALTHY	CACTI relay container is up and forwarding events between the spoke and the Regional Hub.
DEGRADED	Relay is responding but may be experiencing partial failures, high latency, or reconnection loops. Cross-spoke settlements may be slow.
OFFLINE	Relay container is not responding. Cross-spoke settlements will stall. Investigate relay logs and container state immediately.
UNKNOWN	Relay has not been polled yet, or the NOC backend cannot reach the container endpoint.

Pool Stability Statuses

Status	Meaning
STABLE	Pool reserves are within the 70/30 tolerance.
BREACHED	One side of the pool has dropped below 30% of total reserves; transfers on this corridor may fail on price impact.

Topology Link Colours

Colour	Meaning
Green edge	Link between nodes is healthy.
Red edge	Link between nodes is unhealthy.

9. Troubleshooting

Login fails

- Verify the username and password are correct for a `SYS_ADMIN` Keycloak account. Standard bank-portal credentials will not work.
- Confirm the Keycloak service is running (`docker compose ps` in the deploy directory — look for the `keycloak` container).
- Confirm the NOC backend service is running and reachable on port 8091.
- Check the browser console for the Keycloak error description; the portal surfaces the `error_description` field from the Keycloak token endpoint.
- Run `make noc.setup-keycloak` from `scenario-b/` if the `noc-portal` client does not yet exist in Keycloak.

Returned to the login page while working

- The portal renews its token automatically, so this normally only happens when the realm's SSO session limit is reached (10 hours by default) or the session was revoked in Keycloak. The message shown is *"Session expired. Sign in again."*
- If it happens after only a few minutes, the renewal call is failing: check the browser network tab for a rejected request to the Keycloak token endpoint, and confirm the portal's `VITE_KEYCLOAK_URL` points at a realm reachable from the browser.

Status bar shows STALE DATA

- The browser has not completed a backend request for three polling cycles. The monitored components may be perfectly healthy — what failed is the portal's own link to the NOC backend.
- Confirm the NOC backend container is up and answering on its published port.
- Check the browser console for CORS or connection errors; a portal built with the wrong `VITE_NOC_BACKEND_URL` shows exactly this symptom.

Dashboard shows no data or all components show UNKNOWN

- Verify the NOC backend service is running and that the spoke services are up.
- Navigate to **Infrastructure**, select a spoke, and click **Refresh** to trigger a manual health poll.
- Check the NOC backend service logs for database connectivity issues or spoke-endpoint unreachability.
- Run `make noc.setup-agents` from `scenario-b/` if spokes are not yet registered in the NOC backend.

Relay logs are empty while the relay is healthy

- Log collection needs the relay's container name in the entity's deployment manifest (`spec.relay.containerName`). Without it the component is registered with an endpoint only, so there is nothing for the agent to read logs from.
- After adding it, the entity must be re-applied (or its `noc-agent` configuration refreshed) for the change to reach the agent.

Relay page shows an empty table

- A spoke must be selected in the **Spoke Selector** before relay data is shown. Select a specific spoke — the table will populate with `CACTI_RELAY` components for that spoke.

- If the table remains empty after selecting a spoke, no CACTI relay components are registered for that spoke in the NOC backend. Verify the spoke was onboarded correctly.

Pool Stability shows no pools

- Read the page's own message first: it says whether the gateway answered with nothing, could not be read, or was never reached (see [Pool Stability](#)).
- When pairs are listed as unreadable, the NOC backend logs one line per failed pair with the gateway URL it tried: `docker logs <noc-backend container>`. A wrong `AMM_GATEWAY_URL`, or a gateway on a network the backend cannot reach, produces exactly this.
- *"not found among active pairs"* means the gateway is reachable but the configured pair does not exist there. Confirm the pair identifier in `AMM_PAIRS` matches an active pair on the api-gateway.
- If the corridor was never opened, there is genuinely nothing to show: a sovereign pair must be proposed, confirmed and funded with liquidity (done by the central banks from the Governance portal) before any pool exists.

Log Viewer shows no logs or an error

- Verify the component is running and that its endpoint is reachable from the NOC backend host.
- Click **Refresh** to force a new fetch.
- Some components produce minimal stdout output; check the `stderr` stream for error messages.
- If the backend returns an error, check the NOC backend logs for the `/components/:id/logs` request.

Alerts are not updating

- Check the **Polling Interval** in [Settings](#). If set to a high value, new alerts will appear infrequently.
- The WebSocket stream is not yet active — the portal always polls. This is expected behavior and not an error.
- If INFO and WARNING alerts are missing from the Dashboard feed, check whether **Critical Alerts Only** is enabled in [Settings](#). The setting persists across reloads, so it may have been left on by an earlier session in the same browser.
- Click **Refresh** on the Infrastructure page or navigate away and back to the Dashboard to trigger an immediate data pull.

Alert dismissed by mistake

- Dismissed alerts are moved to the **Resolved Alerts** table on the Audit page and cannot be re-activated from the portal.
- Contact the system administrator to re-raise the alert at the backend level if re-activation is required.

Topology diagram shows no nodes or all links red

- Navigate to **Infrastructure** and verify that component data is loading (indicates the backend is reachable and spokes are registered).
- If components load but the topology is empty, the topology endpoint (`/topology`) may have no data registered. Check the NOC backend for topology seeding.
- All-red links indicate that the NOC backend's link health checks are failing for every pair. Check backend logs for the health check job errors.

Settings do not persist after page reload

- Settings are stored in `localStorage` under the key `noc-ui-settings`. Verify the browser is not running in a private/incognito mode (which clears storage on close) or that storage is not being cleared by a browser extension.

End-to-End Walkthrough (Scenario B)

Audience: Anyone running or demonstrating a complete Scenario B cross-border payment — operators, integrators, reviewers, and demo drivers. **Scenario:** Scenario B — International Hub (FXAgreement + AMM + Bridge, with relay and circuit breaker). **Scope:** This document ties the five per-portal manuals together into one “golden path”: the ordered sequence of actions, across every portal, that moves value from a payer at Bank A to a beneficiary at Bank B and settles cleanly.

Each per-portal manual documents its own screens in isolation (see the “Typical Workflows” section in each). This walkthrough is the cross-portal view: who does what, in which portal, in what order, and what a successful result looks like at each hand-off.

This is the **happy path** — the successful case with no exceptions. Timeout, refund, rejection, and circuit-breaker paths are covered per-screen in the individual manuals and in the Troubleshooting section of each.

1. The cast

Actor	Portal	Manual
Central Bank A / B treasury (liquidity, issuance, tokenisation approvals)	Treasury	treasury.md
Central Bank A / B governance (registry, circuit breaker, oversight)	Governance	governance.md
Commercial Bank A operator (payer)	Bank	bank.md
Commercial Bank B operator (beneficiary)	Bank	bank.md
Supervisor / regulator (read-only monitoring)	Supervisor	supervisor.md
Network operations engineer (infrastructure and relay health)	NOC	noc.md

Running example used throughout: **Spoke A settles in BRL, Spoke B settles in ARS**. Bank A pays in its home currency (BRL) and Bank B receives the counterpart currency (ARS). Substitute your own corridor’s currencies as needed.

2. Prerequisites

Before you start, confirm:

- **Access.** You have the portal URL and sign-in credentials for each role you will act as, provided by your administrator or Central Bank. Each manual’s “Access and Login” section describes the sign-in screen for its portal.
- **A healthy network.** The network and relay must be operational so the bridge and commit-reveal steps can settle. A network operations engineer can confirm this in the NOC portal: all spokes and the hub report healthy on the Dashboard, and the relay is running. See [noc.md](#) → [Relay Status](#).

If a gateway or the relay is unavailable, the bridge and liquidity-commit steps stall in a pending state rather than fail loudly — check relay health first.

3. The golden path

The path has seven stages. Stages A–D are one-time set-up per corridor and per bank; Stages E–F are the payment itself; Stage G is routine liquidity management.

3.1 Stage A — Provision Hub liquidity (Central Banks)

The Hub AMM needs a funded pool for the corridor before any bank can transact. Both central banks contribute their side, and the pool activates only once both commits are revealed (2-of-N commit-reveal).

#	Actor	Portal → screen	Action	Expected result
A1	CB-A treasury	Treasury → Liquidity Provisioning	Propose the pair, then seed side A (bridge tCeBM to the Hub, approve the AMM, submit the liquidity commit)	CB-A commit registered
A2	CB-B treasury	Treasury → Liquidity Provisioning	Confirm the pair, then seed side B the same way	CB-B commit registered
A3	Relay	(automatic)	The relay observes both bridge locks and matches the commits	Both commits move to EXECUTED
A4	CB-A / CB-B treasury	Treasury → Liquidity Management	Refresh the pool	Pool status is ACTIVE , with non-zero reserves on both sides

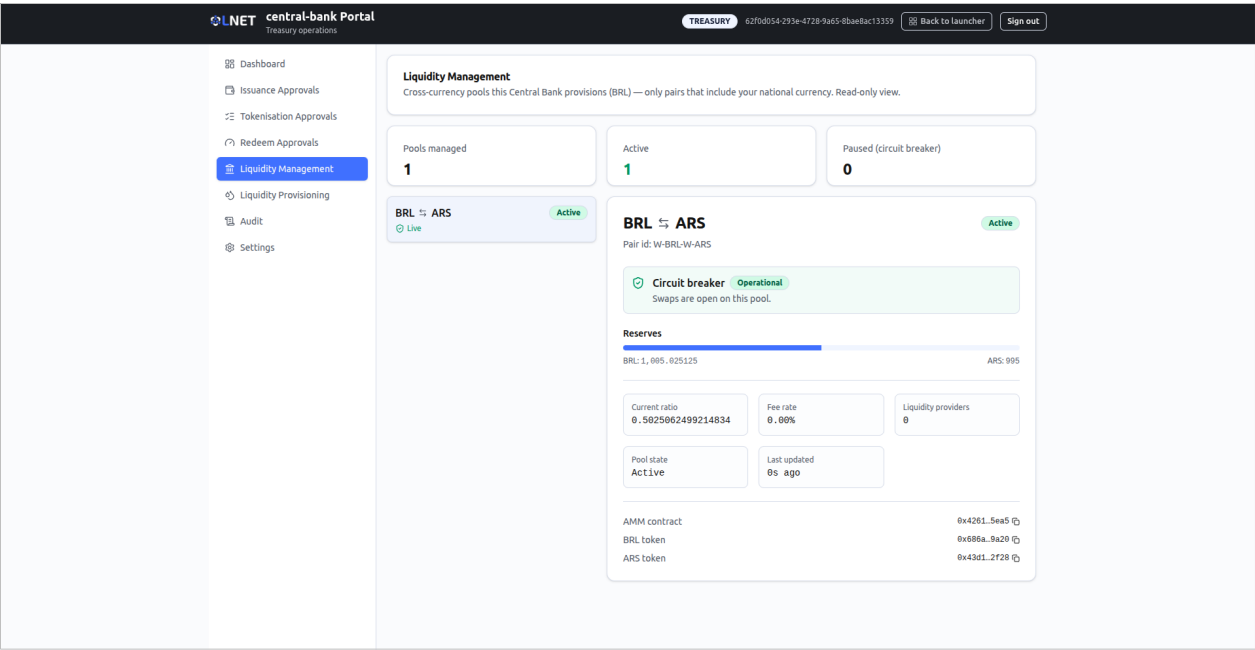


Figure 45: Treasury Liquidity Management with the corridor pool ACTIVE and reserves on both sides

3.2 Stage B — Onboard the commercial banks

Each commercial bank registers with its own central bank. Bank B must be onboarded too, because the payment resolves the beneficiary by bank identity.

#	Actor	Portal → screen	Action	Expected result
B1	Bank A operator	Bank → Onboarding	Complete the registration wizard	Onboarding submitted to CB-A
B2	CB-A governance	Governance → Registry	Approve Bank A's KYC	Bank A status becomes ACTIVE
B3	Bank B operator	Bank → Onboarding	Complete the registration wizard	Onboarding submitted to CB-B
B4	CB-B governance	Governance → Registry	Approve Bank B's KYC	Bank B status becomes ACTIVE

Bank A can now sign in and reach the operational screens (see [bank.md](#) → Access and Login).

3.3 Stage C — Issue fiat reserve (fCeBM)

Bank A converts a real fiat deposit into the on-chain fiat-reserve token (fCeBM).

#	Actor	Portal → screen	Action	Expected result
C1	Bank A operator	Bank → Issuance Requests / Deposits	Submit a deposit (issuance) request with the fiat collateral proof	Request created, status PENDING
C2	CB-A treasury	Treasury → Issuance Approvals	Approve the request	fCeBM minted to Bank A; request AP-PROVED/SETTLED
C3	Bank A operator	Bank → Dashboard	Refresh	Fiat Reserve (fCeBM) balance is non-zero

3.4 Stage D — Tokenise the reserve (tCeBM)

Bank A converts its approved fiat reserve into transferable CBDC (tCeBM). This is an atomic burn-fCeBM / mint-tCeBM operation on the central bank's side.

#	Actor	Portal → screen	Action	Expected result
D1	Bank A operator	Bank → Reserve Tokenisation / Escrows	Request tokenisation of the fCeBM reserve	Escrow created, status PENDING
D2	CB-A treasury	Treasury → Tokenisation Approvals	Approve the escrow	fCeBM burned, tCeBM minted to Bank A (both tx hashes recorded)
D3	Bank A operator	Bank → Dashboard	Refresh	tCeBM balance is non-zero — Bank A is ready to transact

3.5 Stage E — Send the cross-currency payment

This is the payment itself: Bank A spends its home-currency tCeBM and the Hub AMM delivers the counterpart currency toward Bank B. Direction is fixed by the selected pool's sovereignty — the payer does not choose currencies freely.

#	Actor	Portal → screen	Action	Expected result
E1	Bank A operator	Bank → Bridge	Select the corridor pool and request a quote (BRL → ARS)	Quote returned with an effective rate and a valid-for countdown
E2	Bank A operator	Bank → Bridge	Set the max amount in and the beneficiary bank ID, then Execute Bridge	Swap initiated with a swap ID and correlation ID; status begins progressing

#	Actor	Portal → screen	Action	Expected result
E3	Bank A operator	Bank → Bridge	Wait for the progress indicator to complete	Bridge Completed card: status COMPLETED , with amounts, tx hash, and correlation ID
E4	Bank A operator	Bank → Bridge History	Open the history	The operation appears with its route, amounts, rate, and COMPLETED status

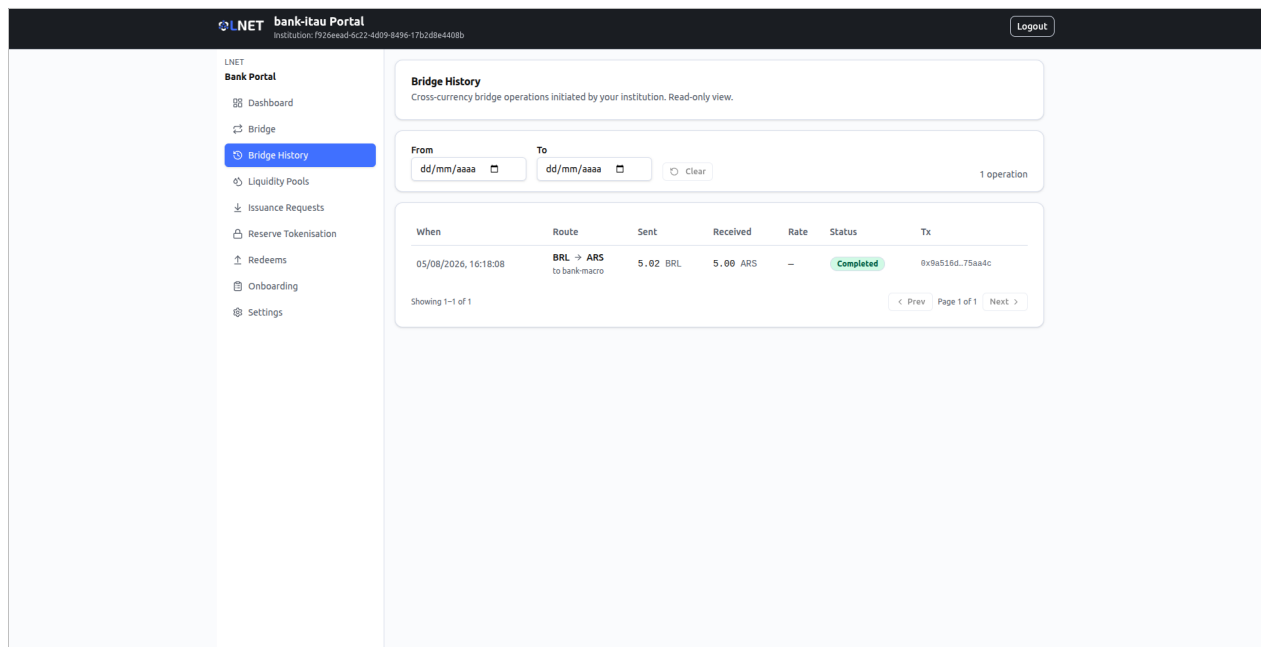


Figure 46: Bank Bridge History showing the cross-currency operation with its route, amounts, transaction hash and COMPLETED status

3.6 Stage F — Beneficiary receipt at Bank B

The Hub side of the swap mints the counterpart tCeBM toward Bank B. This is relay-mediated: the relay burns the Hub position and the beneficiary central bank's bridge delivers tCeBM to Bank B.

#	Actor	Portal → screen	Action	Expected result
F1	Relay	(automatic)	Relay processes the outbound leg; CB-B bridge position reaches BURNED	Counterpart tCeBM (ARS) minted to Bank B

#	Actor	Portal → screen	Action	Expected result
F2	Bank B operator	Bank → Dashboard	Refresh	tCeBM balance is non-zero — the payment is received
F3	CB governance	Governance → Swap Monitor	Review	The swap shows as completed end-to-end

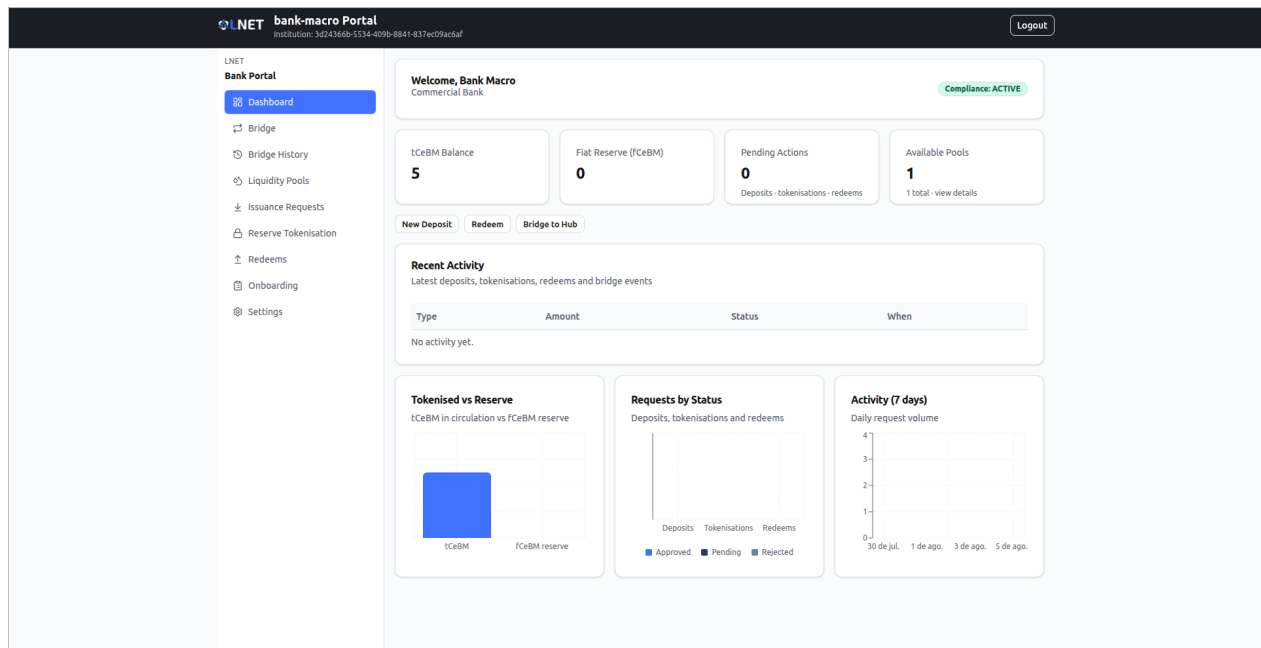


Figure 47: Bank B Dashboard showing the received tCeBM balance after the cross-currency payment settles

3.7 Stage G — Central Bank withdraws liquidity

Routine liquidity management: a central bank can withdraw part of its LP position back to its home currency. A partial withdrawal keeps the position ACTIVE.

#	Actor	Portal → screen	Action	Expected result
G1	CB-A treasury	Treasury → Liquidity Management	Review the on-chain LP-share position	LP shares and pool share are shown
G2	CB-A treasury	Treasury → Liquidity Provisioning	Withdraw a fraction of the position (home-currency exit)	LP shares decrease but remain greater than zero; the position stays ACTIVE

4. Oversight overlay (optional)

Central-bank oversight runs alongside the payment flow. These actions do not block the payment but exercise the control surface:

- **Circuit breaker.** A governance operator can pause AMM swaps for a pair (1-of-N) and resume them with a second signature (2-of-N). On the successful path the breaker stays LIVE throughout. See [governance.md](#) → [Circuit Breaker](#).
 - **Master viewing key disclosure.** A supervisor opens a disclosure request and a second party signs it (2-of-N) to reveal the details of a specific swap for investigation. See [supervisor.md](#) → [Investigation](#) and [governance.md](#) → [Oversight](#).
 - **Continuous monitoring.** Throughout the flow, the Supervisor portal's [Liquidity Monitor](#) and [Stability Insights](#) reflect pool health, and the NOC portal's [Pool Stability](#) view tracks reserve balance.
-

5. Success checklist

The walkthrough is complete when all of the following hold:

- ☐ Corridor pool is **ACTIVE** with non-zero reserves on both sides (Treasury → Liquidity Management).
- ☐ Bank A and Bank B are both **ACTIVE** in the Registry (Governance → Registry).
- ☐ Bank A holds a non-zero **tCeBM** balance after tokenisation (Bank → Dashboard).
- ☐ The swap reaches **COMPLETED** with a tx hash and correlation ID (Bank → Bridge / Bridge History).
- ☐ Bank B holds a non-zero counterpart **tCeBM** balance (Bank → Dashboard).
- ☐ The central bank's LP position remains **ACTIVE** after a partial withdrawal (Treasury → Liquidity Management).
- ☐ No stuck relay events and no unexpected circuit-breaker HALTED state (NOC → Relay Status; Governance → Circuit Breaker).